

Can Platforms Benefit from “Bad” Recommendations? Impact on Prices, Participation, and Welfare

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Recommendation systems are typically designed to optimize for immediate fit or compatibility, overlooking how these decisions influence the long-run strategic behavior of customers and suppliers. We show that recommending best-fit suppliers can backfire by intensifying seller competition and increasing customer selectivity. Instead, maximizing revenue may require strategically recommending lower-fit suppliers. We term this strategy *Scarcity by Design* and demonstrate how it can improve platform and supplier revenues. This paper provides the first equilibrium analysis of recommendation design that jointly incorporates forward-looking customer search, endogenous supplier pricing, and two-sided participation decisions. We derive structural results that restore tractability and characterize the optimal recommendation policy. We then generalize the analysis to personalized recommendations across heterogeneous customer segments. We prove that this high-dimensional problem with endogenous constraints reduces to tractable segment-level decisions. The optimal personalized strategy retains the *Scarcity by Design* structure of the single-segment case, but applies it selectively, tailoring the degree of scarcity to each segment’s characteristics. On the platform and supplier sides, we find that personalization consistently increases revenue. The gains arise from two sources: an allocative gain due to segment-aware allocation of suppliers and a feedback gain through equilibrium price adjustments driven by segment-specific scarcity, where customer patience mediates the relative strength of these forces. On the customer side, however, the aggregate welfare impact is ambiguous: personalization can lower surplus for some segments while raising it for others, generating distributional effects that often harm price-sensitive or more patient customers. This fairness concern suggests a role for regulatory oversight.

Key words: Recommendation, Pricing, Market Design, Stationary Equilibrium.

1. Introduction

Labor and freelance platforms (e.g., Catalant, Toptal, Thumbtack, Angi) rely heavily on recommendation systems to connect customers with providers. These systems create substantial economic value; for example, algorithmic recommendations have been shown to increase job vacancy fill rates by 20% (Horton 2017). By directing customer attention, recommendation systems shape engagement and hiring decisions and play a central role in overall platform performance. As the variety of available services and providers continues to expand, steering users toward the most relevant options becomes even more critical. Platforms, therefore, invest heavily in advanced recommendation

systems to generate precise matches (MacKenzie et al. 2013). For instance, Catalant uses machine-learning algorithms to match enterprise clients with independent consultants from a curated network of over 100,000 professionals¹; clients then evaluate recommended consultants sequentially through interviews before committing to a project engagement (Fuller et al. 2020). Similarly, Toptal combines AI-driven algorithms with a rigorous human screening process to recommend freelancers who match a client’s project requirements (Fuller et al. 2020). Empirical research consistently shows that such high-quality recommendations improve immediate customer outcomes, such as click-through and conversion rates (Ursu 2018a, Pei et al. 2019).

While maximizing immediate outcomes may seem desirable, it can lead to strategic responses that result in unintended long-run consequences. For instance, customers may adjust their search behavior by altering the length of their search and how selective they are. Because recommendations set expectations about customers’ future options, they can shift a customer’s willingness-to-pay (Zhang and Bockstedt 2016). While forward-looking and strategic customer behavior is well-documented in single-firm settings where customers respond to expected future prices, discounts, and availability (Li et al. 2014, Hendel and Nevo 2006, Busse et al. 2013), much less is known about strategic responses to recommendation policies in two-sided platform settings, where the platform’s recommendation policy impacts both customers’ expectations and suppliers’ decisions. On the supplier side, the recommendation policy directly determines the level of exposure for each supplier. In addition, by shaping customers’ search behavior, the policy changes the demand for suppliers’ offerings, which may prompt suppliers to adjust their prices and participation decisions. Therefore, in equilibrium, the recommendation design, customer search behavior, and seller pricing interact, which leads to complex interactions. These long-run equilibrium effects are often overlooked in the prior literature.

This paper provides, to our knowledge, the first formal analysis of optimal recommendation design in a two-sided marketplace when recommendations interact with strategic behavior on both sides of the market in equilibrium. Specifically, we ask: i) How should platforms optimally steer customers with heterogeneous preferences toward suppliers of varying attractiveness? ii) What are the mechanisms driving the optimal strategy? iii) What are its implications for platform revenue, social welfare, and the value of personalized recommendations? To address these questions, we develop a dynamic, infinite-horizon model that explicitly incorporates endogenous entry, strategic pricing by suppliers, and strategic sequential search by customers. We characterize the platform’s revenue-maximizing recommendation strategy in a stationary equilibrium. This framework uncovers long-term interactions between recommendation design, customer search behavior, and seller pricing that are often

¹ See <https://catalant.com>, accessed February 2026.

overlooked in static or myopic models. Our results characterize when maximizing immediate fit is optimal for long-run revenue, and when, instead, deliberately worsening recommendations may benefit one or all parties involved. We uncover the consequences of optimal recommendations for supplier profits and customer surplus. Interestingly, we show that personalization may generate asymmetric welfare effects, leading to systematically poorer outcomes for vulnerable customer segments.

1.1. Model

We analyze an infinite-horizon marketplace managed by a central platform that coordinates two strategic groups: suppliers and customers. Both parties aim to maximize long-term payoffs based on their types. Each period, potential customers enter the market with heterogeneous preferences and private valuations, while suppliers enter with ‘types’ that define their service fit. Entry on both sides is endogenous, driven by expected payoffs. Once participants enter, the platform matches them by recommending active suppliers to customers using segment-specific probability distributions.

In this setting, customers encounter heterogeneous suppliers sequentially, which is a structure common in B2B labor platforms like Toptal. Upon observing a supplier’s price and fit, customers choose to either purchase immediately or wait, balancing urgency against the prospect of better future offers. To reflect impatience and opportunity costs, customers who do not purchase exit with a fixed probability each period. Unlike a one-shot comparison over a simultaneous menu, this sequential search captures high-value service environments where fit must be assessed individually (e.g., through interviews). Suppliers remain active until they transact, setting prices to manage the trade-off between quicker sales and higher per-transaction profits. This combination of forward-looking sequential search, endogenous supplier pricing, and endogenous entry on both sides distinguishes our framework from existing studies that typically assume one-shot choice, myopic customers, or exogenous prices (see Section 2 for a detailed comparison).

The platform chooses a recommendation policy to maximize steady-state commission revenue. To select the optimal recommendation policy, the platform must anticipate the long-term equilibrium consequences of its policy, accounting for endogenous responses: how recommendations influence supplier pricing and entry, as well as customer search and entry. This perspective contrasts with models that optimize for myopic objectives like immediate conversion or match quality. We analyze the system in a steady state where all participant strategies are mutual best responses to the platform’s policy, which ensures a stable market structure and price distribution over time.

1.2. Main Results and Contributions

Our main contribution is to show that recommending best-fit suppliers can backfire in the long run by intensifying seller competition and increasing customer selectivity. Instead, maximizing revenue may require strategically recommending lower-fit suppliers. We term this strategy *Scarcity by Design*

and demonstrate how it can improve platform and supplier revenues. We further show that this strategy can lower customer surplus of some segments while raising the surplus of others, generating asymmetric distributional effects that often harm price-sensitive customers or those with higher patience. This fairness concern suggests a role for regulatory oversight.

To tackle this analytically challenging problem, we first derive structural results that make the equilibrium analysis tractable. We first establish the existence and uniqueness of the stationary equilibrium (Theorem 1), which ensures that comparisons across recommendation policies are well-posed. We then characterize the equilibrium strategies. We show that customer purchase thresholds and willingness-to-pay are strictly increasing in valuations (Proposition 1). We show that suppliers' posted prices increase with fit quality, while, interestingly, the *effective price* defined as price plus misfit disutility (i.e., how poorly the supplier fits the customer preferences) rises with the misfit level (Proposition 2). Leveraging these monotonicity results, we prove that the distribution of customer willingness-to-pay is identical to the arrival valuation distribution, up to a scaling factor (Theorem 2), which is a key result for tractability.

Building on these foundational results, we investigate the optimal recommendation and its implications, starting with a single customer segment to disentangle the fundamental tradeoffs. The central insight, formalized in Theorem 3, is that a revenue-maximizing platform does not always benefit from steering customers exclusively toward their best-fit suppliers. Instead, the optimal policy concentrates exposure on the "best" and "worst" supplier types while excluding the middle of the spectrum.

Next, we uncover the mechanism underlying this counterintuitive result. Propositions 3 and 4 demonstrate that the platform's optimal strategy capitalizes on a fundamental feedback loop: the recommendation policy modulates customer search patterns and selectiveness, which in turn influences the competition effect for suppliers. These endogenous responses alter pricing strategies, further reshaping customer behavior. Specifically, while Proposition 3 isolates the equilibrium effect on willingness-to-pay, Proposition 4 reveals that increasing exposure to lower-fit suppliers can raise both supplier prices and, therefore, platform commissions. Proposition 5 then identifies the market conditions under which restricting exposure to best-fit suppliers is optimal. Regarding welfare, we show that revenue-maximizing steering typically reduces customer surplus while increasing supplier profits (Propositions 6). This highlights a trade-off: incentive alignment among all three parties occurs only when customers are relatively impatient or when the market has limited supplier capacity, i.e., precisely when scarcity injection is unprofitable.

Finally, reflecting the ubiquity of personalized systems and the heterogeneity of customer preferences in practice, Section 6 generalizes the model to an arbitrary number of heterogeneous segments.

This generalization allows customers to vary in preferences, arrival rates, patience, and willingness-to-pay. In the multi-segment setting, the recommendation policy couples segments through the endogenous price and search responses to the recommendation policy, and, in addition, through shared supplier capacity. This coupling creates a high-dimensional allocation problem embedded within the equilibrium framework. Theorem 4 shows that, even with heterogeneous segments and segment-specific personalization, the optimal recommendation policy has the same structure as in the single-segment case (Theorem 3). In particular, it concentrates exposure on the best and the worst supplier types while excluding the middle of the spectrum. This polarized structure holds within each segment’s personalized recommender, and it also holds in the aggregate when exposures are pooled across segments. Key to Theorem 4 is a property we term Segmental-Exclusivity (SE). This property dictates that for any type of supplier, the platform’s optimal strategy is to assign that type’s entire available capacity to just one customer segment. In essence, the optimal allocation becomes locally exclusive. This finding is powerful because it dramatically simplifies a complex, high-dimensional allocation problem with endogenous constraints. Instead of a single, highly coupled decision, the problem elegantly collapses into a collection of one-dimensional choices for each supplier type.

Leveraging this key property and the structure of optimal recommendations, we quantify the value of personalization by comparing the “personalized” and “non-personalized” regimes. In the personalized regime, the platform conditions recommendations on the customer’s true segment; in the non-personalized regime, it pools segments and designs a single recommendation distribution. Proposition 7 provides a canonical decomposition of the value of information from personalization into two mechanisms: an allocative gain, which captures the allocative efficiency from re-routing exposure across segments holding the equilibrium fixed, and a feedback gain, which captures the equilibrium-price mechanism induced by strategic customer response. We show that, while personalization always increases platform revenue, its impact on customer surplus is ambiguous because the allocative gain can be offset by the feedback gain. Our numerical analysis illustrates that personalization allows platforms to tailor diversion based on customer segment, which can lead to distributional welfare effects. For instance, it can disadvantage customers with lower valuations² or higher patience, highlighting equity concerns arising from algorithmically optimized markets.

2. Literature

Our work relates to four streams of literature: (i) rankings and consumer search, (ii) strategic scarcity and rationing, (iii) platform design in two-sided markets, and (iv) recommendation design under strategic seller responses.

² Although we do not directly model socioeconomic status (SES); in many markets, lower observed valuations can reflect tighter budgets and may correlate with lower SES. Hence, our findings suggest that personalization may lead to exploitation of customers with lower SES.

Empirical evidence shows that platform ranking and recommendation choices systematically steer how customers search and allocate attention in online marketplaces (De los Santos and Koulayev 2017, Ghose et al. 2014, Ursu 2018b, Lee and Musolff 2025). These design choices can have economically meaningful downstream effects, potentially on purchases, welfare, and even willingness-to-pay (Ghose et al. 2014, Ursu 2018b, Lee and Musolff 2025, Adomavicius et al. 2018). Several theoretical works have modeled recommendation as an assortment or ranking problem, analyzing how product position, ordering, or framing affects choices. Customers are assumed to inspect displayed products and choose according to stochastic-choice specifications such as random-utility, multinomial-logit, or consideration-set models (Jagabathula and Rusmevichientong 2017, Akchen and Mitrofanov 2025). The platform then optimizes demand or engagement through page-level framing and pricing (Abeliuk et al. 2016, Gallego et al. 2020, Lei et al. 2022), sequential ranking design (Asadpour et al. 2023), or learning-to-rank from clickstream data (Ferreira et al. 2022). These papers typically model the platform’s decision as choosing from a fixed catalog and therefore abstract away from customers’ and suppliers’ long-term strategic reactions.

In single-seller settings, the strategic scarcity and rationing literature (Liu and Van Ryzin 2008, Gallego et al. 2008, Cachon and Swinney 2011) shows that firms can increase profitability by intentionally limiting inventory; this physical scarcity creates a credible threat of stockouts that incentivizes customers to purchase at full price rather than waiting. In a different vein, Balseiro et al. (2024) work on mechanism design with boundedly rational agents. They show that spreading exposure beyond the best match can be optimal when participants are boundedly rational. Although our two-sided market context and equilibrium concept are different, all three settings share a common insight: deliberately withholding the best option can be beneficial. In our model, this principle operates through an equilibrium channel in which the recommendation algorithm rations information rather than physical goods.

In two-sided markets, several related works study various platform design levers. These include pricing under inter-platform competition and congestion (Bernstein et al. 2021), how connectivity shapes competitive outcomes in matching markets (Kanoria et al. 2025), dynamic platform growth through compensation design (Ekbatani and Caldentey 2025), advertising and creator compensation on content platforms (Ma et al. 2025), signaling congestion or queue lengths to market participants (Besbes et al. 2023, Zhu et al. 2025), and routing demand across inventory-holding sellers (Cao and Hu 2025). The mechanism central to our analysis, whereby recommendation design shapes strategic customer search and thereby determines equilibrium pricing and entry, is absent from this literature.

While our research question is distinct, our modeling approach shares core elements with a body of dynamic two-sided matching models. Similar to this literature, we assume a continuum (fluid) model of a platform with strategic agents, focus on stationary equilibria, and analyze how platform design

levers affect equilibrium outcomes. Kanoria and Saban (2021) develop a dynamic fluid model of a two-sided matching platform where agents must bear a cost to screen each potential partner; they characterize evolutionarily stable stationary equilibria and show that restricting which side initiates contact and hiding vertical quality information can significantly improve welfare. Immorlica et al. (2023) consider a fluid matching market where the platform selects meeting rates between types; they characterize the induced stationary equilibrium and design approximately optimal directed search policies, highlighting that substantially limiting choice can still achieve near-optimal outcomes. Aouad et al. (2025) study centralized versus decentralized pricing controls in a fluid dynamic matching platform with strategic agents; their stationary equilibrium features a related willingness-to-pay characterization. Relatedly, Ashlagi et al. (2020) and Rios and Torrico (2023) study match recommendations and display schemes in dynamic matching environments and derive performance guarantees.

There is also a stream of literature on recommendation design under strategic seller responses. Closest to our setting, Papanastasiou and Sekar (2024) studies optimal ranking design with endogenous seller participation and strategic sequential search, but takes prices as exogenous. Zhou and Zou (2023) study the impact of personalized recommendations on seller price competition in a single-period model with one-shot consumer choice, but do not consider optimal recommendation design or endogenous entry. Shi (2024) studies optimal recommendation design with endogenous pricing, a question closer to ours, but models customers as myopic rather than as forward-looking sequential searchers, and takes participation on both sides as given. Our work jointly incorporates all of these ingredients: optimal recommendation design, endogenous pricing, strategic sequential search, and endogenous entry on both sides, within a dynamic stationary equilibrium.

3. Model

We study a marketplace mediated by a central platform over an infinite discrete-time horizon. We call the two sides “supply” and “demand”, with agents referred to as “suppliers” and “customers”. Suppliers and customers make strategic decisions to maximize their expected long-term payoffs based on their costs and valuations, given the platform recommendation policy. At the beginning of each period, a continuum of potential customers arrives, each endowed with a private valuation for the product drawn from a known distribution. Simultaneously, a mass of potential suppliers enters, each characterized by an observed fit level reflecting how closely their product matches customers’ preferences. Customers and suppliers decide independently and endogenously whether to enter the platform based on expected profitability.

Among customers who enter, each is directed by the platform’s recommendation algorithm to inspect a supplier drawn according to a distribution over active suppliers. We model this as a

sequential search process, where customers evaluate options one at a time, though we allow for recall, where customers can choose previously evaluated options. While parallel browsing (often modeled via discrete choice models) is common in settings with standardized products and negligible evaluation time and resources (e.g., online retail), sequential search better captures the dynamics of high-value service and labor markets. For instance, many B2B labor platforms (e.g., Toptal, Catalant) expose freelancers sequentially to their customers. Indeed, in labor and service markets (e.g., hiring a consultant, evaluating a B2B provider), assessing fit and quality requires significant time and cognitive investment, making simultaneous evaluation impractical or prohibitively costly. Upon recommendation, the customer observes the supplier’s price and their realized fit (equivalently, the segment-specific misfit disutility). After observation, each customer decides either to accept the offer (generating an immediate transaction) or to remain unmatched for another period. Although we do not assume an explicit inspection cost, the cost of search in our model is captured implicitly by customers’ impatience: customers who do not purchase leave the platform with some probability at the end of each period.³ Thus, customers strategically balance between accepting an available offer quickly and holding out for a better match and lower-price suppliers in the future. Suppliers, on the other hand, remain active in the market across periods until they successfully sell a single product, after which they exit immediately. We note here that our framework can also accommodate multi-unit suppliers by treating each unit of capacity as a replica supplier of the same type. Each supplier with an unsold product stays in the market for future periods. Thus, suppliers face a trade-off between setting lower prices to achieve a sale versus higher prices to maximize profit per transaction. Importantly, the platform’s choice of recommendation algorithm affects supplier prices through two paths: (i) directly, by determining how often each supplier is exposed to potential customers, and (ii) indirectly, by shaping customers’ purchase decision threshold upon inspecting products.

The platform charges suppliers an ad valorem (commission) fee proportional to the value of each successful transaction. The platform’s objective is to choose its recommendation distribution optimally to maximize its total commission revenue. We analyze the system in steady state, focusing on stationary equilibria in which the inflows and outflows of customers and suppliers are balanced, and each agent’s strategy represents a best response given aggregate market conditions (Hopenhayn 1992). These aggregate conditions include the total number of active customers and suppliers, as well as the distributions governing the strategies of suppliers and customers that also determine their entry, as made formal in Section 3.4.

³ We model the departure probability (formally μ_j in Section 3.1) as exogenous. This allows us to present a clean characterization of the unique steady state and WTP distributions in our dynamic equilibrium problem that underpin the analysis of the platform’s optimization problem. In Appendix K, we extend our model and allow the departure decision to be endogenous. We show that the structural properties of the equilibrium is preserved, and our main qualitative insights are robust to this generalization.

3.1. Customers’ Problem

Arrival process. We consider a setting with J customer segments indexed by $j \in \{1, \dots, J\}$. These segments allow us to capture systematic heterogeneity in customer preferences, valuations, and patience, which is essential for studying the value and impact of personalized recommendations in Section 6. In each period, an exogenous mass λ_j of segment- j *potential* customers arrive. While the arrival rate λ_j is exogenous, the decision to actively participate in the market is endogenous and depends on the expected surplus (detailed below). Each customer has a private valuation v drawn from a segment-specific uniform distribution on $[\underline{v}_j, \bar{v}_j]$, where $0 \leq \underline{v}_j < \bar{v}_j \leq 1$. We allow these valuation ranges to overlap.

Recommendation and customer search. The platform steers customers via the recommendation policy $\mathbf{D} := (D_j)_{j=1}^J$. Here, D_j is a probability distribution over the space of supplier types, $\delta \in [0, 1]$ (with $\int_0^1 D_j(\delta) d\delta = 1$), representing the platform’s recommendation rule for segment j . That is, when a segment- j customer searches, a supplier type δ is drawn from D_j . This segment-specific recommendation captures preference heterogeneity: the same supplier might be perceived differently by different groups. For example, a consultant specializing in rapid prototyping (a specific type δ) might be a great fit for a startup segment (low $d_j(\delta)$) but a poor fit for a large corporation needing scalable solutions (high $d_j(\delta)$). We model this disutility as linear in the supplier type, $d_j(\delta) = a_j\delta + b_j$ with $a_j \neq 0$. The realized net utility from accepting the offer is therefore $v - p(\delta; \mathbf{D}) - d_j(\delta)$ where $p(\delta; \mathbf{D})$ is the price charged by type δ under policy $\mathbf{D}$. Although segments may have distinct preferences, they compete for the same pool of suppliers, coupling their outcomes through the platform’s policy and equilibrium prices. After observing this utility, the customer decides whether to purchase the inspected product or continue searching⁴, comparing the immediate payoff to the continuation value (reward-to-go) from waiting for future opportunities. Customers who do not purchase exit the market at the end of the period with probability μ_j , reflecting limited patience and the implicit cost of waiting and inspection. With probability $1 - \mu_j$, they remain active and are recommended another supplier in the next period according to D_j .

Customer best-response. Given this setting, the customer’s best response can be formulated as a dynamic program. Since each customer is infinitesimal, their actions do not affect the aggregate steady-state; hence, each customer’s problem is an optimal stopping problem with i.i.d. rewards, which is known to admit the threshold policy as optimal. It follows that the continuation value depends on the customer’s valuation v and the recommendation profile $\mathbf{D}$. We represent this dependence by the threshold $\theta_j(v, \mathbf{D})$. This threshold structure implies a simple cutoff rule: customers accept an offer if their realized utility exceeds $\theta_j(v, \mathbf{D})$. This condition is captured by the indicator

⁴ We show in Appendix A.1 that allowing customers to recall previously inspected options does not change the optimal acceptance rule, and hence is without loss of generality.

function $\mathbf{1}\{v - p(\delta, \mathbf{D}) - d_j(\delta) \geq \theta_j(v, \mathbf{D})\}$. Hence, the strategy space can be reduced without loss of generality to valuation-dependent thresholds. Formally, given a threshold $\theta_j(v, \mathbf{D})$ and recommendation distribution D_j , the expected lifetime surplus $\pi_{c,j}(v, \theta_j(v, \mathbf{D}))$ consists of two components. First, the *immediate surplus*, realized if utility exceeds the threshold. Second, if the offer is rejected, the *continuation surplus*, discounted by the probability $(1 - \mu_j)$ that the customer remains active in the market. Accordingly, the Bellman equation is

$$\begin{aligned} \pi_{c,j}(v, \theta_j(v, \mathbf{D})) = & \mathbb{E}_{\delta \sim D_j} \left[(v - d_j(\delta) - p(\delta; \mathbf{D})) \mathbf{1}(v - d_j(\delta) - p(\delta; \mathbf{D}) \geq \theta_j(v, \mathbf{D})) \right] \\ & + (1 - \mu_j) \Pr_{\delta \sim D_j} (v - d_j(\delta) - p(\delta; \mathbf{D}) < \theta_j(v, \mathbf{D})) \pi_{c,j}(v, \theta_j(v, \mathbf{D})), \end{aligned}$$

which can be written explicitly as

$$\pi_{c,j}(v, \theta_j(v, \mathbf{D})) = \frac{\mathbb{E}_{\delta \sim D_j} [(v - d_j(\delta) - p(\delta; \mathbf{D})) \mathbf{1}(v - d_j(\delta) - p(\delta; \mathbf{D}) \geq \theta_j(v, \mathbf{D}))]}{\mu_j + (1 - \mu_j) \Pr_{\delta \sim D_j} (v - d_j(\delta) - p(\delta; \mathbf{D}) \geq \theta_j(v, \mathbf{D}))}.$$

Thus, each potential customer with valuation v chooses their optimal threshold $\theta_j(v, \mathbf{D})$ to maximize this expected lifetime surplus. Crucially, the decision to enter the market (i.e., participate in the search) is endogenous: a customer enters only if their maximized expected surplus is non-negative. Aggregating across all customer valuations, we define $\mathcal{V}_j = \{v \in [0, 1] : \pi_{c,j}(v, \theta_j(v, \mathbf{D})) \geq 0\}$ as the set of customers of segment j who enter the market.

3.2. Suppliers' Problem

Arrival process. At the beginning of each period, an exogenous mass m of *potential* suppliers arrive. Upon arrival, suppliers decide whether to enter the market and, if they do, which price to post. Each supplier is characterized by a public type $\delta \in [0, 1]$, drawn from a distribution F_Δ . The type δ represents the supplier's objective characteristics or specialization (e.g., a freelancer's specific skill set). Importantly, δ is universal across customer segments; however, the heterogeneity in how different segments value this type is captured by the segment-specific misfit disutility $d_j(\delta)$ (Section 3.1). While suppliers know the sale probability given a price they set, which is induced by customer thresholds $\{\theta_j(\cdot, \mathbf{D})\}$, they do not necessarily need knowledge of the customer thresholds.

Exposure and sale probabilities. Let $s(\delta, \mathbf{D})$ denote the steady-state *supplier density* by type, and let $C_j(\mathbf{D})$ denote the steady-state mass of active customers in segment j given $\mathbf{D}$. Then, the per-period exposure probability that a type- δ listing is shown (to any segment) is

$$\psi(\delta, \mathbf{D}) = \min \left\{ 1, \frac{\sum_{j \in \{1, \dots, J\}} C_j(\mathbf{D}) D_j(\delta)}{s(\delta, \mathbf{D})} \right\}.$$

Note that the term $\sum_{j \in \{1, \dots, J\}} C_j(\mathbf{D}) D_j(\delta)$ is the total number of per-period display slots that the policy $\mathbf{D}$ allocates to type- δ suppliers across segments $j \in \{1, \dots, J\}$. A higher supplier density

$s(\delta, \mathbf{D})$ lowers the probability that any individual type- δ listing is shown, since this customer flow must be divided across more suppliers. Similarly, let $\psi_j(\delta, \mathbf{D})$ denote the per-period probability that a type- δ supplier is shown to a customer in segment $j \in \{1, \dots, J\}$. Then we have

$$\psi_j(\delta, \mathbf{D}) = \frac{C_j(\mathbf{D}) D_j(\delta)}{\sum_{j' \in \{1, \dots, J\}} C_{j'}(\mathbf{D}) D_{j'}(\delta)} \psi(\delta; \mathbf{D}) \text{ for } j \in \{1, \dots, J\}.$$

This expression states that the probability a type- δ supplier is shown to segment j equals the overall probability of being shown, $\psi(\delta, \mathbf{D})$, multiplied by segment j 's proportional weight among all segments.

Next, we characterize the probability of sale for a supplier of type δ . Let F_{V_j} denote the steady-state distribution of valuations among active customers in segment j . Remember that if a type- δ supplier charging price p is recommended to a segment- j customer, a sale occurs if and only if $v - p - d_j(\delta) \geq \theta_j(v, \mathbf{D})$. Thus, conditional on recommendation to segment j , the probability of sale is $\Pr_{v \sim F_{V_j}}(v - p - d_j(\delta) \geq \theta_j(v, \mathbf{D}))$. Since a type- δ supplier is recommended to segment j with probability $\psi_j(\delta, \mathbf{D})$, the per-period sale probability is the weighted average of these conditional probabilities across segments $\phi_S(p, \delta, \mathbf{D}) = \sum_{j \in \{1, \dots, J\}} \psi_j(\delta, \mathbf{D}) \Pr_{v \sim F_{V_j}}(v - p - d_j(\delta) \geq \theta_j(v, \mathbf{D}))$. We consider that each supplier supplies a single unit. This framework readily accommodates multi-unit suppliers (i.e., firms selling finitely many units) by treating each unit as a replica supplier of the same type. Under this interpretation, the arrival mass m represents the total arriving *capacity* (mass of units). This approach does not alter the subsequent analysis.

Supplier best-response. The platform charges an ad-valorem commission $\tau \in [0, 1]$, leaving the supplier with revenue $(1 - \tau)p$ upon a sale. Suppliers who do not sell their unit, remain active. Given this, a type δ supplier, posting price p , earns a expected per-period profit of $\pi_s(\delta, p, \mathbf{D}) = (1 - \tau)p \phi_S(p, \delta, \mathbf{D})$. Each supplier of type δ chooses a price to maximize $\pi_s(\delta, p, \mathbf{D})$ and enters the market if and only if the expected profit is positive, that is, if $\pi_s(\delta, p, \mathbf{D}) > 0$. Hence, the equilibrium entry set is determined endogenously as $\mathcal{S} = \{\delta \in [0, 1] : \pi_s(\delta, p, \mathbf{D}) > 0\}$.

Since supplier profit $\pi_s(\delta, p, \mathbf{D})$ depends on the recommendation policy $\mathbf{D}$ through the sale probability $\phi_S(p, \delta, \mathbf{D})$, the algorithm itself becomes a key determinant of suppliers' strategies. First, by allocating recommendation weights $D_j(\delta)$, the platform directly determines the showing probabilities $\psi_j(\delta, \mathbf{D})$. Second, because the distribution of suppliers that customers expect to encounter depends on $\mathbf{D}$, the recommendation policy indirectly shapes customer acceptance thresholds $\theta_j(v, \mathbf{D})$. Together, these direct and indirect paths explain how the platform's recommendation policy influences equilibrium supplier entry and pricing decisions.

3.3. Equilibrium Concept

We now formalize the notion of a stationary equilibrium by combining optimal agent strategies and market-consistency conditions. For a segment- j customer with valuation v , let $q_j(v, \mathbf{D})$ denote the probability of making a purchase in a given period under $\mathbf{D}$. Accordingly, $q_j(v, \mathbf{D}) = \int_{\mathcal{S}(\mathbf{D})} \mathbf{1}\{v - p(\delta, \mathbf{D}) - d_j(\delta) \geq \theta_j(v, \mathbf{D})\} D_j(\delta) d\delta$, where $\mathcal{S}(\mathbf{D})$ denotes the set of suppliers that are active in steady state under $\mathbf{D}$. Furthermore, recall from Section 3.2 that $C_j(\mathbf{D})$ denotes the steady-state mass of active customers in segment j . To make clear how this mass is distributed across valuations, we now introduce the corresponding steady-state density. Let $c_j(v, \mathbf{D})$ denote the steady-state density of active segment- j customers with valuation $v \in \mathcal{V}_j(\mathbf{D})$, so that $C_j(\mathbf{D}) = \int_{\mathcal{V}_j(\mathbf{D})} c_j(v, \mathbf{D}) dv$. Stationarity requires inflows to equal outflows for every customer valuation v and supplier type δ :

$$\text{(Customers)} \quad \frac{\lambda_j}{\bar{v}_j - \underline{v}_j} \mathbf{1}_{\{v \in \mathcal{V}_j(\mathbf{D})\}} = [\mu_j(1 - q_j(v, \mathbf{D})) + q_j(v, \mathbf{D})] c_j(v, \mathbf{D}), \quad \forall j, \forall v \in [\underline{v}_j, \bar{v}_j], \quad (1a)$$

$$\text{(Suppliers)} \quad m f_{\Delta}(\delta) \mathbf{1}_{\{\delta \in \mathcal{S}(\mathbf{D})\}} = s(\delta, \mathbf{D}) \phi_S(p(\delta, \mathbf{D}), \delta, \mathbf{D}), \quad \forall \delta \in [0, 1]. \quad (1b)$$

Equation (1a) balances the flow of segment- j customers with valuation v . Customers arrive to segment j at rate λ_j and valuations are uniform on $[\underline{v}_j, \bar{v}_j]$, which implies an inflow density $\lambda_j/(\bar{v}_j - \underline{v}_j)$. The indicator $\mathbf{1}_{\{v \in \mathcal{V}_j(\mathbf{D})\}}$ equals one when valuation v is active under recommendation profile $\mathbf{D}$ and equals zero otherwise, so it restricts the balance condition to the set of valuations that enter and remain relevant in steady state. The customer balance equation (1a) equates the inflow of customers of valuation v to their outflow, due to either purchasing or exogenous exit.

Equation (1b) balances the flow of suppliers of type δ . A mass $m f_{\Delta}(\delta)$ of type- δ suppliers enters, and the indicator $\mathbf{1}_{\{\delta \in \mathcal{S}(\mathbf{D})\}}$ equals one for types that participate under $\mathbf{D}$ and equals zero otherwise, restricting attention to active types in steady state. The supplier balance equation (1b) states that supplier inflow of type δ equals the outflow resulting from successful sales. With this, we next formally define the stationary equilibrium.

DEFINITION 1 (STATIONARY EQUILIBRIUM GIVEN $\mathbf{D}$). Given a recommendation policy $\mathbf{D}$, a stationary equilibrium is characterized by a tuple of strategies (θ^*, p^*) , entry sets $(\mathcal{V}^*, \mathcal{S}^*)$, and steady-state distributions $(\{c_j^*\}_{j=1}^J, s^*)$ such that:

- (i) **Customer Optimality.** For all j and $v \in \mathcal{V}_j^*$, the strategy $\theta_j^*(v)$ maximizes the customer's expected lifetime surplus $\pi_{c,j}$ (defined in 3.1), taking the equilibrium supplier price p^* and policy $\mathbf{D}$ as given.
- (ii) **Supplier Optimality.** For all $\delta \in \mathcal{S}^*$, the strategy $p^*(\delta)$ maximizes the supplier's expected profit π_s (defined in 3.2), taking the equilibrium customer strategies θ^* and the steady-state distributions (which determine exposure ψ_j and F_{V_j}) as given.

- (iii) **Market Consistency (Steady State).** The distributions $(\{c_j^*\}_{j=1}^J, s^*)$ satisfy the balance equations (1a) and (1b) when evaluated at the strategies (θ^*, p^*) .
- (iv) **Endogenous Entry.** Entry occurs if and only if the expected payoff under the equilibrium strategies is non-negative: $\mathcal{V}_j^* = \{v \mid \pi_{c,j}(v, \theta_j^*(v)) \geq 0\}$ and $\mathcal{S}^* = \{\delta \mid \pi_s(\delta, p^*(\delta)) \geq 0\}$.

For notational simplicity in the definition above and the subsequent analysis, we use the asterisk notation (e.g., θ^*) to denote equilibrium objects, suppressing their explicit dependence on $\mathbf{D}$ unless necessary for clarity.

We next establish the existence and uniqueness of the stationary equilibrium. The proof (Appendix F) constructs a contraction in the vector of mean effective prices $\mathbf{M} = (M_1, \dots, M_J)$ where $M_j = \mathbb{E}_{\delta \sim D_j}[p^*(\delta) + d_j(\delta)]$, and applies Banach’s Fixed-Point Theorem.

THEOREM 1 (Existence & uniqueness of stationary equilibrium). *Given any recommendation policy $\mathbf{D}$, there exists a unique stationary equilibrium.*

Theorem 1 ensures that, for any fixed recommendation profile $\mathbf{D} = (D_j)_{j=1}^J$, there exists a unique induced stationary outcome (acceptance thresholds $\{\theta_j^*\}_{j=1}^J$, optimal prices p^* , active densities (c_j^*, s^*) , and exposure probabilities). Hence, all equilibrium objects can be represented as functions of $\mathbf{D}$, which makes the platform’s optimization problem defined in Section 3.4 well-posed. This existence and uniqueness property allows us to optimize recommendations without equilibrium selection issues and state our structural results globally, rather than equilibrium-by-equilibrium, including monotonicity of strategies and distributional properties of the induced outcomes. This, in turn, enables clean comparative statics and welfare analysis in subsequent sections.

3.4. Platform’s Problem

The platform’s objective is to maximize the commission revenue. Specifically, the platform selects recommendation policies $\mathbf{D} := (D_j)_{j \in \{1, \dots, J\}}$, anticipating the resulting unique stationary equilibrium (as shown in Theorem 1). With commission rate $\tau \in (0, 1)$, the platform earns $\tau p^*(\delta, \mathbf{D})$ from each completed transaction at price $p^*(\delta, \mathbf{D})$. The total mass of transactions occurring per period for suppliers of type δ is given by the steady-state mass of these suppliers, $s^*(\delta, \mathbf{D})$, multiplied by their per-period sale probability, $\phi_S(p^*(\delta, \mathbf{D}), \delta, \mathbf{D})$. Thus, the per-period platform revenue is given by

$$R(\mathbf{D}) = \tau \int_{\mathcal{S}} s^*(\delta, \mathbf{D}) \phi_S(p^*(\delta, \mathbf{D}), \delta, \mathbf{D}) p^*(\delta; \mathbf{D}) d\delta.$$

The integral aggregates the transaction value across all active supplier types $\delta \in \mathcal{S}$. This expression is mathematically equivalent to the customer-side aggregation of expected payments. Hence, the platform chooses $\mathbf{D}$ that maximizes $R(\mathbf{D})$.

While the platform has full control over the recommendation distribution $\mathbf{D}$, it must ensure feasibility: the number of customers directed to any supplier type δ cannot exceed the available

mass of type- δ suppliers in the market, $s^*(\delta, \mathbf{D})$. Specifically, $s^*(\delta, \mathbf{D}) = \frac{m f_{\Delta}(\delta)}{\phi_S(p^*, \delta, \mathbf{D})}$ where $m f_{\Delta}(\delta)$ is the inflow rate of type- δ suppliers and the denominator is the rate at which suppliers of type δ make a sale. This expression follows from the expected lifetime of a supplier under a geometric distribution with success probability $\phi_S(p^*, \delta, \mathbf{D})$. Formally, the recommendation policy $\mathbf{D}$ must satisfy $\sum_{j=1}^J C_j(\mathbf{D}) \cdot D_j(\delta) \leq s^*(\delta, \mathbf{D}) = \frac{m f_{\Delta}(\delta)}{\phi_S(p^*, \delta, \mathbf{D})}$. Under this constraint, the platform solves the optimization problem OPT.

$$\max_{\mathbf{D}: [0,1]^J \rightarrow \mathbb{R}_{\geq 0}} R(\mathbf{D}) \quad (2a)$$

$$\text{(OPT) s.t.} \quad \int_0^1 D_j(\delta) d\delta \leq 1, \quad \forall j \in J, \quad (2b)$$

$$\sum_{j=1}^J C_j(\mathbf{D}) D_j(\delta) \leq s^*(\delta, \mathbf{D}), \quad \forall \delta \in [0, 1]. \quad (2c)$$

The first constraint requires that, for each customer segment, D_j forms a valid probability distribution over supplier types. The second constraint enforces feasibility on the supply side by ensuring that the customer flow directed to any supplier type does not exceed the available mass of that type. The platform chooses a recommendation policy $\mathbf{D}$, denoted by $\mathbf{D}^*$, that solves this maximization problem. Importantly, although the platform maximizes current-period revenue; it is not myopic. The equilibrium approach ensures that the platform fully internalizes the long-term implications of its recommendation policy. Thus, unlike models in which myopic decisions typically discourage diversification and fairness, here an optimal steady-state recommendation incorporates the consequences of deliberately steering customers toward a diverse set of suppliers.

4. Equilibrium Analysis

In this section, we formally analyze the market equilibria for given recommendation distributions $\mathbf{D}$. To describe our result about customer-side strategies, we introduce the functional change of variable $\omega_j(v) = v - \theta_j(v)$, which we call the *willingness-to-pay* of customers of segment j with valuation v . As seen subsequently, equilibrium properties are expressed more conveniently in terms of $\omega_j(\cdot)$ rather than $\theta_j(\cdot)$. Given a stationary equilibrium for recommendation distribution $\mathbf{D}$, remember that $\mathcal{V}_j^* = \{v \in [0, 1] : \pi_{c,j}(v, \theta_j^*(v; \mathbf{D})) \geq 0\}$ denote the set of valuations of segment- j customers who enter the market in equilibrium. We define the upper and lower bounds on the distribution of willingness-to-pay as $\underline{\omega}_j = \inf\{v - \theta_j^*(v) : v \in \mathcal{V}_j^*\}$ and $\bar{\omega}_j = \sup\{v - \theta_j^*(v) : v \in \mathcal{V}_j^*\}$. For simplicity, our notation omits the dependence of $\underline{\omega}_j$ and $\bar{\omega}_j$ on the stationary equilibrium. The upper and lower bounds of willingness-to-pay, $\underline{\omega}_j$ and $\bar{\omega}_j$, are well-defined when a positive mass of customers enters the market. Hence, in our analysis, we disregard trivial equilibria in which no agent participates.

The next two results demonstrate structural facts about the customer willingness-to-pay values and the supplier prices, which are instrumental in subsequent analysis.

PROPOSITION 1 (Monotonicity of optimal customer strategies). *In the unique stationary equilibrium, the acceptance threshold $\theta_j^*(v; \mathbf{D})$ is strictly increasing in v on $\mathcal{V}_j^*$ for any j . The effective willingness to pay $\omega_j(v; \mathbf{D}) = v - \theta_j^*(v; \mathbf{D})$ is also strictly increasing in v for any j .*

Recall that an optimal acceptance threshold balances delaying purchase against the net surplus obtained from an immediate purchase. Proposition 1 shows that equilibrium customer strategies exhibit vertical differentiation. Customers with higher valuations have higher acceptance thresholds, meaning they demand a higher net surplus from successful transactions. They also have higher willingness-to-pay for the product, which grants them a competitive advantage in the market. Continuing with characterizing the customer strategies, Theorem 2 shows that the equilibrium density of customers’ willingness-to-pay preserves the shape of the arrival valuation distribution of customers.

THEOREM 2 (Distribution of willingness-to-pay). *Let $f_{W_j}(\omega)$ denote the steady-state density of willingness-to-pay among active customers in segment j . In the stationary equilibrium, this density is constant over its support $[\underline{\omega}_j, \bar{\omega}_j]$. The density is given by*

$$f_{W_j}(\omega) = \frac{\lambda_j}{\mu_j(\bar{v}_j - \underline{v}_j)}, \quad \omega \in [\underline{\omega}_j, \bar{\omega}_j].$$

Theorem 2 establishes that, at the stationary equilibrium, the shape of the willingness-to-pay distribution matches exactly the distribution of arriving customer valuations (both are uniform). To clarify this result, reconsider the model dynamics: each customer arrives with an initial valuation and either makes an immediate purchase or remains in the market to search further in subsequent periods. Consequently, at any given time, the population of customers of a specific type consists of those who entered in previous periods (or in the current period) and persisted in the marketplace until now. Since the duration that each customer type spends in the system is influenced by type-specific strategies, and these strategies vary across customer types as established in Proposition 1, the equilibrium number and valuations of customers belonging to any particular type are generally expected to differ. Theorem 2 demonstrates that, in the long run, the distribution of customers’ willingness-to-pay (active customer willingness to pay) replicates the initial valuation distribution of customers that arrive at the platform, scaled by a factor related to μ_j and the valuation support length $\bar{v}_j - \underline{v}_j$. In fact, the result can be generalized for any arrival distribution. That is, for any arrival distribution, the equilibrium distribution of willingness-to-pay has the same shape as the arrival distribution, appropriately scaled. This is a powerful result simplifying the subsequent analysis, as knowledge of the valuation arrival distribution and the customer departure rate directly yields the willingness-to-pay distribution at equilibrium.

We now turn to characterizing supplier prices. Recall that $\mathcal{S}^* \subseteq \mathcal{S}$ denotes the set of supplier types that are active in equilibrium (those that receive positive exposure). For ease of exposition of results and presenting fundamental trade-offs clearly, we first analyze the *single customer segment* benchmark. The multi-segment case ($J > 1$) is studied in Section 6.

PROPOSITION 2 (Structure of optimal supplier strategies: single customer segment).

Consider the stationary equilibrium given an optimal recommendation policy D^ when $J = 1$ and $d(\delta) = \delta$. The optimal price mapping $p^*: \mathcal{S}^* \rightarrow \mathbb{R}_+$ satisfies:*

- (i) **Price and profit monotonicity:** $p^*(\delta)$ and the resulting supplier profit $\pi_s^*(\delta; D^*)$ are strictly decreasing in δ on $\mathcal{S}^*$.
- (ii) **Effective-price monotonicity:** the effective price paid by a customer, $X^*(\delta) := p^*(\delta) + \delta$, is strictly increasing in δ on $\mathcal{S}^*$.

Intuitively, the proposition illustrates an important economic insight. When the level of misfit between a supplier's product/service and a customer's preferences is small, the product closely aligns with the customer's needs, and the supplier can post a relatively high price without distressing demand. As misfit disutility δ increases, the customer's perceived value decreases, and the supplier has to reduce their posted price to remain competitive in the market. Yet these reductions never fully offset the customer's growing misfit disutility. Consequently, the effective price, which is the posted price plus the misfit disutility, rises with increasing misfit disutility. Effectively, this result implies that competitive pressure marginalizes niche suppliers; customers naturally gravitate toward suppliers offering the lowest effective prices, thus favoring those with lower misfit disutilities.

5. Scarcity by Design is Optimal: The Case of a Single Customer Segment

In this section, we study a single customer segment where customers prefer low- δ suppliers with $d(\delta) = \delta$. Studying a single market segment allows us to highlight the core trade-offs clearly before generalizing our results in Section 6. We characterize the optimal policy, the mechanism driving it, and the resulting welfare implications.

5.1. Revenue-maximizer recommendation policy

We begin by presenting our main result: the revenue-maximizing recommendation policy concentrates customer traffic on two groups of suppliers positioned at opposite extremes of the supplier-type spectrum. We subsequently dissect the logic behind this result, providing insights into the underlying mechanism, supported by formal results.

THEOREM 3 (Structure of the revenue-maximizing recommendation). *Let D^* denote the optimal recommendation distribution for the single-segment model with $d(\delta) = \delta$. Then:*

(i) **Structure.** There exist unique thresholds $0 \leq \delta_0 \leq \delta_1 \leq \bar{\delta}$ such that

$$D^*(\delta) = \begin{cases} \frac{s^*(\delta; D^*)}{C(D^*)}, & \delta \in [0, \delta_0] \cup (\delta_1, \bar{\delta}], \\ 0, & \delta \in (\delta_0, \delta_1]. \end{cases}$$

Hence, the recommendation policy is unique and saturates the capacity of suppliers on two extreme blocks while allocating zero recommendations to the suppliers in the middle.

(ii) **Thick-Market Limit.** As the supplier arrival rate m increases, both intervals $[0, \delta_0]$ and $[\delta_1, \bar{\delta}]$ monotonically shrink in length. In the limit as $m \rightarrow \infty$, D^* converges weakly to a two-point distribution (a mixture of two Dirac measures) concentrated at 0 and $\bar{\delta}$.

Proof sketch for Theorem 3. Establishing Theorem 3 is technically challenging because the platform optimizes over an infinite-dimensional object (a density D), while both the objective (revenue) and the capacity constraints are endogenous equilibrium outcomes of the policy D itself.

The key geometric insight is that the platform’s marginal value of directing exposure to supplier type δ is *U-shaped*: it is highest at the extremes (δ near 0 or $\bar{\delta}$) and lowest for intermediate types. Intuitively, best-fit suppliers ($\delta \approx 0$) generate high surplus per match, while worst-fit suppliers ($\delta \approx \bar{\delta}$) create the most scarcity value by raising equilibrium prices (Proposition 4); intermediate types do neither well. Because this marginal value is strictly convex in δ , the platform’s optimum has a two-threshold structure: there exist cutoffs δ_0 and δ_1 such that the platform saturates capacity on $[0, \delta_0] \cup [\delta_1, \bar{\delta}]$ and assigns zero exposure on the interior gap (δ_0, δ_1) .

The main technical challenge is that equilibrium responds endogenously to the policy D through the mean effective price $M(D) := \mathbb{E}_{\delta \sim D}[p^*(\delta) + \delta]$, which is the unique fixed point of a stable contraction (Theorem 1). We address this by introducing M as an auxiliary decision variable and imposing equilibrium consistency as an explicit constraint, which yields an equivalent augmented problem whose optimality (KKT) conditions internalize the equilibrium feedback. Strict convexity of the resulting marginal value then delivers the two-threshold structure. As market thickness m grows, capacity slack increases among high-misfit suppliers, and the optimal allocation converges to a two-point distribution on $\{0, \bar{\delta}\}$. Proposition 5 characterizes when the upper (high-misfit) interval is nonempty. See Appendix C for details. $\square$

Theorem 3 establishes that the platform optimally assigns positive recommendation probability only to two groups of suppliers: the very best ($\delta \in [0, \delta_0]$) and the worst ($\delta \in (\delta_1, \bar{\delta}]$) fits among those that profitably participate. Suppliers in both groups fully utilize their capacities. Furthermore, as the market thickens ($m \rightarrow \infty$), the platform becomes more selective, concentrating recommendations entirely on the extremes ($\delta = 0$ and $\delta = \bar{\delta}$). On the other hand, for sufficiently small m , it may be optimal to have $\delta_0 = \delta_1$, meaning the gap disappears and all active suppliers are utilized.

It may seem paradoxical that *intentionally* diverting customers away from their best-fit suppliers could increase revenue. The explanation hinges upon the strategic responses of customers and suppliers. To illustrate this mechanism, we formally introduce a baseline recommender and a shifted recommender.

Baseline and shifted recommenders. We define a *baseline recommender* D^B that efficiently uses capacity for good fit suppliers. Specifically, D^B completely *saturates* available supplier capacity within a best-fit segment $[0, \delta_0]$, i.e., $C(D^B) D^B(\delta) = s^*(\delta, D^B)$ for all $\delta \in [0, \delta_0]$. In contrast, capacity is *slack* for the remainder, i.e., $C(D^B) D^B(\delta) < s^*(\delta, D^B)$ for $\delta \in (\delta_0, \bar{\delta}]$. In other words, capacity is fully utilized among better-matches (low- δ), while undesirable (higher- δ) suppliers' capacity is not fully utilized. Let $\omega_B(v)$ and $p_B^*(\delta)$ denote the induced equilibrium willingness-to-pay and supplier prices under D^B .

To examine the impact of steering customers toward less desirable suppliers, we select a *source set* $I_\ell \subseteq [0, \delta_0]$ and a *destination set* $I_h \subset (\delta_0, \bar{\delta}]$. For a sufficiently small $\varepsilon > 0$, we define the *shifted recommender* D^ε as:

$$D^\varepsilon(\delta) := D^B(\delta) - \varepsilon \frac{\mathbf{1}_{I_\ell}(\delta)}{|I_\ell|} + \varepsilon \frac{\mathbf{1}_{I_h}(\delta)}{|I_h|}.$$

This construction reallocates an ε -fraction of recommendation mass from best match suppliers (in I_ℓ) to less desirable suppliers (in I_h). Let $\omega_\varepsilon(v)$ and $p_\varepsilon^*(\delta)$ denote the equilibrium outcomes under D^ε . The following results characterize the strategic responses of customers (Proposition 3) and suppliers (Proposition 4) to this shift.

PROPOSITION 3 (Scarcity softens search). *Steering any positive mass of customers from a baseline recommender D^B toward less desirable suppliers raises each customer's willingness-to-pay. In particular, $\omega_\varepsilon(v) > \omega_B(v)$ for every $v \in \mathcal{V}^*$ and every sufficiently small $\varepsilon > 0$.*

Proposition 3 establishes that introducing a probability of encountering misfit suppliers increases each customer's reservation willingness-to-pay $\omega(v)$. Since $\omega(v)$ is the acceptance cutoff that equates accepting an offer with continuing to search, a higher $\omega(v)$ implies fewer rejections and a larger set of offers that are accepted, making the prospect of future search less attractive and thereby softening search. The mechanism is that rejecting an offer exposes the customer to the risk of drawing a lower-quality fit, which, by Proposition 2, is effectively more expensive. Hence, it results in a lower continuation value of searching and higher willingness to accept. This lower interest in searching decreases the perceived price elasticity of demand by suppliers. The following proposition formalizes the suppliers' strategic response.

PROPOSITION 4 (Scarcity raises prices). *Steering any positive mass of customers from a baseline recommender D^B toward less desirable suppliers raises each supplier's price. In particular, $p_\varepsilon^*(\delta) > p_B^*(\delta)$ for every $\delta \in \mathcal{S}^*$ and every sufficiently small $\varepsilon > 0$.*

Proposition 4 clarifies the core tradeoff behind the strategic injection of scarcity. On one hand, diverting customers to higher misfit suppliers softens price competition, enabling higher markups (Propositions 3 and 4). On the other hand, this diversion increases the expected search duration, risking lost sales as impatient customers exit the market before purchasing.⁵ The following proposition resolves this fundamental tradeoff, identifying how to perform the diversion and the conditions under which the revenue gained from higher margins outweighs the loss from increased churn.

PROPOSITION 5 (When and how to allocate high misfit recommendations). *Suppose diverting a positive fraction of customer traffic from a baseline recommendation policy D^B toward less desirable suppliers increases platform revenue. Then, revenue is maximized by allocating the entire diverted mass to the supplier with the highest available misfit disutility, i.e., to the supplier type $\delta = \sup\{\delta' : D^B(\delta') < s(\delta')/C(D^B)\}$.*

Moreover, such a diversion increases platform revenue if and only if the following conditions are satisfied simultaneously:

1. *Customers are sufficiently patient: $\mu_1 < \bar{\mu}(m)$, where $\bar{\mu}(m)$ is a critical threshold. In the limit of abundant supply ($m \rightarrow \infty$), the threshold converges to $\bar{\mu}(\infty) = \frac{1}{3}$.*
2. *Supplier capacity is sufficiently large: $m > m^\dagger$ for some critical market size $m^\dagger > 0$.⁶ Equivalently, $m^\dagger$ is the smallest m such that the interval $(\delta_1, \bar{\delta}]$ in Theorem 3 has positive measure.*

Proposition 5 establishes that if scarcity injection is profitable, it should be done using the least desirable suppliers available (highest $\delta \in \mathcal{S}^*$). The mechanism driving this result is the strict convexity of the platform’s marginal value of exposure at optimality, as a function of supplier misfit type. Intuitively, recommendation algorithms navigate a trade-off between two primary drivers: *direct commission revenue*, which decays quadratically as the supplier-consumer ‘misfit’ increases, and the *equilibrium effect*, which scales linearly with that misfit. This linear scaling occurs because recommending lower-quality (higher-misfit) suppliers exerts upward pressure on prices as shown in Proposition 4). Consequently, exposure to intermediate misfit types represents the “worst of both

⁵ One might expect that if customer departure (μ_j) were endogenous (i.e., customer becomes more likely to exit upon receiving a bad recommendation, potentially due to an outside option), this mechanism would be significantly dampened by increased abandonment (*Attrition Effect*). However, our analysis in Appendix K reveals a counteracting effect. Indeed, when the platform worsens recommendations, the departure rate increases. This shortens the expected duration of search, making customers effectively more impatient. In a sequential search environment, however, higher impatience reduces the *option value of waiting* for a better future match (lower continuation value). Consequently, customers become less selective and increase their willingness-to-pay to secure a transaction quickly and avoid the heightened risk of the search terminating (the *Impatience Effect*) at equilibrium. This allows suppliers to raise prices even further when scarcity is induced. Our analysis shows that the revenue gain from this Impatience Effect dominates the revenue loss from the Attrition Effect, thereby reinforcing the platform’s incentive to induce scarcity.

⁶ *Definition of the critical capacity level $m^\dagger$.* Let $\delta_0(m)$ be the threshold type up to which capacity is fully utilized under the best-match policy given m . We define the *critical market size $m^\dagger$* as the lowest supplier-arrival rate at which the capacity of best-fit suppliers is sufficient to serve the entire customer base: $m^\dagger = \inf\left\{m > 0 : \int_0^{\delta_0(m)} \frac{s^*(\delta; m)}{C(m)} d\delta = 1\right\}$.

worlds”: they sacrifice significant direct revenue compared to best matches, yet fail to generate the strong equilibrium effect on prices provided by the worst matches. This creates a U-shaped marginal value profile, implying that revenue is maximized by polarizing exposure between the best matches and the highest-misfit suppliers.⁷

Moreover, Proposition 5 states when this strategy is profitable, balancing the benefit of higher prices against the risk of losing customers. First, the strategy requires patient customers (small μ_1). Patient customers are likely to continue searching after encountering a poor recommendation. This provides a safety net, allowing the platform to induce scarcity without losing many customers. Impatient customers, conversely, make the strategy too risky. Second, the diversion is profitable only when the market is sufficiently thick ($m > m^\dagger$). If the market is thin (small m), the optimization problem is trivial: the platform must utilize all available suppliers to meet demand, leaving no scope for strategic diversion. Only when supply exceeds the critical threshold $m^\dagger$ does the platform have the flexibility to strategically allocate demand toward niche suppliers.

5.2. Welfare Trade-offs and Policy Performance

Section 5.1 reveals that the platform may optimally induce scarcity to boost revenue. We now examine the welfare implications of this strategy, demonstrating a fundamental trade-off between platform revenue/supplier profit and customer surplus. First, we show that any feasible diversion toward higher- δ suppliers reduces both total customer surplus and total transaction volume.

PROPOSITION 6 (Scarcity reduces customer surplus and transaction volume). *Let $CS(\varepsilon)$ denote the equilibrium customer surplus and $Q(\varepsilon)$ the total transaction volume (mass of matches) under the shifted recommender D^ε . Then, for every baseline D^B and every feasible diversion (I_ℓ, I_h) , we have $\left. \frac{dCS}{d\varepsilon} \right|_{\varepsilon=0} < 0$ and $\left. \frac{dQ}{d\varepsilon} \right|_{\varepsilon=0} < 0$. In other words, every shift toward high misfit suppliers strictly lowers total customer surplus and the total volume of transactions.*

When the platform steers customers toward undesirable suppliers, equilibrium prices increase (Proposition 4), which raises the commission earned per transaction. At the same time, the probability of a successful match, and hence total transaction volume, falls (Proposition 6). Because platform revenue is an ad-valorem commission on total transaction value, scarcity injection raises revenue only if the price increase is large enough to outweigh the loss in volume. As shown earlier in Proposition 5, this can occur only when customers are sufficiently patient and supplier capacity is sufficiently large.

⁷ Formally, the marginal value density is $\hat{H}^*(\delta) = C(M^*)r(\delta; M^*) + \xi^*X(\delta; M^*)$. The effective price $X(\delta)$ is affine in δ ; however, the per-exposure revenue kernel $r(\delta)$ is quadratic and strictly convex due to the linear demand structure (due to uniformity of WTP by Theorem 2). Consequently, $\hat{H}^*(\delta)$ is strictly convex. We provide the derivation of $\hat{H}''(\delta) > 0$ in Appendix C.5.

Table 1 Baseline vs. Scarcity outcomes in Example 1.

Metric	Baseline ($\alpha = 0$)	Scarcity ($\alpha = 0.35$)	Impact
Max WTP ($\bar{\omega}$)	0.333	0.435	$\uparrow$
Good Price (p_g^*)	0.167	0.217	$\uparrow$
Volume (Q)	0.833	0.707	$\downarrow$
Revenue (R)	0.0139	0.0154	$\uparrow (+10.6\%)$
Customer Surplus	0.347	0.277	$\downarrow (-20.4\%)$

Moreover, the interests of the platform and the suppliers are perfectly aligned regarding the strategic use of scarcity. Because the platform collects an ad-valorem commission, platform revenue and total supplier profit are proportional for every ε . Consequently, whenever inducing scarcity is profitable, it benefits both the platform and suppliers while hurting customers (Proposition 6). Customer surplus, supplier profit, and platform revenue are therefore simultaneously maximized by D^* if and only if $\mu_1 \geq \bar{\mu}(m)$ or $m \leq m^\dagger$; that is, precisely when the conditions for profitable scarcity (Proposition 5) fail. The following example provides a closed-form illustration of the welfare implications of scarcity injection.

EXAMPLE 1 (WELFARE TRADE-OFFS UNDER SCARCITY INJECTION). Consider a thick market ($m \rightarrow \infty$) with customer valuations $v \sim U[0, 1]$, impatience $\mu = 0.2$, and two supplier types: Good ($\delta_g = 0$) and Bad ($\delta_b = 0.6$). The platform’s policy $\alpha \in [0, 1]$ is the probability of recommending the Bad type. We compare a Baseline policy ($\alpha = 0$) with a Scarcity policy for illustration ($\alpha = 0.35$).

The mechanism is as follows: increasing α lowers the probability of receiving an attractive offer, which reduces the continuation value of search and raises customers’ willingness-to-pay $\bar{\omega}$ (customer becomes less selective). In addition, the change in the willingness-to-pay distribution decreases the price elasticity. Suppliers respond by increasing prices (p_g^*). Table 1 reports the resulting outcomes: transaction volume (Q), platform revenue (R), and aggregate customer surplus flow (CS). Platform revenue rises by 10.6% under the Scarcity policy, while customer surplus falls by 20.4%, illustrating the fundamental tension identified in Propositions 5 and 6. The closed-form derivations appear in Appendix D.5.

5.3. Numerical Results and Illustrations of Main Findings

In this section, we use numerical experiments to illustrate the performance of the optimal recommendation policy characterized in Section 5.1 and to compare it with two natural benchmarks across a range of market conditions. The benchmark policies are (i) *best-match* (BM), and (ii) *uniform random* (U). Best-match recommends the lowest-misfit suppliers and assigns them as much traffic as their steady-state capacity allows. Formally, $D_{BM}(\delta) = \frac{s^*(\delta, D_{BM})}{C(D_{BM})}$ for $\delta \leq \delta_{BM}$, and $D_{BM}(\delta) = 0$ otherwise, where $s^*(\delta, D_{BM})$ is the steady-state mass (capacity) of type- δ suppliers and $C(D_{BM})$ is

the steady-state mass of customers under BM. The cutoff δ_{BM} is chosen so that D_{BM} is a valid distribution, i.e., $\int_0^\infty D_{BM}(\delta) d\delta = 1$. In contrast, Uniform Random assigns customers uniformly across the spectrum of active supplier types, such that $D_U(\delta) = 1/\bar{\delta}$ for $\delta \in [0, \bar{\delta}]$. We use R^* , R_{BM} , and R_U to denote the revenues under the optimal, best-match, and uniform random policies, respectively.

Note that this numerical analysis is enabled by our theoretical results, which significantly simplify the required computations. By Theorem 3, finding the optimal policy reduces to determining two cutoffs, δ_0 and δ_1 . We compute δ_0 and δ_1 from a closed-form fixed-point equation that our equilibrium characterization yields (Appendix E).

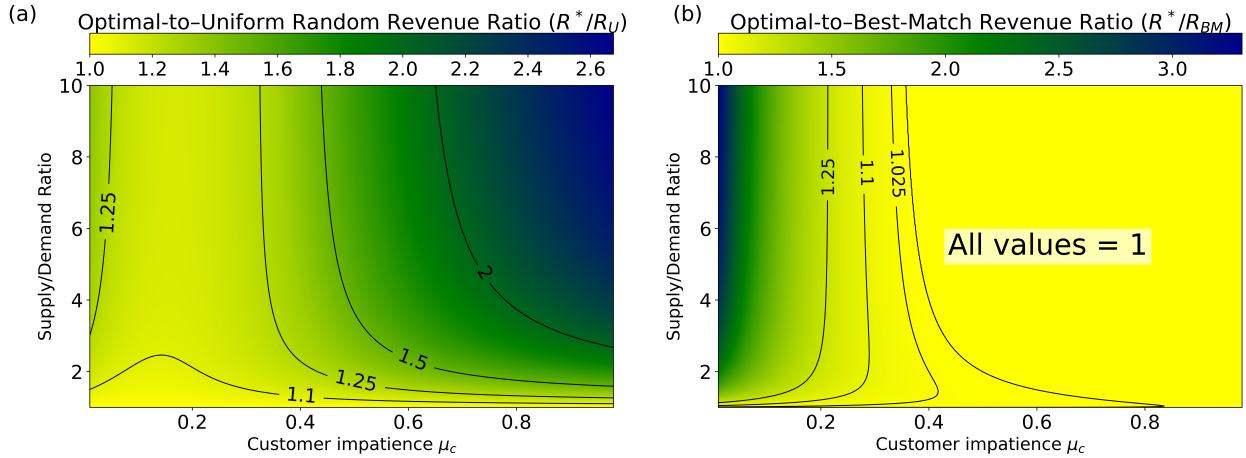


Figure 1 Revenue ratios across supply–demand ratio (market thickness) and customer impatience μ . The x -axis varies customer impatience μ . The y -axis varies the supply-demand ratio by keeping $C(D)$ constant and varying $s(\delta, D) = s$ (assuming uniform supplier density for visualization).

Figure 1 compares revenues as market thickness (supply-demand ratio) and customer impatience (μ) vary. Panel (a) shows the revenue gain over not using a recommendation algorithm (i.e, Uniformly Random), R^*/R_U . The optimal policy consistently outperforms Uniform Random, especially when customers are impatient (high μ), where the ratio often exceeds 2. Uniform Random performs poorly because it spreads recommendations too widely, leading to excessive customer exit. Interestingly, for very low μ , Uniform Random performs reasonably well relative to Best-Match (compare panels a and b). This is because when μ is low, injecting scarcity is beneficial (Proposition 5), and Uniform Random does inject scarcity, albeit “randomly” and suboptimally. However, for large μ , injecting any scarcity is suboptimal, which is where Uniform performs poorly because the very act of injecting scarcity is harmful. Panel (b) shows the revenue benefit of the optimal recommender compared to the Best-Match recommender (R^*/R_{BM}). The advantage depends on customer patience. When customers are patient (low μ), strategic diversion is optimal as shown in Proposition 5. The gains can well exceed 25%. When customers are impatient (high μ), the risk of exit dominates, and the

optimal policy converges to Best-Match ($R^*/R_{BM} = 1$), illustrating that alignment among all parties emerges once the conditions for profitable scarcity fail. The gains also require sufficient market thickness, as thin markets leave no room for diversion ($m \leq m^\dagger$).

6. The Value of Personalization

In practice, platforms observe rich heterogeneity in customer preferences, patience, and willingness-to-pay. Hence, a natural question is whether and how the platform should tailor its recommendation policy to different customer segments. Addressing this, in this section, we generalize our analysis to a setting with J heterogeneous customer segments, where the platform can deploy segment-specific recommendation policies. We solve OPT as stated in Section 3.4 to characterize the structure of optimal personalized recommendations for each segment. As discussed in Section 3.1, we allow segments to differ in their arrival rates λ_j , valuation distributions $v \sim U[\underline{v}_j, \bar{v}_j]$, exit probabilities μ_j , and how they perceive supplier types through segment-specific misfit costs $d_j(\delta)$.

We next characterize the structure of optimal personalized recommendations, establish a key structural property (Segmental-Exclusivity) that simplifies the high-dimensional allocation problem, quantify the Value of Information from personalization, and analyze distributional welfare effects through numerical experiments and a formal proposition.

6.1. The Structure of Optimal Personalized Recommendations

The optimization problem studied in this section is significantly more complex than the single-segment case studied in Section 5. In addition to the equilibrium effects of the recommendation policy on prices and search behavior, the segments are now coupled through shared supplier capacity (constraint OPT-(2c)). This creates a high-dimensional, tightly coupled allocation problem embedded within an equilibrium framework.

Theorem 4 is one of our main results stating the key structural result that tames this complexity.

THEOREM 4 (Structure of the Personalized Recommendation). *Let $\mathbf{D}^*$ be an optimal solution to OPT. Let $[\underline{\delta}^*, \bar{\delta}^*]$ be the convex hull of the active suppliers $\mathcal{S}^*$. The optimal policy exhibits the following structure:*

- (i) **Segmental-Exclusivity (SE):** *Let $\mathcal{B} \subseteq [\underline{\delta}^*, \bar{\delta}^*]$ be the set where the capacity constraint (2c) binds. For almost every $\delta \in \mathcal{B}$, the capacity $s^*(\delta; \mathbf{D}^*)$ is allocated exclusively to a single customer segment j .*
- (ii) **Personalized Structure (Bind-Gap-Bind):** *Let $[\underline{\delta}_j^*, \bar{\delta}_j^*]$ be the convex hull of the support of D_j^* . There exist cutoffs $\underline{\delta}_j^* \leq \delta_{j,0} \leq \delta_{j,1} \leq \bar{\delta}_j^*$ such that $D_j^*(\delta) = 0$ almost everywhere on the interior gap $(\delta_{j,0}, \delta_{j,1})$. On the two extreme blocks $[\underline{\delta}_j^*, \delta_{j,0}]$ and $[\delta_{j,1}, \bar{\delta}_j^*]$, the capacity constraint binds and (by SE) segment j is the unique segment receiving type δ for almost every δ on those blocks; hence $C_j(\mathbf{D}^*) D_j^*(\delta) = s^*(\delta; \mathbf{D}^*)$ for almost everywhere $\delta \in [\underline{\delta}_j^*, \delta_{j,0}] \cup [\delta_{j,1}, \bar{\delta}_j^*]$.*

(iii) **Aggregate Structure (Bind-Gap-Bind):** There exist unique cutoffs $\underline{\delta}^* \leq \delta_0 \leq \delta_1 \leq \bar{\delta}^*$ such that the optimal aggregate exposure $y^*(\delta) = \sum_j C_j(\mathbf{D}^*) D_j^*(\delta)$ saturates the total capacity $s^*(\delta; \mathbf{D}^*)$ (binding) on $[\underline{\delta}^*, \delta_0] \cup [\delta_1, \bar{\delta}^*]$, and $y^*(\delta) = 0$ a.e. on the interior gap (δ_0, δ_1) .

The SE property (i) is fundamental for analytical tractability. Despite the complexity of J heterogeneous segments competing for shared capacity, the optimal solution results in *local specialization*. At almost any δ , the platform allocates it entirely to the single segment generating the highest marginal revenue. This result effectively decomposes the coupled optimization problem. Instead of a complex, multi-segment allocation decision for every unit of capacity, the problem elegantly decomposes into a collection of simpler, one-dimensional decisions (one for each supplier type). This allows equilibrium prices $p^*(\delta)$ to be determined locally. The Personalized Structure (ii) shows that the ‘‘Scarcity by Design’’ strategy is applied at the most granular level. For each segment, the platform concentrates recommendations on the extremes—the very best and the very worst *available* suppliers. This follows because, within each segment, the marginal value of directing an additional unit of exposure to supplier type δ is strictly convex in δ , which mirrors the U-shaped geometry that drives the single-segment result (see Appendix G for details). The Aggregate Structure (iii) demonstrates that the strategic polarization persists at the aggregate level. If we pool exposure across segments, $y^*(\delta) = \sum_{j=1}^J C_j(\mathbf{D}^*) D_j^*(\delta)$, the aggregate allocation still takes a *bind-gap-bind* form. There exist cutoffs $\delta_0 \leq \delta_1$ such that the platform sends zero total exposure to supplier types in the interior region (δ_0, δ_1) , while outside this interval it fully utilizes available capacity. Intuitively, personalization changes which segments receive which types of suppliers, but it does not eliminate overall polarization.

6.2. Revenue and Welfare Implications of Personalization

We now quantify the value of segment-level information for recommendation design by comparing two regimes. (i) *Personalized (P)*: The platform observes each customer’s segment j and deploys segment-specific recommendation policies $\{D_j\}_{j=1}^J$. We let R^P denote the optimal steady-state revenue and E^P the induced equilibrium. (ii) *Non-Personalized (NP)*: The platform pools segments and chooses a single recommendation distribution D^{NP} applied to all customers. We let R^{NP} denote the optimal steady-state revenue under this restriction and E^{NP} the induced equilibrium.

The Canonical Decomposition. We define the *total Value of Information* as the revenue gain from personalization: $\text{VoI}^{\text{Total}} := R^P - R^{NP}$. To disentangle the drivers of this gain, we construct an intermediate benchmark that isolates the re-routing benefit from the equilibrium-feedback effect.

Let $y^{NP}(\delta) := \sum_{j=1}^J C_j(E^{NP}) D^{NP}(\delta)$ denote the aggregate steady-state flow of customers recommended to supplier type δ under the NP policy. We define the reallocation benchmark R^{Re} as the maximum revenue achievable by optimally reassigning this fixed aggregate exposure profile across

segments, while holding the equilibrium environment frozen at E^{NP} , which includes prices $p^{\text{NP}}(\delta)$, WTP distributions, and steady-state segment masses $C_j(E^{\text{NP}})$. In this counterfactual, the platform can change *which* segments generate traffic to each supplier type, but cannot change how much total traffic each type receives. Formally,

$$R^{\text{Re}} := \max_{\mathbf{D} \in \mathcal{D}} R(\mathbf{D}; E^{\text{NP}}) \quad \text{s.t.} \quad \sum_{j=1}^J C_j(E^{\text{NP}}) D_j(\delta) = y^{\text{NP}}(\delta) \quad \text{for a.e. } \delta \in [0, \bar{\delta}], \quad (3)$$

where $\mathcal{D}$ is the feasible policy set defined in the platform problem (OPT) and $R(\cdot; E^{\text{NP}})$ denotes revenue evaluated at the frozen equilibrium E^{NP} .

Let $\mathbf{D}^{\text{Re}}$ denote the policy attaining the maximum in (3). Because the frozen objective decomposes pointwise by δ and the revenue kernels are generically distinct across segments, the maximizer is unique for a.e. δ , so $\mathbf{D}^{\text{Re}}$ is well-defined up to measure-zero modifications. Define $CS^{\text{Re}} := CS(\mathbf{D}^{\text{Re}}; E^{\text{NP}})$ as the customer surplus it induces at the frozen equilibrium. Write $\Delta CS^{\text{Total}} := CS^{\text{P}} - CS^{\text{NP}}$. Adding and subtracting the reallocation benchmark yields the decompositions:

$$\text{VoI}^{\text{Total}} = \underbrace{(R^{\text{Re}} - R^{\text{NP}})}_{\text{VoI}^{\text{Allocative}}} + \underbrace{(R^{\text{P}} - R^{\text{Re}})}_{\text{VoI}^{\text{Feedback}}}, \quad (4)$$

$$\Delta CS^{\text{Total}} = \underbrace{(CS^{\text{Re}} - CS^{\text{NP}})}_{\Delta CS^{\text{Allocative}}} + \underbrace{(CS^{\text{P}} - CS^{\text{Re}})}_{\Delta CS^{\text{Feedback}}}. \quad (5)$$

The *Allocative* terms capture the impact of re-routing exposure across segments in the frozen NP environment. The *Feedback* terms capture the additional impact once equilibrium prices and search incentives adjust endogenously and the platform is no longer constrained to the aggregate profile y^{NP} . The following proposition formalizes the decomposition and its welfare implications.

PROPOSITION 7 (Value of Information: Decomposition and Welfare). *The decomposition in (4)–(5) satisfies:*

- (i) **Allocative Gain.** $\text{VoI}^{\text{Allocative}} \geq 0$: re-routing exposure to better-matched segments weakly increases revenue. However, $\Delta CS^{\text{Allocative}}$ is ambiguous in sign.
- (ii) **Feedback Gain.** $\text{VoI}^{\text{Feedback}}$ is ambiguous in sign; equilibrium adjustments may raise or lower revenue relative to the reallocation benchmark. However, $\Delta CS^{\text{Feedback}} \leq 0$: the equilibrium response weakly reduces customer surplus. Moreover, the magnitude of this loss is increasing in customer patience (i.e., decreasing in μ_j).
- (iii) **Total Gain.** $\text{VoI}^{\text{Total}} \geq 0$: personalization weakly increases platform revenue. However, ΔCS^{Total} is ambiguous in sign.

Proposition 7 establishes that personalization weakly benefits the platform ($\text{VoI}^{\text{Total}} \geq 0$) but creates a fundamental tension for customer welfare. Notably, even the Allocative Gain, which improves immediate revenue by better matching segments to suppliers, need not raise customer surplus. We now detail the driving mechanisms.

The Allocative Gain. $\text{VoI}^{\text{Allocative}}$ captures the revenue benefit from re-routing exposure across segments while holding aggregate supplier traffic and equilibrium prices fixed. Under NP, the platform optimizes for the average customer, which can mismatch premium suppliers with low-WTP segments. Personalization may correct this by directing high-margin suppliers toward high-WTP segments, raising conversion rates and revenue. However, revenue-maximizing reallocation need not raise customer surplus: if traffic shifts toward segments where the platform extracts nearly all the match surplus, aggregate customer welfare can fall even as transaction volume rises (Appendix H).

The Feedback Gain. $\text{VoI}^{\text{Feedback}}$ captures the equilibrium effect of personalization: the revenue change when the platform reshapes aggregate exposure and prices adjust endogenously, rather than merely re-routing a fixed profile. The platform strategically introduces segment-specific scarcity (Theorem 4), exploiting the positive feedback loop in which higher expected future prices raise current willingness-to-pay. This equilibrium adjustment weakly harms customers ($\Delta CS^{\text{Feedback}} \leq 0$). Patient customers, who weigh future search opportunities most heavily, are particularly vulnerable to this channel (Appendix H).

Personalization weakly raises platform revenue, with the Allocative component always non-negative. For customers, only the Feedback channel has an unambiguous sign: it reduces surplus, with the loss amplified by patience. The net welfare effect ΔCS^{Total} is theoretically ambiguous and depends on whether the Allocative Gain offsets the Feedback loss.

From a *platform-design* perspective, Proposition 7 implies that personalization is unambiguously profitable, but the welfare implications for customers are most concerning when customers are patient, since the surplus loss from equilibrium feedback is amplified in that regime. From a *policy* perspective, the decomposition highlights a trade-off: restricting personalization would eliminate both the Allocative Gain and the Feedback Gain, so blunt prohibitions may sacrifice allocative efficiency alongside consumer protection. We quantify these trade-offs numerically in Section 6.3.

6.3. Numerical Study: The Value of Information in Practice

We use numerical experiments to quantify the VoI from personalization and analyze its underlying drivers. Specifically, we study how these effects vary with segment heterogeneity in customer patience (μ) and valuations (v). In line with Theorems 3 and 4, we restrict attention to a two-point policy class that places all recommendation mass on the best match ($\delta = 0$) and the worst active match ($\bar{\delta}$). This class is parameterized by a scarcity level $\alpha_j \in [0, 1]$, where α_j is the probability that segment j is shown the worst match (and $1 - \alpha_j$ the best match). Under NP, we impose the restriction $\alpha_j \equiv \alpha$ for all j . We describe how we calculate the optimal α_j values in Appendices I and J.

In Experiment 1, we study the value of information about segment patience μ_j in the thick-market limit ($m \rightarrow \infty$). In this regime, the gap between P and NP is driven primarily by the *Feedback Gain*.

In Experiment 2, we study the value of information about segment valuations v_j under finite market thickness ($m < \infty$). Because best-fit supply is scarce, the gap between P and NP is driven primarily by the *Allocative Gain* from reallocating limited high-quality exposure across segments.

6.3.1. Experiment 1: Patience Heterogeneity and the Feedback Gain. We analyze a thick market ($m \rightarrow \infty$) with two customer segments that are identical in valuations but differ in patience: a *Patient* segment (L) with lower impatience μ_L , and an *Impatient* segment (H) with higher impatience μ_H . Both segments draw valuations $v \sim U[0, 1]$, and they are equal in size, i.e., $\lambda_L = \lambda_H = 0.5$. Holding fixed the average impatience at $\bar{\mu} = 0.5$, we vary the polarization gap $g \in [0, 1]$ so that $\mu_H = \bar{\mu} + g/2$ and $\mu_L = \bar{\mu} - g/2$. In this setting, capacity constraints are negligible, so the VoI from segment information about μ_j operates through the *Feedback Gain*. Figure 2 reports how outcomes vary with polarization by plotting (i) the revenue ratio R^P/R^{NP} , (ii) the optimal scarcity choices under personalization and under NP, and (iii) segment-level customer surplus (Patient and Impatient), all as functions of g in panels (a)–(c), respectively.

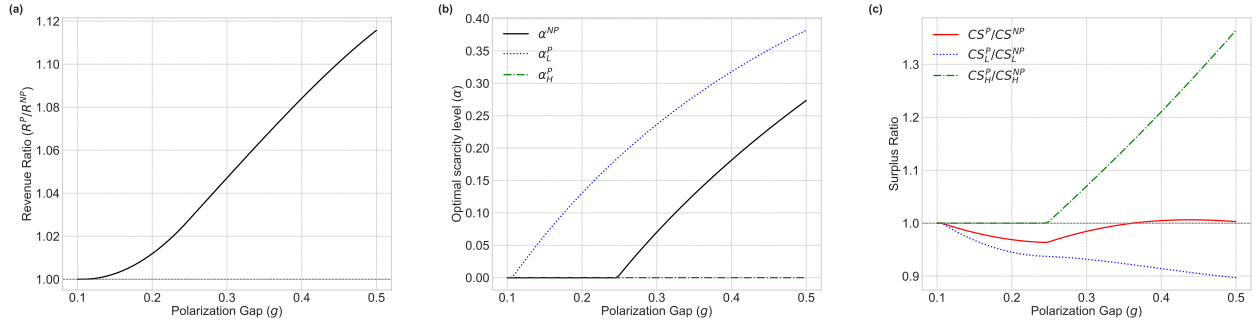


Figure 2 Experiment 1: Patience Heterogeneity and abundant suppliers ($m \rightarrow \infty$). Impact of polarization gap g on (a) Revenue Ratio R^P/R^{NP} ; (b) Optimal Scarcity Levels; (c) Segment-level Surplus Ratios CS^P/CS^{NP} .

Revenue Gains. Figure 2(a) shows that as heterogeneity rises, the NP policy, which must apply a common scarcity level α^{NP} , becomes a poor compromise across segments. Personalization instead tailors scarcity by segment, yielding revenue gains that exceed 10% when polarization is high.

Targeted Scarcity. Figure 2(b) shows that, under NP, the platform selects a compromise α^{NP} that lies between the segment-specific optima. Under personalization, the platform sets $\alpha_H^P = 0$ for the Impatient segment (high μ_H), since inducing scarcity is too risky for customers who exit quickly (consistent with Proposition 5), while it increases α_L^P sharply with g for the Patient segment. The mechanism is the Feedback Gain: patient customers place more weight on the future search environment, so concentrating scarcity on them raises effective prices and revenues.

Distributional Welfare Effects. Let CS_j^P and CS_j^{NP} denote the steady-state customer surplus of segment $j \in \{L, H\}$ under the Personalized and Non-Personalized regimes, respectively, and let CS^P and CS^{NP} denote the corresponding aggregate customer surplus across segments (weighted by population shares). Figure 2(c) plots the surplus ratios and shows that personalization benefits the Impatient segment ($CS_H^P/CS_H^{NP} > 1$) while strictly harming the Patient segment ($CS_L^P/CS_L^{NP} < 1$), and that the patient-segment loss becomes larger as g increases. The reason is that personalization spares the Impatient segment from scarcity while subjecting the Patient segment to higher induced scarcity and, consequently, higher equilibrium prices. Aggregate surplus changes little in comparison, so the main welfare implication is distributional: personalization reallocates surplus from patient to impatient customers while allowing the platform to extract more revenue by concentrating scarcity on the segment most tolerant to it.

6.3.2. Experiment 2: Valuation Heterogeneity and the Allocative Gain. We study the value of information regarding v_j . Crucially, we introduce a *finite capacity* constraint ($m < \infty$) on best-fit suppliers. This is necessary because, in a thick market ($m \rightarrow \infty$) with ad-valorem fees and homogeneous patience, the optimal scarcity level depends primarily on μ , not valuation bounds (Appendix I). If capacity were infinite, P and NP policies would converge, yielding zero VoI. Finite capacity creates resource contention, activating the Allocative Gain by forcing the platform to prioritize the allocation of scarce high-quality supply. We retain the two-point structure (allocation to the extremes) for analytical tractability, which captures the essential trade-off in allocation. Appendix J details the capacity-coupled optimization. Specifically, we consider two segments with homogeneous patience (μ) but different valuations: a Budget segment (B) with $v_B \sim U[0, \bar{v}_B]$ and a General/Premium segment (G) with $v_G \sim U[0, \bar{v}_G]$, with $\bar{v}_G > \bar{v}_B$. We analyze a scenario where the relative capacity $M_{rel} = 0.75$ (available capacity is 75% of the maximum potential demand, see Appendix J for details) and $\mu = 0.5$ (an impatient market). This setting allows us to isolate the Allocative Gain cleanly. Since $\mu = 0.5$ is relatively high, the incentive to strategically induce scarcity for the Feedback Gain is naturally muted (Proposition 5). Thus, the VoI in this setting is driven primarily by the efficient allocation of constrained capacity.

Before analyzing the numerical results, we establish a key theoretical result regarding how the platform optimally allocates scarce capacity when the capacity constraint binds under the market primitives of Experiment 2.

PROPOSITION 8 (Prioritization of High-Value Segments). *Given a Budget segment (B) with $v_B \sim U[0, \bar{v}_B]$ and a General/Premium segment (G) with $v_G \sim U[0, \bar{v}_G]$, with $\bar{v}_G > \bar{v}_B$ and homogeneous patience ($\mu_G = \mu_B$), the personalized policy (P) consistently imposes higher scarcity on the lower-valuation segment. That is, $\alpha_G^P < \alpha_B^P$.*

We provide the proof in Appendix J. Proposition 8 formalizes the Allocative Gain under constraints. Because the General segment generates higher expected revenue per transaction, the opportunity cost of allocating scarce best-match capacity to the Budget segment is high. The platform optimally responds by rationing the Budget segment more severely (higher α_B^P) to prioritize the General segment (lower α_G^P). This structural result confirms that the patterns observed in our representative scenario (Figure 3) are general properties of the optimal policy in this setting.

Next, we vary the market share of the General segment, η . As in Experiment 1, we trace how the value of personalization evolves with market composition by plotting (i) the revenue ratio R^P/R^{NP} (VoI in revenue terms), (ii) the optimal scarcity choices under NP and P, and (iii) segment-level surplus ratios, all as functions of η in Figure 3.

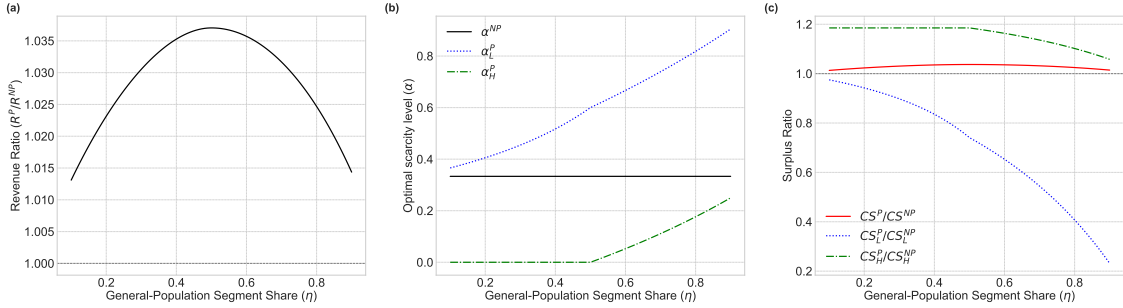


Figure 3 Experiment 2: Valuation Heterogeneity (Finite Capacity, $\mu = 0.50$, $M_{rel} = 0.75$, $\bar{v}_G = 1$, $\bar{v}_B = 0.5$). (a) Revenue ratio R^P/R^{NP} , (b) optimal scarcity levels, and (c) segment-level surplus ratios, plotted against the General/Premium segment share η .

Revenue Gains. Figure 3(a) plots the revenue ratio R^P/R^{NP} against the General-segment share η . The ratio exceeds one throughout, so personalization increases revenue relative to NP. The gain is non-monotone in η : it rises from low η , peaks around $\eta = 0.5$, and then declines as η approaches one. At its peak, the revenue ratio is about 1.035–1.04 (roughly a 3.5–4% gain). This pattern is driven by composition under a binding capacity constraint. When η is close to 0 or 1, the market is effectively single-segment, leaving little scope to reallocate scarce best-fit exposure, so P and NP are similar. When η is intermediate, the pooled NP policy must apply a single scarcity level that is a poor compromise across segments. Personalization can instead differentiate scarcity by imposing higher scarcity on B to prioritize G , which delivers the largest Allocative Gain. Since $\mu = 0.5$ is relatively high, the Feedback Gain is muted, and the resulting VoI is primarily allocative.

Targeted Scarcity. Figure 3(b) plots the optimal scarcity levels as the General-segment share η varies. Under NP, the capacity constraint pins down a single scarcity level α^{NP} , which is constant in η . Under P, the platform can tailor scarcity by segment: it consistently imposes higher scarcity on the low-value/Budget segment than on the high-value/General segment, i.e., $\alpha_B^P > \alpha_G^P$ for all η

(Proposition 8). When the General segment is relatively small, the platform keeps it essentially free of scarcity ($\alpha_G^P = 0$) and satisfies the capacity constraint by raising scarcity for the Budget segment. As η increases, the shadow price of capacity rises (Appendix J), requiring higher scarcity for both segments to satisfy the constraint; nevertheless, the priority structure remains, with most of the increase coming through the Budget segment.

Distributional Welfare Effects. Figure 3(c) plots how customer surplus changes under personalization relative to a pooled, non-personalized policy as the market composition varies. Specifically, we plot surplus ratios against the General-segment share η . The welfare impacts starkly highlight the consequences of revenue-maximizing allocation. The Budget segment is unambiguously harmed by personalization ($CS_B^P/CS_B^{NP} < 1$). They face higher scarcity under P than under NP across nearly all market compositions, as the platform reallocates best-match exposure toward the General segment. Conversely, the General segment benefits significantly, particularly when it is a minority (low η), as personalization shields it from the scarcity required under the pooled NP policy.

7. Conclusion and Discussion

This paper investigates the optimal design of recommendation systems in two-sided marketplaces, moving beyond immediate match quality to analyze long-term equilibrium effects. We develop a dynamic model that integrates strategic customer search, endogenous supplier pricing, and participation decisions in environments where evaluation is sequential. Our analysis reveals that maximizing short-term metrics does not necessarily align with long-term platform health.

The central insight is that a revenue-maximizing platform may strategically degrade recommendation quality, a strategy we term “Scarcity by Design.” By occasionally steering customers toward less-preferred suppliers, the platform makes the search environment riskier, prompting customers to become less selective. Simultaneously, this strategy softens competition among top suppliers. Together, these equilibrium forces enable higher prices, boosting platform revenue and supplier profits, particularly when customers are patient and the market is thick. However, this strategy comes at the expense of customer surplus and overall transaction volume, highlighting a fundamental tension in platform objectives.

Our analysis of personalization reveals that granular customer data amplifies the platform’s ability to manage this trade-off. We established the Segmental-Exclusivity (SE) property, which simplifies the complex allocation problem, and demonstrated that personalization yields value through two channels: improved allocative efficiency (Allocative Gain) and enhanced strategic manipulation of the equilibrium (Feedback Gain). Perhaps counterintuitively, even the purely allocative component of personalization can reduce customer surplus. Because the platform re-routes exposure to maximize revenue rather than welfare, some segments receive worse matches even when aggregate efficiency

improves. While personalization always increases revenue, it can have distributional consequences that harm vulnerable customer segments.

Our findings offer actionable insights for platform managers. In environments characterized by sequential search and customer patience (e.g., B2B services, freelance labor), strategically inducing scarcity can be a powerful lever for enhancing margins. Furthermore, when a high-quality supply is constrained, personalization is critical for optimizing allocative efficiency. However, managers must balance these revenue gains against the potential risks of reduced customer satisfaction. The distributional effects uncovered in our analysis raise significant policy considerations. Platforms can tailor scarcity to specific segments, often exploiting patient customers (bargain hunters) or disadvantaging budget-constrained users when capacity is limited. This highlights the need for transparency in recommendation design and suggests a role for regulatory oversight to ensure fairness, concerning how platforms allocate visibility.

Our model relies on several assumptions that suggest avenues for future research. We consider a sequential search process, which is appropriate for high-value decisions. Exploring how recommendation design interacts with strategic behavior under alternative search protocols (e.g., simultaneous evaluation) would be valuable. Our analysis focuses on a stationary equilibrium. This approach is best suited for evaluating the long-term impact of persistent platform policies, such as those updated periodically (e.g., monthly), rather than online policies that change with every action. Future work could explore learning dynamics, investigating how platforms and users adapt their strategies in non-stationary environments.

Finally, we consider revenue maximization via ad-valorem commissions, which is common for digital platforms. This commission structure directly links platform revenue to equilibrium prices, which is what creates the incentive for Scarcity by Design. Future research could explore how this incentive shifts under monetization strategies prioritizing transaction volume (e.g., per-transaction or subscription fees), or under hybrid monetization models, such as the interaction between organic recommendations and promoted listings (advertising), where platforms must balance organic scarcity with the incentive to sell visibility.

References

- Abeliuk A, Berbeglia G, Cebrian M, Van Hentenryck P (2016) Assortment optimization under a multinomial logit model with position bias and social influence. *4OR* 14:57–75.
- Adomavicius G, Bockstedt JC, Curley SP, Zhang J (2018) Effects of online recommendations on consumers’ willingness to pay. *Information Systems Research* 29(1):84–102.
- Akchen YC, Mitrofanov D (2025) Consider or choose? the role and power of consideration sets. *Management Science* .

- Aouad A, Sarıtaç Ö, Yan C (2025) Centralized versus decentralized pricing controls for dynamic matching platforms, working paper. Available at SSRN: 4453799 (DOI: 10.2139/ssrn.4453799).
- Asadpour A, Niazadeh R, Saberi A, Shameli A (2023) Sequential submodular maximization and applications to ranking an assortment of products. *Operations Research* 71(4):1154–1170.
- Ashlagi I, Braverman M, Kanoria Y, Shi P (2020) Clearing matching markets efficiently: informative signals and match recommendations. *Management Science* 66(5):2163–2193.
- Balseiro SR, Besbes O, Castro F (2024) Mechanism design under approximate incentive compatibility. *Operations Research* 72(1):355–372.
- Bernstein F, DeCroix GA, Keskin NB (2021) Competition between two-sided platforms under demand and supply congestion effects. *Manufacturing & Service Operations Management* 23(5):1043–1061.
- Besbes O, Fonseca Y, Lobel I, Zheng F (2023) Signaling competition in two-sided markets. Available at SSRN 4451693 .
- Busse MR, Knittel CR, Zettelmeyer F (2013) Are consumers myopic? evidence from new and used car purchases. *American Economic Review* 103(1):220–256, URL <http://dx.doi.org/10.1257/aer.103.1.220>.
- Cachon GP, Swinney R (2011) The value of fast fashion: Quick response, enhanced design, and strategic consumer behavior. *Management science* 57(4):778–795.
- Cao S, Hu M (2025) Inventory competition under algorithmic demand allocation. Available at SSRN 5661610 .
- De los Santos B, Koulayev S (2017) Optimizing click-through in online rankings with endogenous search refinement. *Marketing Science* 36(4):542–564.
- Ekbatani F, Caldentey R (2025) A stochastic growth model for online platforms. Available at SSRN .
- Ferreira KJ, Parthasarathy S, Sekar S (2022) Learning to rank an assortment of products. *Management Science* 68(3):1828–1848.
- Fuller JB, Raman M, Palano J, Bailey A, Vaduganathan N, Kaufman E, Laverdière R, Lovett S (2020) Building the on-demand workforce. Published by Harvard Business School and BCG .
- Gallego G, Li A, Truong VA, Wang X (2020) Approximation algorithms for product framing and pricing. *Operations Research* 68(1):134–160.
- Gallego G, Phillips R, Şahin Ö (2008) Strategic management of distressed inventory. *Production and Operations Management* 17(4):402–415.
- Ghose A, Ipeiritos PG, Li B (2014) Examining the impact of ranking on consumer behavior and search engine revenue. *Management Science* 60(7):1632–1654.
- Hendel I, Nevo A (2006) Measuring the implications of sales and consumer inventory behavior. *Econometrica* 74(6):1637–1673, URL <http://dx.doi.org/10.1111/j.1468-0262.2006.00721.x>.

- Hopenhayn HA (1992) Entry, exit, and firm dynamics in long-run equilibrium. *Econometrica* 60(5):1127–1150, URL <http://dx.doi.org/10.2307/2951541>.
- Horton JJ (2017) The effects of algorithmic labor market recommendations: Evidence from a field experiment. *Journal of Labor Economics* 35(2):345–385.
- Immorlica N, Lucier B, Manshadi V, Wei A (2023) Designing approximately optimal search on matching platforms. *Management Science* 69(8):4609–4626, URL <http://dx.doi.org/10.1287/mnsc.2022.4601>.
- Jagabathula S, Rusmevichientong P (2017) A nonparametric joint assortment and price choice model. *Management Science* 63(9):3128–3145.
- Kanoria Y, Min S, Qian P (2025) The competition for partners in matching markets. *Management Science* 71(6):5230–5250.
- Kanoria Y, Saban D (2021) Facilitating the search for partners on matching platforms: Restricting agent actions. *Management Science* 67(10):5990–6029, URL <http://dx.doi.org/10.1287/mnsc.2020.3794>.
- Lee KH, Musolf L (2025) Two-sided markets shaped by platform-guided search.’.
- Lei Y, Jasin S, Uichanco J, Vakhtinsky A (2022) Joint product framing (display, ranking, pricing) and order fulfillment under the multinomial logit model for e-commerce retailers. *Manufacturing & Service Operations Management* 24(3):1529–1546.
- Li J, Granados N, Netessine S (2014) Are consumers strategic? Structural estimation from the air-travel industry. *Management Science* 60(9):2114–2137, URL <http://dx.doi.org/10.1287/mnsc.2013.1860>.
- Liu Q, Van Ryzin GJ (2008) Strategic capacity rationing to induce early purchases. *Management Science* 54(6):1115–1131.
- Ma R, Mai Y, Hu B (2025) User-generated content platforms: Managing the relationship triangle. *Manufacturing & Service Operations Management* .
- MacKenzie I, Meyer C, Noble S (2013) How retailers can keep up with consumers. Industry report, McKinsey & Company, URL <https://www.mckinsey.com/industries/retail/our-insights/how-retailers-can-keep-up-with-consumers>.
- Papanastasiou Y, Sekar S (2024) Designing pandora’s box: Product rankings and marketplace participation. Available at SSRN: <https://ssrn.com/abstract=5006666> or <http://dx.doi.org/10.2139/ssrn.5006666>.
- Pei C, Yang X, Cui Q, Lin X, Sun F, Jiang P, Ou W, Zhang Y (2019) Value-aware recommendation based on reinforcement profit maximization. *The World Wide Web Conference*, 3123–3129, WWW ’19 (New York, NY, USA: Association for Computing Machinery), ISBN 9781450366748, URL <http://dx.doi.org/10.1145/3308558.3313404>.
- Rios I, Torrico A (2023) Platform design in curated dating markets. *arXiv preprint arXiv:2308.02584* .
- Shi P (2024) Optimal match recommendations in two-sided marketplaces with endogenous prices. *Management Science* .

- Ursu RM (2018a) The power of rankings in online consumer search. *Marketing Science* 37(4):555–578.
- Ursu RM (2018b) The power of rankings: Quantifying the effect of rankings on online consumer search and purchase decisions. *Marketing Science* 37(4):530–552.
- Zhang M, Bockstedt JC (2016) Complements and substitutes in product recommendations: The differential effects on consumers’ willingness-to-pay. *Proceedings of the 2016 Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS 2016)*, volume 1679 of *CEUR Workshop Proceedings*, 36–43 (Boston, MA, USA: CEUR-WS.org), URL <http://ceur-ws.org/Vol-1679/paper6.pdf>.
- Zhou B, Zou T (2023) Competing for recommendations: The strategic impact of personalized product recommendations in online marketplaces. *Marketing Science* 42(2):360–376.
- Zhu D, Minner S, Bichler M (2025) Pricing policy and queue-length disclosure in on-demand service platforms. *Transportation Research Part E: Logistics and Transportation Review* 204:104422.

Appendix A: Proof of Theorem 2

In this section, we fix an arbitrary customer segment. Further, for ease of exposition, we drop any index indicating the customer segment. Consider a customer with valuation $v \in [\underline{v}, 1]$ who faces a sequence of suppliers drawn independently each period from the distribution D . The customer follows a stationary accept-reject threshold strategy, denoted by $\theta(v)$. Recall the instantaneous net utility from matching with a supplier of type δ is given by:

$$u_c(v, \delta) = v - p(\delta) - \delta.$$

Recall the acceptance probability:

$$q_c(v) = \Pr_{\delta \sim D}(u_c(v, \delta) \geq \theta(v)),$$

which is the probability that the customer meets a supplier providing at least the threshold utility $\theta(v)$. Then, we have the following result that characterizes the customer’s lifetime surplus.

LEMMA 1 (Customer lifetime surplus). *The expected lifetime surplus of the customer can be expressed as:*

$$\pi_c(v, \theta(v)) = \frac{\mathbb{E}_{\delta \sim D}[u_c(v, \delta) \mathbf{1}\{u_c(v, \delta) \geq \theta(v)\}]}{\mu + (1 - \mu)q_c(v)} = \frac{\theta(v)}{1 - \mu}.$$

Furthermore, the threshold $\theta(v)$ satisfies the indifference condition:

$$\theta(v) = (1 - \mu)\pi_c(v, \theta(v)).$$

Proof of Lemma 1 The first expression for the expected lifetime surplus,

$$\pi_c(v, \theta(v)) = \frac{\mathbb{E}_{\delta \sim D}[u_c(v, \delta) \mathbf{1}\{u_c(v, \delta) \geq \theta(v)\}]}{\mu + (1 - \mu)q_c(v)},$$

is derived directly from the Bellman equation characterizing the customer’s dynamic program, as established in Section 3.1.

Next, we explicitly characterize the threshold condition. By definition, at the threshold utility $\theta(v)$, the customer is exactly indifferent between accepting the marginal offer and rejecting to continue. If the customer rejects the offer, they remain active with probability $(1 - \mu)$, preserving the continuation value $\pi_c(v, \theta(v))$. This implies the threshold must satisfy:

$$\theta(v) = (1 - \mu)\pi_c(v, \theta(v)),$$

reflecting equality between immediate utility at the margin and the discounted expected future surplus. Rearranging this threshold condition directly yields:

$$\pi_c(v, \theta(v)) = \frac{\theta(v)}{1 - \mu},$$

establishing precisely the second equality of the lemma. Lastly, we verify that the denominator $\mu + (1 - \mu)q_c(v)$ is strictly positive. Since $\mu \in (0, 1)$, it immediately follows that this expression is strictly bounded away from zero, ensuring that the customer lifetime surplus π_c is well-defined and unique. This concludes the proof of Lemma 1. $\square$

A.1. Allowing recall does not change customer best responses

LEMMA 2 (Recall is without loss of generality). *Fix (j, v) and a stationary environment induced by $\mathbf{D}$ as in Section 3.1. If the customer can purchase any previously inspected option at its posted price (full recall), then an optimal policy purchases at the first inspection whose realized net utility exceeds $\theta_j(v; \mathbf{D})$. Consequently, $q_{c,j}(v; \mathbf{D})$ and $\pi_{c,j}(v)$ (hence all equilibrium objects) coincide with the baseline model.*

Proof. Fix $(j, v, \mathbf{D})$. Let U denote the i.i.d. net-utility draw from a single inspection in Section 3.1, and set $\tilde{U} := U \vee 0$. With recall, after each inspection the customer's feasible purchase payoff is the running maximum $m \geq 0$ of previously observed $\tilde{U}$'s. Let $V(m)$ be the optimal value at this post-inspection decision epoch. Then

$$V(m) = \max\{m, (1 - \mu_j) \mathbb{E}[V(m \vee \tilde{U})]\}, \quad m \geq 0. \quad (6)$$

The Bellman operator in (6) is a $(1 - \mu_j)$ -contraction on bounded functions; moreover, starting from $V_0(m) = m$, iterates preserve monotonicity and 1-Lipschitzness, so the fixed point V is increasing and 1-Lipschitz. Define $\Delta(m) := m - (1 - \mu_j) \mathbb{E}[V(m \vee \tilde{U})]$. For $m' > m$, Lipschitzness implies $\mathbb{E}[V(m' \vee \tilde{U})] - \mathbb{E}[V(m \vee \tilde{U})] \leq m' - m$, hence $\Delta(m') - \Delta(m) \geq \mu_j(m' - m) > 0$. Thus Δ crosses 0 once; letting m^* solve $\Delta(m^*) = 0$, stopping is optimal iff $m \geq m^*$. At m^* , indifference gives $m^* = (1 - \mu_j) \mathbb{E}[V(m^* \vee \tilde{U})]$, and since $m^* \vee \tilde{U} \geq m^*$ lies in the stopping region, $V(m^* \vee \tilde{U}) = m^* \vee \tilde{U}$, so

$$m^* = (1 - \mu_j) \mathbb{E}[m^* \vee \tilde{U}]. \quad (7)$$

In the baseline (no-recall) problem, the Bellman equation in Section 3.1 together with $\theta_j(v; \mathbf{D}) = (1 - \mu_j) \pi_{c,j}(v)$ (Lemma 1) implies the same fixed point $\theta_j(v; \mathbf{D}) = (1 - \mu_j) \mathbb{E}[\theta_j(v; \mathbf{D}) \vee \tilde{U}]$. The map $x \mapsto (1 - \mu_j) \mathbb{E}[x \vee \tilde{U}]$ is a contraction, hence has a unique fixed point, so $m^* = \theta_j(v; \mathbf{D})$.

Starting from empty history, $m = 0 < \theta_j(v; \mathbf{D})$, so the stopping time is $\inf\{t : \tilde{U}_t \geq \theta_j(v; \mathbf{D})\}$; at that time the running maximum crosses the cutoff and equals the current draw, so previously rejected options are never chosen. Therefore the customer's purchase rule and $q_{c,j}(v; \mathbf{D})$ coincide with the baseline model, leaving supplier demand, equilibrium prices, and all downstream results unchanged.

A.2. Proof of Proposition 1: Envelope derivative and slope bounds

In this subsection, we show that, in any stationary equilibrium, the threshold $\theta(v)$ is absolutely continuous, and its derivative satisfies, for almost every valuation v :

$$\theta'(v) = \frac{(1 - \mu) q_c(v)}{\mu + (1 - \mu) q_c(v)} \in (0, 1). \quad (8)$$

Consequently, the map $g(v) := v - \theta(v)$ is strictly increasing and invertible.

Proof. We structure the proof in several steps, and start by defining the key quantities. We begin by defining two critical equilibrium-related terms. First, let

$$A(v) := \mathbb{E}_\delta [(v - p(\delta) - \delta) \cdot \mathbf{1}\{v - p(\delta) - \delta \geq \theta(v)\}],$$

represent the expected surplus conditional on acceptance. Second, define the customer's effective departure rate at valuation v by

$$B(v) := \mu + (1 - \mu) q_c(v),$$

accounting for both impatience (μ) and acceptance probability ($q_c(v)$). In equilibrium, the customer’s value function satisfies the Bellman equation, yielding the lifetime surplus:

$$\pi_c(v) = \frac{A(v)}{B(v)}. \quad (9)$$

At the acceptance threshold $\theta(v)$, the customer is precisely indifferent between accepting and continuing to wait. Thus, we must have:

$$A(v) = \frac{\theta(v)}{1-\mu} B(v). \quad (10)$$

We next differentiate $A(v)$ with respect to valuation v . By changing variables and applying Leibniz’s rule for differentiation under the integral sign, we obtain:

$$A'(v) = q_c(v) + \theta(v) q'_c(v). \quad (11)$$

Here, $q_c(v)$ appears naturally as the integrand evaluated at the boundary defined by $g(v) = v - \theta(v)$. Using equations (9) and (11), along with $B'(v) = (1-\mu)q'_c(v)$, we apply the quotient rule to differentiate the lifetime surplus $\pi_c(v)$:

$$\pi'_c(v) = \frac{A'(v)B(v) - A(v)B'(v)}{B(v)^2} = \frac{[q_c(v) + \theta(v)q'_c(v)]B(v) - A(v)(1-\mu_1)q'_c(v)}{B(v)^2}. \quad (12)$$

Substitute the equilibrium indifference condition from (10), $A(v) = \frac{\theta(v)}{1-\mu_1} B(v)$, into the second term of the numerator:

$$A(v)(1-\mu_1)q'_c(v) = \left(\frac{\theta(v)}{1-\mu_1} B(v) \right) (1-\mu_1)q'_c(v) = \theta(v)B(v)q'_c(v).$$

Substituting this back into the numerator of (12), the terms involving $q'_c(v)$ cancel exactly:

$$[q_c(v) + \theta(v)q'_c(v)]B(v) - \theta(v)B(v)q'_c(v) = q_c(v)B(v).$$

This yields a clean, intuitive expression for the derivative of the value function:

$$\pi'_c(v) = \frac{q_c(v)B(v)}{B(v)^2} = \frac{q_c(v)}{\mu_1 + (1-\mu_1)q_c(v)}. \quad (13)$$

Lastly, recall that at equilibrium, the threshold $\theta(v)$ is directly related to the surplus $\pi_c(v)$ by:

$$\theta(v) = (1-\mu)\pi_c(v).$$

Differentiating this relationship with respect to v and substituting (13) yields:

$$\theta'(v) = (1-\mu)\pi'_c(v) = \frac{(1-\mu)q_c(v)}{\mu + (1-\mu)q_c(v)}, \quad (14)$$

Since $0 < \mu < 1$ and $0 < q_c(v) < 1$, it follows directly that

$$0 < \theta'(v) < 1,$$

establishing absolute continuity and strict monotonicity of $g(v) = v - \theta(v)$, as required. This completes the proof. $\square$

A.3. Steady-State Densities and Proof of Theorem 2

Let the arrival valuation distribution be uniform on $[\underline{v}, \bar{v}]$, with arrival density $f_V^{\text{arr}}(v) = \frac{1}{\bar{v}-\underline{v}}$ on this support.

Step 1: Entry Decision and Inflow. Since the lifetime surplus $\pi_c(v)$ is strictly increasing in v (Appendix A.2), let v_e be the valuation such that $\pi_c(v_e) = 0$. The set of entering customers is $\mathcal{V}^* = \{v \in [\underline{v}, \bar{v}] : v \geq v_e\}$. The effective entry cutoff is $v_e^* = \max\{\underline{v}, v_e\}$. The per-type inflow of entrants is:

$$\Lambda_{in}(v) = \lambda f_V^{\text{arr}}(v) \mathbf{1}\{v \geq v_e\}. \quad (15)$$

Step 2: Steady-state density of valuations. The exit hazard rate for a customer of valuation v is $B(v) = \mu + (1 - \mu)q_c(v)$. The steady-state flow balance requires inflow to equal outflow at every valuation level. Thus, the steady-state density of active customers $c_V(v)$ satisfies $c_V(v)B(v) = \Lambda_{in}(v)$.

$$c_V(v) = \frac{\Lambda_{in}(v)}{B(v)} = \frac{\lambda f_V^{\text{arr}}(v) \mathbf{1}\{v \geq v_e\}}{\mu + (1 - \mu)q_c(v)}. \quad (16)$$

Step 3: Transformation to willingness-to-pay. We apply the change of variables $\omega = g(v) = v - \theta(v)$. By Proposition 1, $g(v)$ is strictly increasing; let $h = g^{-1}$ be the inverse function. The density of WTP, $f_W(\omega)$, is given by the standard change-of-variables formula:

$$f_W(\omega) = \frac{c_V(h(\omega))}{g'(h(\omega))}. \quad (17)$$

From Appendix A.2 (Eq. (14)), we derived the derivative of the threshold $\theta'(v)$. The derivative of the mapping $g(v)$ is:

$$g'(v) = 1 - \theta'(v) = 1 - \frac{(1 - \mu)q_c(v)}{\mu + (1 - \mu)q_c(v)} = \frac{(\mu + (1 - \mu)q_c(v)) - (1 - \mu)q_c(v)}{\mu + (1 - \mu)q_c(v)} = \frac{\mu}{\mu + (1 - \mu)q_c(v)} = \frac{\mu}{B(v)}. \quad (18)$$

We now substitute $c_V(v)$ from (16) and $g'(v)$ from (18) into the transformation (17). Let $v = h(\omega)$.

$$f_W(\omega) = c_V(v) \cdot \frac{1}{g'(v)} = \left(\frac{\lambda f_V^{\text{arr}}(v) \mathbf{1}\{v \geq v_e\}}{B(v)} \right) \cdot \left(\frac{B(v)}{\mu} \right). \quad (19)$$

The term $B(v)$ cancels exactly, yielding the general relationship:

$$f_W(\omega) = \frac{\lambda}{\mu} f_V^{\text{arr}}(h(\omega)) \mathbf{1}\{h(\omega) \geq v_e\}. \quad (20)$$

This establishes the fundamental result that the equilibrium WTP density perfectly preserves the shape of the arrival valuation density, scaled by the factor λ/μ .

Step 4: Uniform Arrivals (Proof of Theorem 2). If arrivals are uniform on $[\underline{v}, \bar{v}]$, $f_V^{\text{arr}}(v) = \frac{1}{\bar{v} - \underline{v}}$. Substituting this specific density into the general result (20):

$$f_W(\omega) = \frac{\lambda}{\mu} \cdot \frac{1}{\bar{v} - \underline{v}} = \frac{\lambda}{\mu(\bar{v} - \underline{v})}. \quad (21)$$

This density is constant on the support of WTP, which is $[\underline{\omega}, \bar{\omega}] = g([v_e, \bar{v}])$. This completes the proof of Theorem 2. $\square$

Appendix B: Equilibrium Pricing Analysis and Proof of Proposition 2

We develop the framework for supplier optimization in the general J -segment equilibrium. We first characterize the optimal pricing strategy (Section B.1). We then analyze the monotonicity and convexity properties of these prices under the optimal policy D^* , and prove Proposition 2 (Section B.2). Finally, we establish key auxiliary equilibrium relationships (Section B.3).

B.1. Supplier Optimization in the J -Segment Equilibrium

We analyze the pricing problem of a type- δ supplier in a stationary equilibrium. By Theorem 2, segment- j willingness-to-pay (WTP) is uniform on $[\underline{\omega}_j, \bar{\omega}_j]$ with span $\Delta_j := \bar{\omega}_j - \underline{\omega}_j$.

B.1.1. Effective Parameters and Demand Aggregation

Exposure and segment mix. Let $\psi(\delta) \leq 1$ denote the per-period exposure probability of a type- δ supplier (Section 3.2). Conditional on exposure, the probability that the customer is from segment j is $\beta_j(\delta)$, with $\sum_{j=1}^J \beta_j(\delta) = 1$.

Sale probabilities. Given price p and misfit disutility $d_j(\delta)$ for segment j , uniform WTP implies the conditional sale probability

$$\sigma_j(p, \delta) = \frac{\bar{\omega}_j - d_j(\delta) - p}{\Delta_j}, \quad (22)$$

on the interior region where $0 < \sigma_j < 1$.⁸ Aggregating across segments, the normalized sale probability is $\sigma(p, \delta) = \sum_{j=1}^J \beta_j(\delta) \sigma_j(p, \delta)$, and the per-period transaction rate is $\phi_S(p, \delta) = \psi(\delta) \sigma(p, \delta)$.

Effective parameters. It is convenient to represent the aggregated demand as if the supplier faced a single “effective” segment.

DEFINITION 2 (EFFECTIVE PARAMETERS). Define the effective WTP span

$$\Delta_{\text{eff}}(\delta) := \left(\sum_{j=1}^J \frac{\beta_j(\delta)}{\Delta_j} \right)^{-1}, \quad (23)$$

and the effective upper WTP and misfit disutility

$$\bar{\omega}_{\text{eff}}(\delta) := \Delta_{\text{eff}}(\delta) \sum_{j=1}^J \frac{\beta_j(\delta) \bar{\omega}_j}{\Delta_j}, \quad d_{\text{eff}}(\delta) := \Delta_{\text{eff}}(\delta) \sum_{j=1}^J \frac{\beta_j(\delta) d_j(\delta)}{\Delta_j}.$$

LEMMA 3 (EQUIVALENT SALE PROBABILITY). The normalized sale probability can be written as

$$\sigma(p, \delta) = \frac{\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta) - p}{\Delta_{\text{eff}}(\delta)}.$$

Proof of Lemma 3. Expand $\sigma(p, \delta) = \sum_j \beta_j(\delta) \sigma_j(p, \delta)$ using (22) and collect terms in $\sum_j \beta_j(\delta) / \Delta_j$, $\sum_j \beta_j(\delta) \bar{\omega}_j / \Delta_j$, and $\sum_j \beta_j(\delta) d_j(\delta) / \Delta_j$; substituting Definition 2 yields the claim.

B.1.2. Optimal Pricing and Profit Characterization We normalize the ad-valorem commission to $\tau = 0$, since it scales profits but does not affect the supplier’s optimal price. A type- δ supplier chooses p to maximize expected per-period profit $\pi_s(p, \delta) = p \phi_S(p, \delta) = \psi(\delta) p \sigma(p, \delta)$. Using Lemma 3, this becomes a concave quadratic in p :

$$\pi_s(p, \delta) = \frac{\psi(\delta)}{\Delta_{\text{eff}}(\delta)} \left(p(\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta)) - p^2 \right).$$

PROPOSITION 9 (OPTIMAL PRICING). The supplier’s optimal price is

$$p^*(\delta) = \frac{\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta)}{2}. \quad (24)$$

Moreover, at the optimum the (effective) residual willingness-to-pay equals the posted price:

$$\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta) - p^*(\delta) = p^*(\delta). \quad (25)$$

⁸ Outside the interior, σ_j is truncated to 0 or 1; these boundary cases are irrelevant for the interior pricing characterization used in the equilibrium analysis.

Proof of Proposition 9. Differentiate the quadratic objective in p and set the derivative to zero: $\partial\pi_s/\partial p \propto (\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta)) - 2p = 0$. Concavity implies the solution is the unique maximizer, yielding (24). Identity (25) follows by substitution.

COROLLARY 1 (Optimal Profit). *At the optimum,*

$$\pi_s^*(\delta) = \psi(\delta) \frac{(p^*(\delta))^2}{\Delta_{\text{eff}}(\delta)} = \psi(\delta) \frac{(\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta))^2}{4\Delta_{\text{eff}}(\delta)}.$$

Proof. Substitute $p^*(\delta)$ into $\pi_s(p, \delta) = \psi(\delta)p\sigma(p, \delta)$ and use

$$\sigma(p^*(\delta), \delta) = \frac{\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta) - p^*(\delta)}{\Delta_{\text{eff}}(\delta)} = \frac{p^*(\delta)}{\Delta_{\text{eff}}(\delta)},$$

where the last equality follows from (25).

B.2. Monotonicity, Convexity, and Proof of Proposition 2

We analyze the optimal supplier price function $p^*(\delta)$ in the single-segment case ($J = 1$) with $d(\delta) = \delta$, under the equilibrium induced by the optimal recommendation policy D^* .

B.2.1. Simplification in the single-segment equilibrium By Lemma 3 specialized to $J = 1$, the (per-exposure) sale probability is $\sigma(p, \delta) = (\bar{\omega} - \delta - p)/\Delta$. A type- δ supplier chooses p to maximize per-period expected profit $\psi(\delta)p\sigma(p, \delta)$. Since $\psi(\delta)$ is multiplicative and does not depend on p , the optimal price solves the static quadratic problem and is given by

$$p^*(\delta) = \frac{\bar{\omega} - \delta}{2}, \quad p_e^*(\delta) := p^*(\delta) + \delta = \frac{\bar{\omega} + \delta}{2}. \quad (26)$$

LEMMA 4 (Full exposure at the optimum (single segment)). *Under the optimal recommendation policy D^* , the exposure probability satisfies $\psi(\delta) = 1$ for almost all active suppliers $\delta \in \mathcal{S}^*$.*

Proof. Work at the optimal equilibrium induced by D^* and write $C^* := C(M^*)$. Feasibility is the pointwise ceiling $C^*D^*(\delta) \leq s^*(\delta)$ (a.e.), with multiplier $\eta^*(\delta) \geq 0$ and complementary slackness $\eta^*(\delta)(s^*(\delta) - C^*D^*(\delta)) = 0$. The KKT stationarity condition for D (Appendix C.4) can be written pointwise as $\hat{H}^*(\delta) - \nu^* - C^*\eta^*(\delta) + \zeta^*(\delta) = 0$ a.e., where ν^* is the mass-constraint multiplier, $\zeta^*(\delta) \geq 0$ is the nonnegativity multiplier, and $\zeta^*(\delta)D^*(\delta) = 0$ a.e. Hence, whenever $D^*(\delta) > 0$ we have $\zeta^*(\delta) = 0$ and therefore $\hat{H}^*(\delta) = \nu^* + C^*\eta^*(\delta)$.

If the capacity constraint were slack on a set of positive measure inside $\text{supp}(D^*)$, then $\eta^*(\delta) = 0$ there and thus $\hat{H}^*(\delta) = \nu^*$ would be constant on that set. But in the present single-segment model, $\hat{H}^*(\delta) = C^*r(\delta) + \xi^*X^*(\delta)$ (Appendix C.4), where $X^*(\delta) = p^*(\delta) + \delta$ is affine in δ by (26), and $r(\delta) = p^*(\delta)\sigma(p^*(\delta), \delta) = p^*(\delta)^2/\Delta$ is strictly convex in δ because $p^*(\delta)$ is affine and non-constant. Adding an affine function preserves strict convexity, so $\hat{H}^*$ cannot be constant on a set of positive measure. This contradiction implies the ceiling binds a.e. on $\text{supp}(D^*)$, i.e., $C^*D^*(\delta) = s^*(\delta)$ for a.e. $\delta \in \mathcal{S}^*$. By definition $\psi(\delta) = \min\{1, C^*D^*(\delta)/s^*(\delta)\}$, so $\psi(\delta) = 1$ for a.e. $\delta \in \mathcal{S}^*$. $\square$

REMARK 1 (WHERE THE MULTI-SEGMENT FULL-EXPOSURE STATEMENT IS PROVED). In the generalized (J -segment) optimal policy, the analogous “full exposure on the used types” statement follows from residual-capacity saturation and the aggregate bind-gap-bind structure proved in Appendix G; see Corollary 2.

B.2.2. Proof of Proposition 2 From (26), the posted price is affine with slope $dp^*/d\delta = -\frac{1}{2} < 0$, so $p^*(\delta)$ is strictly decreasing in δ . Moreover, for $\delta \in \mathcal{S}^*$ we have $\psi(\delta) = 1$ a.e. by Lemma 4, so the equilibrium profit is proportional to $r(\delta) = p^*(\delta)^2/\Delta$ and is strictly decreasing in δ as well. Finally, the effective price $p_e^*(\delta) = (\bar{\omega} + \delta)/2$ satisfies $dp_e^*/d\delta = \frac{1}{2} > 0$, hence it is strictly increasing in δ . This completes the proof. $\square$

B.3. Auxiliary Equilibrium Relationships

We establish fundamental relationships between the segment- j WTP bounds and the mean effective price

$$M_j := \mathbb{E}_{\delta \sim D_j} [p^*(\delta) + d_j(\delta)],$$

allowing for general $U[\underline{v}_j, \bar{v}_j]$ primitives and an arbitrary minimum misfit $d_j^{\min}$.

LEMMA 5 (Generalized WTP Bounds and Feedback Identity). *Fix any stationary equilibrium. Let $\Delta_j := \bar{\omega}_j - \underline{\omega}_j$ and let*

$$d_j^{\min} := \inf_{\delta \in \text{supp}(D_j)} d_j(\delta)$$

be the infimum misfit disutility among supplier types recommended to segment j .

(i) **Feedback Identity:** $\bar{\omega}_j(M_j) = \mu_j \bar{v}_j + (1 - \mu_j)M_j$.

(ii) **WTP Bounds Relationship:** *Suppose the infimum $d_j^{\min}$ is attained by an active supplier $\delta_j^* \in \text{supp}(D_j)$ and that this supplier (a) has full exposure and (b) is exposed only to segment j customers (conditions that hold on the binding set under the optimal generalized policy by Theorem 4 and Lemma 14 in Appendix G). Then*

$$\Delta_j = \underline{\omega}_j - d_j^{\min} \quad \left(\text{equivalently, } \underline{\omega}_j = \frac{\bar{\omega}_j + d_j^{\min}}{2} \right).$$

(iii) **Explicit Bound:** *Under the conditions of (ii), the lowest WTP satisfies*

$$\underline{\omega}_j(M_j) = \frac{\mu_j \bar{v}_j + (1 - \mu_j)M_j + d_j^{\min}}{2}.$$

Proof. Step 1: Flow balance and Feedback Identity (i). Let $L_j = \bar{v}_j - \underline{v}_j$. By Theorem 2, the active WTP density is $f_{W_j}(\omega) = \lambda_j/(\mu_j L_j)$ on $[\underline{\omega}_j, \bar{\omega}_j]$, so the active mass is $C_j = \lambda_j \Delta_j/(\mu_j L_j)$. The inflow rate of customers who enter the active pool is $F_j = \lambda_j(\bar{v}_j - \underline{\omega}_j)/L_j$.

Let $X_j(\delta) := p^*(\delta) + d_j(\delta)$ denote the segment- j effective price of type δ . Conditional on a match with type δ , uniform WTP on $[\underline{\omega}_j, \bar{\omega}_j]$ implies the purchase probability

$$\sigma_j(\delta) = \Pr(\omega \geq X_j(\delta)) = \frac{\bar{\omega}_j - X_j(\delta)}{\Delta_j},$$

on the interior region. Therefore, the average purchase probability among active segment- j customers is $q_j = \mathbb{E}_{\delta \sim D_j} [\sigma_j(\delta)] = (\bar{\omega}_j - M_j)/\Delta_j$. Stationarity implies $F_j = (\mu_j + (1 - \mu_j)q_j)C_j$.

Step 2: Relationship between Δ_j and $d_j^{\min}$ (ii). Let δ_j^* attain $d_j^{\min}$. Under the stated conditions, δ_j^* is fully exposed and faces only segment j , so it solves a standard static pricing problem against $\sigma_j(p, \delta_j^*) = (\bar{\omega}_j - d_j^{\min} - p)/\Delta_j$. Maximizing $p\sigma_j(p, \delta_j^*)$ yields $p^*(\delta_j^*) = (\bar{\omega}_j - d_j^{\min})/2$. Because under (b) the segment- j effective price is $X_j(\delta) = (\bar{\omega}_j + d_j(\delta))/2$ (so it is increasing in $d_j(\delta)$), the type attaining $d_j^{\min}$ also attains

the lowest effective price faced by segment j , hence $p^*(\delta_j^*) + d_j^{\min} = \underline{\omega}_j$. Combining these equalities gives $\underline{\omega}_j = (\bar{\omega}_j + d_j^{\min})/2$ and hence

$$\Delta_j = \bar{\omega}_j - \underline{\omega}_j = \bar{\omega}_j - d_j^{\min}.$$

Step 3: Explicit bound (iii). From Step 2, $\bar{\omega}_j = 2\underline{\omega}_j - d_j^{\min}$. Equating this with the Feedback Identity (i) and solving for $\underline{\omega}_j$ gives $2\underline{\omega}_j = \mu_j \bar{v}_j + (1 - \mu_j)M_j + d_j^{\min}$, which is exactly (iii). $\square$

LEMMA 6 (Strict convexity of the revenue kernel under single-segment exposure). *Fix a segment j and consider the single-segment demand environment with linear misfit $d_j(\delta) = a_j\delta + b_j$ where $a_j \neq 0$. On the interior region, the (per-exposure) sale probability takes the linear form*

$$\sigma_j(p, \delta) = \frac{\bar{\omega}_j - d_j(\delta) - p}{\Delta_j},$$

where $(\underline{\omega}_j, \bar{\omega}_j)$ and $\Delta_j = \bar{\omega}_j - \underline{\omega}_j$ are held fixed in δ . Then the optimal per-exposure revenue kernel $r_j(\delta) := p_j^*(\delta) \sigma_j(p_j^*(\delta), \delta)$ is strictly convex in δ .

Proof. The supplier's myopic best response solves a concave quadratic and is $p_j^*(\delta) = (\bar{\omega}_j - d_j(\delta))/2$ (cf. Proposition 9 applied to a single segment). At the optimum, $\sigma_j(p_j^*(\delta), \delta) = p_j^*(\delta)/\Delta_j$, so $r_j(\delta) = \frac{(p_j^*(\delta))^2}{\Delta_j} = \frac{(\bar{\omega}_j - d_j(\delta))^2}{4\Delta_j}$. Since $d_j(\delta)$ is affine with nonzero slope a_j , $r_j(\delta)$ is a quadratic in δ with

$$r_j''(\delta) = \frac{a_j^2}{2\Delta_j} > 0,$$

establishing strict convexity. $\square$

LEMMA 7 (Comparative statics in M (under single-segment exposure)). *Under the conditions of Lemma 5(ii)–(iii) for a given segment j (so the WTP bounds are pinned down by M_j and the best-fit active supplier is fully exposed and faces only segment j), fix a supplier type δ . Then the equilibrium price $p^*(\delta)$ and revenue kernel $r(\delta)$ satisfy*

$$0 < \frac{\partial p^*(\delta)}{\partial M_j} < 1 - \mu_j, \quad \frac{\partial r(\delta)}{\partial M_j} > 0.$$

Proof. Fix a segment and drop the subscript j for readability. Holding δ fixed, treat M as the scalar parameter shifting the equilibrium WTP bounds.

Step 1: WTP bounds as functions of M . Lemma 5(i) gives

$$\bar{\omega}(M) = \mu \bar{v} + (1 - \mu)M. \tag{27}$$

Moreover, Lemma 5(ii) implies $\underline{\omega} = (\bar{\omega} + d^{\min})/2$ and hence

$$\Delta(M) = \bar{\omega}(M) - \underline{\omega}(M) = \frac{\bar{\omega}(M) - d^{\min}}{2}. \tag{28}$$

(As in the main text, one may normalize $d^{\min} = 0$ without affecting any derivative with respect to M .)

Step 2: Price. Against linear demand $\sigma(p, \delta) = (\bar{\omega}(M) - d(\delta) - p)/\Delta(M)$, the static best response is $p^*(\delta; M) = (\bar{\omega}(M) - d(\delta))/2$. Since $d\bar{\omega}/dM = 1 - \mu$ from (27), we obtain

$$\frac{\partial p^*}{\partial M} = \frac{1}{2} \frac{d\bar{\omega}}{dM} = \frac{1 - \mu}{2} > 0,$$

and in particular $0 < \partial p^* / \partial M < (1 - \mu)$.

Step 3: Revenue kernel. At the optimal price, $\sigma(p^*, \delta) = p^*(\delta; M) / \Delta(M)$, so $r(\delta; M) = (p^*(\delta; M))^2 / \Delta(M)$. Equivalently, $r(\delta; M) = (\bar{\omega}(M) - d(\delta))^2 / (2(\bar{\omega}(M) - d^{\min}))$ using (28). Differentiating with respect to $\bar{\omega}$ yields

$$\frac{\partial r}{\partial \bar{\omega}} = \frac{(\bar{\omega} - d(\delta))(\bar{\omega} + d(\delta) - 2d^{\min})}{2(\bar{\omega} - d^{\min})^2}.$$

For an active supplier, $\bar{\omega} - d(\delta) > 0$, and $\bar{\omega} + d(\delta) - 2d^{\min} = (\bar{\omega} - d^{\min}) + (d(\delta) - d^{\min}) > 0$. Thus $\partial r / \partial \bar{\omega} > 0$, and since $d\bar{\omega} / dM = 1 - \mu > 0$, we conclude $\partial r / \partial M > 0$. $\square$

Appendix C: Proof of Theorem 3

Before we provide the fully technical arguments, we start by providing a high-level proof sketch.

C.1. Proof sketch

We first define the marginal-revenue density (Gateaux derivative) $H(\delta; D)$,⁹ which measures the first-order change in platform revenue from shifting an infinitesimal amount of recommendation mass toward supplier type δ when the current recommendation density is D . We show the key geometric fact is that $\delta \mapsto H(\delta; D)$ is U-shaped: it decreases up to a unique interior minimizer $\delta_{\min}$ and increases thereafter. Intuitively, intermediate- δ suppliers are neither “good enough” matches to generate high surplus nor “niche enough” to create much scarcity value, so the marginal value of sending exposure to them is lowest; by contrast, the extremes have the highest marginal value. With this U-shape in hand, we write down the Karush–Kuhn–Tucker (KKT) conditions for the platform’s problem with the endogenous capacity ceiling $D(\delta) \leq \bar{D}(\delta; D)$, where $\bar{D}(\delta; D) = s(\delta; D) / C(D)$. In the augmented formulation used in the proof, the KKT system compares the *KKT-relevant marginal value density* $\hat{H}^*(\delta)$ to the multiplier ν^* on the mass constraint $\int D^* = 1$: if the ceiling binds at δ , then $\hat{H}^*(\delta) \geq \nu^*$; if $D^*(\delta) = 0$, then $\hat{H}^*(\delta) \leq \nu^*$; and equality holds only where the ceiling is slack (i.e., where $0 < D^*(\delta) < \bar{D}(\delta; D^*)$).

Combining this logic with the U-shape yields a two-threshold structure. There exist cutoffs $0 \leq \delta_0 \leq \delta_{\min} \leq \delta_1 \leq \bar{\delta}$ such that the ceiling is saturated on the extremes $[0, \delta_0] \cup [\delta_1, \bar{\delta}]$, while no probability is placed in the middle gap (δ_0, δ_1) . Since the ceiling ratio $s(\delta; D^*) / C(D^*)$ is strictly decreasing, the cutoffs are pinned down uniquely by the mass-balance condition $\int_0^{\bar{\delta}} D^*(\delta) d\delta = 1$. Finally, we relate (δ_0, δ_1) to market thickness m , which scales supplier capacity. When m is small, capacity is tight, and the solution uses all suppliers ($\delta_1 = \bar{\delta}$). As m increases, slack capacity appears among high-misfit suppliers and an additional upper block emerges. In the thick-market limit $m \rightarrow \infty$, the optimal allocation converges to the two-point distribution supported on the extremes $\{0, \bar{\delta}\}$.

C.2. The Main Challenges in Analysis

The proof sketch above captures the main geometry (a U-shaped marginal value combined with KKT inequalities), but the full problem is more delicate because we optimize over a distribution D and equilibrium responds endogenously through the fixed point $M(D)$. The detailed sections address three challenges that are essential for a complete proof.

⁹ Formally, $H(\delta; D)$ is the Gateaux derivative of the reduced revenue functional $R(D)$, so it already internalizes how changing D shifts the equilibrium fixed point and all equilibrium objects (prices, customer thresholds, and the total customer mass $C(D)$).

Challenge 1: Handling endogenous (state-dependent) constraints (Appendix C.4). The capacity ceiling is itself an equilibrium object: $\bar{D}(\delta; D) = s(\delta; D)/C(D)$. Changing D shifts the equilibrium fixed point $M(D)$ and therefore shifts prices, willingness-to-pay bounds, and the steady-state masses C and s , moving the ceiling. To apply KKT theory without treating the ceiling as fixed, we introduce the fixed point M as an explicit decision variable and impose the equilibrium identity $\Psi(D, M) := \int_0^{\bar{\delta}} X(\delta; M) D(\delta) d\delta - M = 0$. Conditional on M , feasibility becomes the standard box constraint $0 \leq D(\delta) \leq \bar{D}(\delta; M) := s(\delta; M)/C(M)$ a.e., and the multiplier on $\Psi = 0$ is precisely what internalizes equilibrium feedback in the KKT system.

Challenge 2: Existence and uniqueness of the cutoffs (Appendix C.6). Even if the relevant marginal value is strictly convex, we still need to show that the implied bind-or-zero policy is feasible and uniquely determined. The KKT conditions imply that the platform uses capacity where the KKT-relevant marginal value $\hat{H}^*(\delta)$ exceeds the shadow price of mass ν^* . Appendix C.6 formalizes this by defining the induced mass map $\Phi(\nu)$ (the total mass allocated at level ν) and showing that Φ is continuous and strictly decreasing. The Intermediate Value Theorem then yields a unique ν^* clearing $\int D^* = 1$, which uniquely pins down (δ_0, δ_1) .

Challenge 3: Comparative statics in market thickness (Appendix C.8). Theorem 3 also characterizes how the optimal policy changes with supplier capacity m . Appendix C.8 makes this comparative-statics argument rigorous and establishes the thick-market limit: as m grows, the policy becomes increasingly polarized and converges to the two-point distribution supported on the extremes.

We now proceed with the detailed arguments. We use the following notation conventions: $r(\delta; M)$ is the revenue kernel per exposure, $X(\delta; M)$ is the effective price, $H(\delta; D)$ is the Gateaux derivative of $R(D)$, and $\hat{H}^*(\delta)$ is the KKT-relevant marginal value density (the pointwise derivative of the augmented Lagrangian w.r.t. D at the optimum).

C.3. Augmented Formulation and Equilibrium Feedback

We begin by isolating the precise channel through which recommendation policies affect equilibrium. The central difficulty is that the platform's choice of D changes the equilibrium fixed point $M(D)$, which changes equilibrium prices $p^*(\cdot; M(D))$ and therefore effective prices $X(\cdot; M(D))$, which in turn changes market size $C(M(D))$ and supplier mass $s(\cdot; M(D))$. The capacity ceiling $\bar{D}(\delta; D) = s(\delta; D)/C(D)$ therefore moves endogenously with the platform's decision.

Step 1: The original optimization problem and endogenous equilibrium objects. We work with recommendation densities D on $[0, \bar{\delta}]$ satisfying

$$D(\delta) \geq 0 \text{ a.e.}, \quad \int_0^{\bar{\delta}} D(\delta) d\delta = 1.$$

Given such a D , the equilibrium is summarized by the scalar $M(D)$ that satisfies the fixed-point identity

$$M(D) = \int_0^{\bar{\delta}} X(\delta; M(D)) D(\delta) d\delta. \quad (29)$$

Here $X(\delta; M) = p^*(\delta; M) + \delta$ is the effective price, and $p^*(\delta; M)$ is the equilibrium price posted by type δ given the equilibrium scalar M (derived earlier in Appendix B). Equilibrium customer mass and supplier mass are denoted $C(M)$ and $s(\delta; M)$ respectively, and the per-exposure revenue kernel is $r(\delta; M)$ (All of

these objects are endogenous and depend on the platform decision D through the induced equilibrium scalar $M(D)$). The platform’s per-period revenue under D is

$$R(D) = C(M(D)) \int_0^{\bar{\delta}} r(\delta; M(D)) D(\delta) d\delta. \quad (30)$$

The platform also faces the endogenous capacity ceiling:

$$0 \leq D(\delta) \leq \bar{D}(\delta; D) := \frac{s(\delta; D)}{C(D)} \quad \text{a.e.} \quad (31)$$

Because the equilibrium depends on D only through $M(D)$ (Appendix F), we may equivalently write

$$\bar{D}(\delta; D) = \bar{D}(\delta; M(D)) \quad \text{where} \quad \bar{D}(\delta; M) := \frac{s(\delta; M)}{C(M)}.$$

Step 2: The equilibrium constraint $\Psi(D, M) = 0$ and why it captures feedback. Define the equilibrium residual

$$\Psi(D, M) := \int_0^{\bar{\delta}} X(\delta; M) D(\delta) d\delta - M. \quad (32)$$

The equilibrium condition (29) is equivalent to the constraint $\Psi(D, M(D)) = 0$. Crucially, $\Psi(D, M) = 0$ is the mathematical object that ensures we do *not* treat prices, willingness-to-pay, or market sizes as fixed when D changes: a perturbation in D changes the left-hand side of (32), forcing M to adjust endogenously, which then changes $p^*(\cdot; M)$ and $X(\cdot; M)$, and therefore changes $C(M)$ and $s(\cdot; M)$.

Step 3: Uniqueness of the fixed point (well-defined $M(D)$). We recall the key equilibrium stability property established earlier: for all (δ, M) in the relevant domain,

$$0 < \frac{\partial X}{\partial M}(\delta; M) < 1, \quad (33)$$

so that equilibrium feedback is stable (prices respond to M but not one-for-one). This implies that, for each fixed D , the fixed-point equation (29) has a unique solution.

LEMMA 8 (Existence and uniqueness of the equilibrium scalar). *Fix any admissible density D with $\int_0^{\bar{\delta}} D(\delta) d\delta = 1$ and $D \geq 0$ a.e. Define*

$$F_D(M) := \int_0^{\bar{\delta}} X(\delta; M) D(\delta) d\delta.$$

Then the fixed-point equation $M = F_D(M)$ admits a unique solution.

Proof of Lemma 8. Step 1: Existence. Because primitives are bounded and prices are nonnegative, there exists a finite constant $\bar{M} < \infty$ such that

$$0 \leq X(\delta; M) \leq \bar{M} \quad \text{for all } (\delta, M) \text{ in the relevant domain.}$$

(For example, if valuations are bounded above by $\bar{v}$ and $\delta \in [0, \bar{\delta}]$, one may take $\bar{M} = \bar{v} + \bar{\delta}$, since any $p > \bar{v}$ yields zero demand and is (weakly) dominated by $p = 0$; hence $p^*(\delta; M) \leq \bar{v}$ and $X(\delta; M) = p^*(\delta; M) + \delta \leq \bar{v} + \bar{\delta}$.) It follows that $F_D(M) \in [0, \bar{M}]$ for all $M \in [0, \bar{M}]$. Define $G_D(M) := F_D(M) - M$. By continuity of $X(\delta; M)$ in M and dominated convergence, $F_D(\cdot)$ (hence $G_D(\cdot)$) is continuous on $[0, \bar{M}]$. Moreover,

$$G_D(0) = F_D(0) \geq 0, \quad G_D(\bar{M}) = F_D(\bar{M}) - \bar{M} \leq 0.$$

Therefore, by the Intermediate Value Theorem, there exists $M^* \in [0, \bar{M}]$ such that $G_D(M^*) = 0$, i.e., $M^* = F_D(M^*)$.

Step 2: Uniqueness. By differentiability of $X(\delta; M)$ in M and dominated convergence,

$$F'_D(M) = \int_0^{\bar{\delta}} X_M(\delta; M) D(\delta) d\delta.$$

Using (33) and $D \geq 0$ with $\int_0^{\bar{\delta}} D(\delta) d\delta = 1$, we obtain

$$0 < F'_D(M) < 1.$$

Hence $G'_D(M) = F'_D(M) - 1 < 0$, so G_D is strictly decreasing and can cross zero at most once. Therefore the fixed point $M = F_D(M)$ is unique. $\square$

Step 4: The augmented optimization problem. We now treat M as an explicit decision variable and impose the equilibrium identity as a constraint. Define the fixed- M ceiling

$$\bar{D}(\delta; M) := \frac{s(\delta; M)}{C(M)}.$$

Consider the augmented problem:

$$(P_{\text{aug}}) \quad \max_{D, M} \quad C(M) \int_0^{\bar{\delta}} r(\delta; M) D(\delta) d\delta \quad (34)$$

$$\text{s.t.} \quad \int_0^{\bar{\delta}} D(\delta) d\delta = 1, \quad 0 \leq D(\delta) \leq \bar{D}(\delta; M) \quad \text{a.e.}, \quad (35)$$

$$\Psi(D, M) = 0. \quad (36)$$

This formulation is *not* a relaxation and does *not* hold equilibrium objects fixed: constraint (36) forces M to adjust consistently with D , and all equilibrium objects in (34)–(35) are evaluated at this endogenous M .

LEMMA 9 (Equivalence of the original and augmented problems). *Let (P) denote the original problem of choosing D to maximize $R(D)$ in (30) subject to (31) and $\int D = 1$, $D \geq 0$. Let (P_{aug}) be defined by (34)–(36). Then:*

1. *For any feasible D in (P) , the pair $(D, M(D))$ is feasible for (P_{aug}) and achieves the same objective value.*
2. *For any feasible (D, M) in (P_{aug}) , the density D is feasible for (P) and achieves the same objective value.*

Consequently, (P) and (P_{aug}) have the same set of optimal recommendation policies D^ .*

Proof of Lemma 9. We show the feasible sets correspond one-to-one via $D \leftrightarrow (D, M(D))$, with identical objective values.

(i) From (P) to (P_{aug}) . Take any feasible D in (P) and let $M(D)$ be the unique solution to (29) (Lemma 8), equivalently $\Psi(D, M(D)) = 0$. Because all equilibrium objects and the ceiling depend on D only through $M(D)$, feasibility of D under the original endogenous ceiling (31) is exactly feasibility under the box constraint $0 \leq D(\delta) \leq \bar{D}(\delta; M(D))$ in (P_{aug}) . The objective values coincide by (30) and (34).

(ii) **From (P_{aug}) to (P) .** Take any feasible (D, M) in (P_{aug}) . The constraint $\Psi(D, M) = 0$ implies that M solves the fixed point equation for D , hence $M = M(D)$ by Lemma 8. Substituting $M(D)$ into the box constraint gives $0 \leq D(\delta) \leq \bar{D}(\delta; M(D)) = \bar{D}(\delta; D)$, so D is feasible for (P) . The objective values again coincide.

Therefore the two problems have the same optimal recommendation policies D^* .

Step 5: Differentiating the equilibrium constraint (explicit feedback computation). Fix a feasible density D and a mass-preserving perturbation $D_\varepsilon = D + \varepsilon h$ with $\int_0^{\bar{\delta}} h(\delta) d\delta = 0$. Let M_ε be the unique scalar satisfying $\Psi(D_\varepsilon, M_\varepsilon) = 0$, i.e.,

$$0 = \Psi(D + \varepsilon h, M_\varepsilon) = \int_0^{\bar{\delta}} X(\delta; M_\varepsilon)(D(\delta) + \varepsilon h(\delta)) d\delta - M_\varepsilon.$$

Differentiate both sides with respect to ε at $\varepsilon = 0$ (implicit differentiation). This gives

$$0 = \int_0^{\bar{\delta}} X(\delta; M(D)) h(\delta) d\delta + \left(\int_0^{\bar{\delta}} X_M(\delta; M(D)) D(\delta) d\delta - 1 \right) \cdot \left. \frac{dM_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0}.$$

Rearranging yields the feedback formula

$$\left. \frac{dM_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} = \frac{\int_0^{\bar{\delta}} X(\delta; M(D)) h(\delta) d\delta}{1 - \int_0^{\bar{\delta}} X_M(\delta; M(D)) D(\delta) d\delta}, \quad (37)$$

where the denominator is strictly positive by (33). This is the same feedback channel that will appear in the KKT system through the multiplier on $\Psi(D, M) = 0$.

C.4. KKT Conditions and the Bind–Or–Zero Rule

We now derive the Karush–Kuhn–Tucker (KKT) conditions for the augmented problem (P_{aug}) defined in (34)–(36). Importantly, because equilibrium feedback is imposed explicitly by the constraint $\Psi(D, M) = 0$, the KKT system we derive here *fully internalizes* how a change in D changes the equilibrium fixed point M , which changes equilibrium prices $p^*(\cdot; M)$, effective prices $X(\cdot; M)$, and therefore market sizes $C(M)$ and supplier masses $s(\cdot; M)$.

The Lagrangian and the role of the equilibrium multiplier. Introduce the following multipliers: $\nu \in \mathbb{R}$ for the mass constraint $\int D = 1$; $\xi \in \mathbb{R}$ for the equilibrium constraint $\Psi(D, M) = 0$; $\zeta(\delta) \geq 0$ for the lower bound $D(\delta) \geq 0$ (a.e.); $\eta(\delta) \geq 0$ for the upper bound $\bar{D}(\delta; M) - D(\delta) \geq 0$ (a.e.). Then, the Lagrangian is

$$\begin{aligned} \mathcal{L}(D, M; \nu, \xi, \eta, \zeta) = & C(M) \int_0^{\bar{\delta}} r(\delta; M) D(\delta) d\delta - \nu \left(\int_0^{\bar{\delta}} D(\delta) d\delta - 1 \right) \\ & + \xi \left(\int_0^{\bar{\delta}} X(\delta; M) D(\delta) d\delta - M \right) + \int_0^{\bar{\delta}} \zeta(\delta) D(\delta) d\delta \\ & + \int_0^{\bar{\delta}} \eta(\delta) (\bar{D}(\delta; M) - D(\delta)) d\delta. \end{aligned} \quad (38)$$

The key point is the following: the equilibrium constraint enters *linearly* in D via the term $\int X(\delta; M) D(\delta) d\delta$, so its multiplier ξ will appear in the stationarity condition for D multiplied by $X(\delta; M)$. This is precisely the formal mathematical representation of equilibrium feedback: shifting recommendation mass toward types with larger $X(\delta; M)$ tends to increase the equilibrium fixed point M , thereby shifting prices and market sizes, and this effect is priced by ξ .

Step 3: Stationarity with respect to D (pointwise derivation). We compute the directional derivative of $\mathcal{L}$ with respect to D . Fix $(M, \nu, \xi, \eta, \zeta)$, and consider a perturbation $D_\varepsilon = D + \varepsilon h$ where $h \in L^\infty$ is arbitrary.¹⁰ We compute $\frac{d}{d\varepsilon} \mathcal{L}(D_\varepsilon, M; \nu, \xi, \eta, \zeta) \big|_{\varepsilon=0}$ term by term.

Since D appears linearly inside the integral, we evaluate the derivative at $\varepsilon = 0$ for each component term-by-term:

(i) The derivative of the objective term yields $C(M) \int_0^{\bar{\delta}} r(\delta; M) h(\delta) d\delta$, (ii) the mass constraint term contributes $-\nu \int_0^{\bar{\delta}} h(\delta) d\delta$, (iii) in the equilibrium constraint term, only the integral depends on D , resulting in $\xi \int_0^{\bar{\delta}} X(\delta; M) h(\delta) d\delta$, (iv) the lower-bound multiplier term adds $\int_0^{\bar{\delta}} \zeta(\delta) h(\delta) d\delta$, (v) in the upper-bound multiplier term, since only the $-D_\varepsilon$ part depends on D , we obtain $-\int_0^{\bar{\delta}} \eta(\delta) h(\delta) d\delta$. Summing the derivatives from (i)–(v) gives

$$\frac{d}{d\varepsilon} \mathcal{L}(D_\varepsilon, M; \nu, \xi, \eta, \zeta) \big|_{\varepsilon=0} = \int_0^{\bar{\delta}} \left(C(M)r(\delta; M) + \xi X(\delta; M) - \nu + \zeta(\delta) - \eta(\delta) \right) h(\delta) d\delta. \quad (39)$$

At an optimal solution (D^*, M^*) with associated multipliers $(\nu^*, \xi^*, \eta^*, \zeta^*)$, stationarity with respect to D requires that the directional derivative (39) equals zero for all directions h .¹¹ Because h is arbitrary, the integrand must be zero almost everywhere: if it were strictly positive on a set of positive measure, choosing h supported on that set would make (39) strictly positive; if it were strictly negative on such a set, choosing h with opposite sign would do the same. Hence we obtain the pointwise stationarity condition

$$C(M^*)r(\delta; M^*) + \xi^* X(\delta; M^*) - \nu^* + \zeta^*(\delta) - \eta^*(\delta) = 0 \quad \text{for a.e. } \delta \in [0, \bar{\delta}]. \quad (40)$$

Step 4: Stationarity with respect to M (explicit computation). We next compute $\partial \mathcal{L} / \partial M$. At a KKT point, stationarity requires $\frac{\partial}{\partial M} \mathcal{L}(D^*, M; \nu^*, \xi^*, \eta^*, \zeta^*) \big|_{M=M^*} = 0$. We again differentiate term by term, treating D^* and the multipliers as fixed during this partial differentiation.

(i) *Objective term.* Let $\bar{r}(D^*, M) := \int_0^{\bar{\delta}} r(\delta; M) D^*(\delta) d\delta$. Then the objective term is $C(M)\bar{r}(D^*, M)$. Differentiate using the product rule:

$$\begin{aligned} \frac{\partial}{\partial M} \left(C(M)\bar{r}(D^*, M) \right) &= C_M(M)\bar{r}(D^*, M) + C(M) \frac{\partial}{\partial M} \left(\int_0^{\bar{\delta}} r(\delta; M) D^*(\delta) d\delta \right) \\ &= C_M(M) \int_0^{\bar{\delta}} r(\delta; M) D^*(\delta) d\delta + C(M) \int_0^{\bar{\delta}} r_M(\delta; M) D^*(\delta) d\delta. \end{aligned}$$

(ii) *Mass and lower-bound terms.* The terms $-\nu^*(\int D^* - 1)$ and $\int \zeta^* D^*$ do not depend on M , so their M -derivatives are zero.

(iii) *Equilibrium constraint term.* Differentiate $\xi^*(\int X(\delta; M) D^*(\delta) d\delta - M)$ with respect to M :

$$\frac{\partial}{\partial M} \left[\xi^* \left(\int_0^{\bar{\delta}} X(\delta; M) D^*(\delta) d\delta - M \right) \right] = \xi^* \left(\int_0^{\bar{\delta}} X_M(\delta; M) D^*(\delta) d\delta - 1 \right).$$

¹⁰ At the stationarity step we differentiate the Lagrangian in an *unconstrained* direction h ; feasibility of D_ε is enforced separately via complementary slackness and dual feasibility.

¹¹ Formally, under standard constraint qualifications (e.g., a Slater-type condition), KKT multipliers exist for a local maximizer and stationarity holds; we use the pointwise form below because the direction h is arbitrary.

(iv) *Upper-bound term.* Only $\bar{D}(\delta; M)$ depends on M :

$$\frac{\partial}{\partial M} \left[\int_0^{\bar{\delta}} \eta^*(\delta) (\bar{D}(\delta; M) - D^*(\delta)) d\delta \right] = \int_0^{\bar{\delta}} \eta^*(\delta) \bar{D}_M(\delta; M) d\delta.$$

Collecting (i)–(iv). Evaluating at $M = M^*$, the M -stationarity condition is

$$\begin{aligned} 0 = & C_M(M^*) \int_0^{\bar{\delta}} r(\delta; M^*) D^*(\delta) d\delta + C(M^*) \int_0^{\bar{\delta}} r_M(\delta; M^*) D^*(\delta) d\delta \\ & + \xi^* \left(\int_0^{\bar{\delta}} X_M(\delta; M^*) D^*(\delta) d\delta - 1 \right) + \int_0^{\bar{\delta}} \eta^*(\delta) \bar{D}_M(\delta; M^*) d\delta. \end{aligned} \quad (41)$$

Equation (41) pins down the equilibrium shadow price ξ^* . Because $0 < X_M(\delta; M^*) < 1$ for all δ (Appendix C.3, Eq. (33)), and D^* is a probability density ($D^* \geq 0$ and $\int D^* = 1$), we have $0 < \int_0^{\bar{\delta}} X_M(\delta; M^*) D^*(\delta) d\delta < 1$, so the coefficient on ξ^* in (41) is strictly negative. Equivalently, we can rearrange (41) as

$$\xi^* = \frac{C_M(M^*) \int_0^{\bar{\delta}} r(\delta; M^*) D^*(\delta) d\delta + C(M^*) \int_0^{\bar{\delta}} r_M(\delta; M^*) D^*(\delta) d\delta + \int_0^{\bar{\delta}} \eta^*(\delta) \bar{D}_M(\delta; M^*) d\delta}{1 - \int_0^{\bar{\delta}} X_M(\delta; M^*) D^*(\delta) d\delta}. \quad (42)$$

The denominator in (42) is strictly positive, which shows equilibrium stability.

Step 5: Complementary slackness and dual feasibility. In addition to primal feasibility of (D^*, M^*) (i.e., the constraints of (P_{aug})), the KKT system requires dual feasibility:

$$\eta^*(\delta) \geq 0, \quad \zeta^*(\delta) \geq 0 \quad \text{for a.e. } \delta,$$

and complementary slackness:

$$\zeta^*(\delta) D^*(\delta) = 0, \quad \eta^*(\delta) (\bar{D}(\delta; M^*) - D^*(\delta)) = 0 \quad \text{for a.e. } \delta. \quad (43)$$

Step 6: The KKT-relevant marginal value density and the bind-or-zero rule. Define the *KKT-relevant marginal value density* as the pointwise derivative of the Lagrangian with respect to D , excluding inequality multipliers:

$$\hat{H}^*(\delta) := C(M^*) r(\delta; M^*) + \xi^* X(\delta; M^*). \quad (44)$$

This is the correct pointwise object to compare to the mass shadow price ν^* . It differs conceptually from the Gateaux derivative of the *unconstrained* reduced objective $R(D)$ because here we are characterizing an optimum under *state-dependent feasibility* through the explicit constraint $\Psi(D, M) = 0$ and the box constraints $0 \leq D \leq \bar{D}(\cdot; M)$. Using (40), we now obtain the standard bind-or-zero implications.

Case 1: $\hat{H}^*(\delta) > \nu^*$. Then $\eta^*(\delta) - \zeta^*(\delta) > 0$, hence $\eta^*(\delta) > 0$. By complementary slackness in (43), $\eta^*(\delta) > 0$ implies $\bar{D}(\delta; M^*) - D^*(\delta) = 0$, so $D^*(\delta) = \bar{D}(\delta; M^*)$ for a.e. δ such that $\hat{H}^*(\delta) > \nu^*$.

Case 2: $\hat{H}^*(\delta) < \nu^*$. Then $\eta^*(\delta) - \zeta^*(\delta) < 0$, hence $\zeta^*(\delta) > 0$. By complementary slackness in (43), $\zeta^*(\delta) > 0$ implies $D^*(\delta) = 0$, so $D^*(\delta) = 0$ for a.e. δ such that $\hat{H}^*(\delta) < \nu^*$.

Case 3: $0 < D^*(\delta) < \bar{D}(\delta; M^*)$ (*interior point*). If $D^*(\delta) > 0$, then $\zeta^*(\delta) = 0$ by $\zeta^* D^* = 0$. If $D^*(\delta) < \bar{D}(\delta; M^*)$, then $\eta^*(\delta) = 0$ by $\eta^*(\bar{D} - D^*) = 0$. Substituting $\eta^*(\delta) = \zeta^*(\delta) = 0$ into (40) yields $\hat{H}^*(\delta) = \nu^*$.

The three cases above are the precise *bind-or-zero rule*:

- where the KKT marginal value $\hat{H}^*$ is strictly above the mass price ν^* , the platform saturates the ceiling (upper bound binds);
- where $\hat{H}^*$ is strictly below ν^* , the platform assigns zero recommendation mass;
- equality $\hat{H}^* = \nu^*$ can occur only at points where both bounds are slack.

Remark. The only difference between (44) and a “naïve” pointwise revenue term $C(M^*)r(\delta; M^*)$ is the additional term $\xi^* X(\delta; M^*)$. This term appears *solely* because the equilibrium identity $\Psi(D, M) = 0$ must hold: shifting mass in D changes the mean effective price $\int X(\delta; M)D(\delta)d\delta$, so M must adjust and equilibrium prices change. The multiplier ξ^* is the shadow price of this equilibrium adjustment. Thus, the KKT system already incorporates the feedback effect and does not treat prices or WTP as fixed.

In the next subsection, we prove that $\hat{H}^*(\cdot)$ is strictly convex in δ . Once strict convexity is established, the bind-or-zero rule above implies the two-threshold structure in the theorem statement, and it also implies that the interior set $\{\delta : 0 < D^*(\delta) < \bar{D}(\delta; M^*)\}$ cannot have positive measure (since $\hat{H}^*$ cannot be constant on a set of positive measure).

C.5. Strict Convexity of the KKT-Relevant Marginal Value Density

We prove that the pointwise KKT-relevant marginal value density $\delta \mapsto \hat{H}^*(\delta)$ is strictly convex on $[0, \bar{\delta}]$. This curvature property is the only geometric input needed to turn the bind-or-zero KKT rule into a two-threshold (bind-gap-bind) structure.

Object of interest. At the endogenous optimum (D^*, M^*) , we have $\hat{H}^*(\delta) := C(M^*)r(\delta; M^*) + \xi^* X(\delta; M^*)$, as before, where $X(\delta; M) = p^*(\delta; M) + \delta$ and $r(\delta; M)$ is the per-exposure revenue kernel.

Closed forms. Fix M and suppress the dependence on M where convenient. By (24), a type- δ supplier chooses p to maximize $p(\bar{\omega}(M) - \delta - p)/\Delta(M)$, which is strictly concave in p and yields the unique best response $p^*(\delta; M) = \frac{\bar{\omega}(M) - \delta}{2}$. Hence, the effective price is

$$X(\delta; M) = p^*(\delta; M) + \delta = \frac{\bar{\omega}(M) + \delta}{2}, \quad (45)$$

and the per-exposure revenue kernel is

$$r(\delta; M) = p^*(\delta; M) \sigma(p^*(\delta; M), \delta; M) = \frac{(\bar{\omega}(M) - \delta)^2}{4\Delta(M)}. \quad (46)$$

Curvature. Holding M fixed, $\bar{\omega}(M)$ and $\Delta(M)$ are constants in δ . Differentiating (46) twice yields $\frac{\partial^2 r}{\partial \delta^2}(\delta; M) = \frac{1}{2\Delta(M)} > 0$, while (45) implies $X_{\delta\delta}(\delta; M) = 0$. Therefore,

$$\hat{H}^{*''}(\delta) = C(M^*) \frac{\partial^2 r}{\partial \delta^2}(\delta; M^*) + \xi^* \frac{\partial^2 X}{\partial \delta^2}(\delta; M^*) = C(M^*) \cdot \frac{1}{2\Delta(M^*)} + \xi^* \cdot 0 = \frac{C(M^*)}{2\Delta(M^*)}.$$

Since $C(M^*) > 0$ in any active market and $\Delta(M^*) > 0$ by definition, $\hat{H}^{*''}(\delta) > 0$ for all $\delta \in [0, \bar{\delta}]$. Thus, $\hat{H}^*(\cdot)$ is strictly convex on $[0, \bar{\delta}]$, and in particular it admits a unique minimizer $\delta_{\min} \in [0, \bar{\delta}]$.

C.6. Uniqueness of Optimal Solution (Existence and Uniqueness of the Two Cutoffs)

This subsection turns the bind-or-zero KKT rule (Appendix C.4) into a unique two-threshold (bind-gap-bind) structure by exploiting strict convexity of $\hat{H}^*$ (Appendix C.5).

Recall $\hat{H}^*(\delta) = C(M^*)r(\delta; M^*) + \xi^* X(\delta; M^*)$, which already internalizes equilibrium feedback through the multiplier ξ^* on $\Psi(D, M) = 0$.

Step 1: Geometry (strict convexity and a unique minimizer). By Appendix C.5, we have that $\hat{H}^{*''}(\delta) > 0$ for all $\delta \in [0, \bar{\delta}]$. Hence $\hat{H}^*$ is continuous and strictly convex on the compact interval $[0, \bar{\delta}]$, so it has a unique minimizer $\delta_{\min} := \arg \min_{\delta \in [0, \bar{\delta}]} \hat{H}^*(\delta)$, and $\hat{H}^*$ is strictly decreasing on $[0, \delta_{\min}]$ and strictly increasing on $[\delta_{\min}, \bar{\delta}]$. Define the following relevant range to use in Step 3:

$$\nu_{\min} := \hat{H}^*(\delta_{\min}), \quad \nu_{\max} := \max_{\delta \in [0, \bar{\delta}]} \hat{H}^*(\delta), \quad \mathcal{N} := [\nu_{\min}, \nu_{\max}].$$

Step 2: Cutoff maps via inverse functions. Fix $\nu \in \mathcal{N}$ and define the superlevel set $\mathcal{A}(\nu) := \{\delta \in [0, \bar{\delta}] : \hat{H}^*(\delta) \geq \nu\}$. Strict convexity implies $\mathcal{A}(\nu)$ is the union of (at most) two extreme intervals. To describe its endpoints cleanly, consider the restrictions of $\hat{H}^*$ to the two monotone branches:

$$\hat{H}_L := \hat{H}^*|_{[0, \delta_{\min}]} \quad (\text{continuous, strictly decreasing}), \quad \hat{H}_R := \hat{H}^*|_{[\delta_{\min}, \bar{\delta}]} \quad (\text{continuous, strictly increasing}).$$

A continuous strictly monotone function has a continuous inverse on its range. Using this fact, define $\alpha(\nu) := \hat{H}_L^{-1}(\nu) \in [0, \delta_{\min}]$, and $\beta(\nu) := \hat{H}_R^{-1}(\nu) \in [\delta_{\min}, \bar{\delta}]$. Then $\alpha(\cdot)$ is continuous and (weakly) decreasing in ν , while $\beta(\cdot)$ is continuous and (weakly) increasing in ν . Moreover, we that the superlevel set is the following:

$$\mathcal{A}(\nu) = [0, \alpha(\nu)] \cup [\beta(\nu), \bar{\delta}].$$

Step 3: Mass map and uniqueness of the shadow price. At the optimum the capacity ceiling is evaluated at M^* :

$$\overline{D}^*(\delta) := \overline{D}(\delta; M^*) = \frac{s(\delta; M^*)}{C(M^*)}.$$

Given $\nu \in \mathcal{N}$, the bind-or-zero rule implies that the total mass used when the binding region is $\mathcal{A}(\nu)$ equals

$$\Phi(\nu) := \int_{\mathcal{A}(\nu)} \overline{D}(\delta; M^*) d\delta = \int_0^{\alpha(\nu)} \overline{D}(\delta; M^*) d\delta + \int_{\beta(\nu)}^{\bar{\delta}} \overline{D}(\delta; M^*) d\delta.$$

Because α decreases and β increases with ν , the set $\mathcal{A}(\nu)$ shrinks as ν increases. Since $\overline{D}(\delta; M^*) \geq 0$ a.e. and is positive a.e. on the relevant support, $\Phi(\nu)$ is strictly decreasing on $\mathcal{N}$:

$$\nu_1 < \nu_2 \implies \Phi(\nu_1) > \Phi(\nu_2). \quad (47)$$

Continuity of Φ follows from continuity of $\alpha(\cdot)$ and $\beta(\cdot)$ and integrability of $\overline{D}(\cdot; M^*)$:

$$\Phi(\cdot) \text{ is continuous on } \mathcal{N}. \quad (48)$$

At $\nu = \nu_{\min}$ we have $\mathcal{A}(\nu_{\min}) = [0, \bar{\delta}]$, hence $\Phi(\nu_{\min}) = \int_0^{\bar{\delta}} \overline{D}(\delta; M^*) d\delta$. At $\nu = \nu_{\max}$ the superlevel set consists only of maximizers of $\hat{H}^*$; strict convexity implies this set has Lebesgue measure zero, hence

$$\Phi(\nu_{\max}) = 0. \quad (49)$$

Feasibility of (D^*, M^*) implies $\int D^* = 1 \leq \int \overline{D}(\delta; M^*) d\delta = \Phi(\nu_{\min})$, i.e., $\Phi(\nu_{\min}) \geq 1$. By (48)–(49) and the Intermediate Value Theorem, there exists $\nu^* \in \mathcal{N}$ such that $\Phi(\nu^*) = 1$; by strict monotonicity (47), this ν^* is unique.

Step 4: The unique cutoffs and the bind-gap-bind form. Define $\delta_0 := \alpha(\nu^*)$ and $\delta_1 := \beta(\nu^*)$. Then $\mathcal{A}(\nu^*) = [0, \delta_0] \cup [\delta_1, \bar{\delta}]$, and by the bind-or-zero rule in Appendix C.4 we obtain, for almost every δ ,

$$D^*(\delta) = \begin{cases} \overline{D}(\delta; M^*) & \text{for a.e. } \delta \in [0, \delta_0] \cup [\delta_1, \bar{\delta}], \\ 0 & \text{for a.e. } \delta \in (\delta_0, \delta_1). \end{cases} \quad (50)$$

Uniqueness of ν^* implies uniqueness of (δ_0, δ_1) ; and (50) pins down D^* uniquely up to changes on a null set.

Remark (single-block boundary case). If capacity is tight enough that the upper block is empty, then $\delta_1 = \bar{\delta}$ and the same argument applies with $\Phi(\nu) = \int_0^{\alpha(\nu)} \overline{D}(\delta; M^*) d\delta$.

C.7. Boundary-Slope Verification (U-shape)

Appendix C.5 proves $\hat{H}^*$ is strictly convex, hence it has a unique minimizer $\delta_{\min}$. Here we record the slope formula and an explicit interior-minimizer condition.

Recall $\hat{H}^*(\delta) = C(M^*)r(\delta; M^*) + \xi^*X(\delta; M^*)$. Appendix C.5 gives

$$r(\delta; M^*) = \frac{(\bar{\omega}(M^*) - \delta)^2}{4\Delta(M^*)}, \quad X(\delta; M^*) = \frac{\bar{\omega}(M^*) + \delta}{2}.$$

Differentiating with respect to δ (holding M^* fixed) yields

$$r_\delta(\delta; M^*) = -\frac{\bar{\omega}(M^*) - \delta}{2\Delta(M^*)}, \quad X_\delta(\delta; M^*) = \frac{1}{2},$$

so $\hat{H}^{*'}(\delta) = -C(M^*)\frac{\bar{\omega}(M^*) - \delta}{2\Delta(M^*)} + \frac{\xi^*}{2}$. Since $\hat{H}^*$ is strictly convex, the minimizer is interior (i.e., $\delta_{\min} \in (0, \bar{\delta})$) if and only if $\hat{H}^{*'}(0) < 0 < \hat{H}^{*'}(\bar{\delta})$, which is equivalent to

$$C(M^*)\frac{\bar{\omega}(M^*) - \bar{\delta}}{\Delta(M^*)} < \xi^* < C(M^*)\frac{\bar{\omega}(M^*)}{\Delta(M^*)}. \quad (51)$$

When (51) holds, the unique minimizer solves $\hat{H}^{*'}(\delta) = 0$, i.e., $\delta_{\min} = \bar{\omega}(M^*) - \frac{\xi^* \Delta(M^*)}{C(M^*)}$, truncated to $[0, \bar{\delta}]$ if necessary.

C.8. Comparative Statics in the Supplier-Arrival Rate m

We show how the supplier-arrival rate m affects feasibility and therefore the optimal recommendation policy. Throughout, equilibrium feedback remains fully internalized via the constraint $\Psi(D, M) = 0$ in (P_{aug}) : m affects the problem only through the capacity ceiling, not through the equilibrium identity itself.

Step 1: m enters only through the ceiling. In (P_{aug}) , the capacity ceiling is

$$\bar{D}(\delta; M, m) := \frac{s(\delta; M, m)}{C(M)}. \quad (52)$$

The objective and the equilibrium identity $\Psi(D, M) = 0$ do not contain m directly; m changes the feasible set through $\bar{D}(\cdot; M, m)$.

Step 2: Linear scaling of the ceiling in m (holding (δ, M) fixed). Fix (δ, M) . Steady-state flow balance for suppliers of type δ gives

$$mf_\Delta(\delta) = s(\delta; M, m)\phi_S(p^*(\delta; M), \delta; M),$$

so

$$s(\delta; M, m) = \frac{mf_\Delta(\delta)}{\phi_S(p^*(\delta; M), \delta; M)}. \quad (53)$$

Substituting (53) into (52) yields

$$\bar{D}(\delta; M, m) = m \cdot \underbrace{\frac{f_\Delta(\delta)}{C(M)\phi_S(p^*(\delta; M), \delta; M)}}_{=: \bar{D}_1(\delta; M)}. \quad (54)$$

In particular, we have that $\frac{\partial \bar{D}}{\partial m}(\delta; M, m) = \bar{D}_1(\delta; M) > 0$, so that increasing m relaxes the ceiling pointwise.

Step 3: Structure of $D^(m)$.* For each feasible m , let $(D^*(m), M^*(m))$ be an optimal solution. Appendix C.4 implies the bind-or-zero rule with the KKT-relevant marginal value density

$$\hat{H}_m^*(\delta) = C(M^*(m))r(\delta; M^*(m)) + \xi^*(m)X(\delta; M^*(m)),$$

and Appendix C.5 implies $\hat{H}_m^*(\cdot)$ is strictly convex. Therefore the binding region is a union of two extreme intervals, so there exist cutoffs $0 \leq \delta_0(m) \leq \delta_{\min}(m) \leq \delta_1(m) \leq \bar{\delta}$ such that, almost everywhere,

$$D^*(m, \delta) = \begin{cases} \bar{D}(\delta; M^*(m), m), & \delta \in [0, \delta_0(m)] \cup [\delta_1(m), \bar{\delta}], \\ 0, & \delta \in (\delta_0(m), \delta_1(m)). \end{cases}$$

Step 4: Mass balance and the role of scaling. Since $D^*(m, \cdot)$ is a density, the mass constraint gives $1 = \int_0^{\delta_0(m)} \bar{D}(\delta; M^*(m), m) d\delta + \int_{\delta_1(m)}^{\bar{\delta}} \bar{D}(\delta; M^*(m), m) d\delta$. Using (54), this can be rewritten as

$$\int_0^{\delta_0(m)} \bar{D}_1(\delta; M^*(m)) d\delta + \int_{\delta_1(m)}^{\bar{\delta}} \bar{D}_1(\delta; M^*(m)) d\delta = \frac{1}{m}. \quad (55)$$

Thus, as m increases, the required *normalized* capacity mass on the binding region must decrease (the right-hand side is $1/m$). This is the formal mechanism behind the widening gap region in thick markets.

Step 5: Thick-market limit $m \rightarrow \infty$. We show that as $m \rightarrow \infty$, the optimal recommendations concentrate on the two extremes $\{0, \bar{\delta}\}$ in the weak sense.

Regularity. Because M is a mean of effective prices and primitives are bounded, the equilibrium scalars $\{M^*(m)\}$ lie in some compact interval $\mathcal{M} \subset \mathbb{R}$.¹² Moreover, $\bar{D}_1(\delta; M)$ is continuous in (δ, M) and strictly positive on $[0, \bar{\delta}] \times \mathcal{M}$ under the maintained positivity assumptions on f_Δ , $C(\cdot)$, and $\phi_S(\cdot)$ on the active region.

Step 5.1 (support shrinkage). Fix any $\varepsilon \in (0, \bar{\delta}/2)$. Define the positive constants $c_L(\varepsilon) := \min_{M \in \mathcal{M}} \int_0^\varepsilon \bar{D}_1(\delta; M) d\delta > 0$ and $c_U(\varepsilon) := \min_{M \in \mathcal{M}} \int_{\bar{\delta}-\varepsilon}^{\bar{\delta}} \bar{D}_1(\delta; M) d\delta > 0$. If $\delta_0(m) \geq \varepsilon$, then $\int_0^{\delta_0(m)} \bar{D}_1(\delta; M^*(m)) d\delta \geq c_L(\varepsilon)$, contradicting (55) for all sufficiently large m since $1/m \rightarrow 0$. Hence $\delta_0(m) < \varepsilon$ for all large m , and because ε is arbitrary this implies $\delta_0(m) \rightarrow 0$. The same argument on the upper block implies $\bar{\delta} - \delta_1(m) \rightarrow 0$ (and this holds trivially if the upper block is inactive so that $\delta_1(m) = \bar{\delta}$).

Step 5.2 (weak convergence to a two-point distribution). Step 5.1 above shows that the support of $D^*(m, \cdot)$ collapses into shrinking neighborhoods of the two extremes. This does *not* by itself identify a limit distribution because the *split of total mass* between the two neighborhoods can vary with m (and may oscillate in theory). Step 5.2 therefore converts the support-shrinkage statement into a precise weak convergence statement by testing against continuous functions.

Let μ_m be the probability measure on $[0, \bar{\delta}]$ with density $D^*(m, \cdot)$, and define the left-block mass $\pi_m := \mu_m([0, \delta_0(m)]) \in [0, 1]$. Since μ_m is supported on $[0, \delta_0(m)] \cup [\delta_1(m), \bar{\delta}]$, for any bounded continuous g with modulus of continuity $\omega_g(\varepsilon) := \sup\{|g(x) - g(y)| : |x - y| \leq \varepsilon\}$ we have

$$\left| \int g(\delta) d\mu_m - (g(0)\pi_m + g(\bar{\delta})(1 - \pi_m)) \right| \leq \omega_g(\delta_0(m)) + \omega_g(\bar{\delta} - \delta_1(m)) \xrightarrow{m \rightarrow \infty} 0,$$

¹² This follows from earlier bounds on equilibrium prices/WTP.

because $\delta_0(m) \rightarrow 0$ and $\bar{\delta} - \delta_1(m) \rightarrow 0$ by Step 5.1. Since (π_m) is bounded in $[0, 1]$, along any subsequence with $\pi_m \rightarrow \pi \in [0, 1]$ we obtain

$$\mu_m \Rightarrow \pi \delta_0 + (1 - \pi) \delta_{\bar{\delta}}.$$

Thus every weak limit point of $D^*(m)$ as $m \rightarrow \infty$ is a (possibly degenerate) two-point distribution supported on $\{0, \bar{\delta}\}$, establishing the polarized thick-market limit claimed in Theorem 3.

Appendix D: Other proofs from Section 5

D.1. Proof of Proposition 3

Let D^ε be the shifted recommender and write $D^\varepsilon = D^B + \varepsilon h$ with $h := |I_h|^{-1} \mathbf{1}_{I_h} - |I_\ell|^{-1} \mathbf{1}_{I_\ell}$. Let M_ε denote the induced mean effective price and, for $v \in \mathcal{V}^*$, let $\theta_\varepsilon(v)$ be the equilibrium acceptance threshold with $\omega_\varepsilon(v) := v - \theta_\varepsilon(v)$.

By (37) and (33),

$$\left. \frac{dM_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} = \frac{\int_0^{\bar{\delta}} X(\delta; M_0) h(\delta) d\delta}{1 - \int_0^{\bar{\delta}} X_M(\delta; M_0) D^B(\delta) d\delta} > 0,$$

since I_h lies to the right of I_ℓ and $X(\cdot; M_0)$ is strictly increasing (Proposition 2). Hence

$$\left. \frac{dM_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} > 0. \quad (56)$$

Fix $v \in \mathcal{V}^*$ and write $(x)_+ := \max\{x, 0\}$. Lemma 1 is equivalent to

$$\mu_1 \theta_\varepsilon(v) = (1 - \mu_1) \int_0^{\bar{\delta}} (v - \theta_\varepsilon(v) - X(\delta; M_\varepsilon))_+ D^\varepsilon(\delta) d\delta. \quad (57)$$

Let $\theta(v; D, M)$ solve (57) for fixed (D, M) , and let $\delta^* = \delta^*(v; D, M)$ satisfy $X(\delta^*; M) = v - \theta(v; D, M)$; implicit differentiation (Leibniz; boundary term = 0 at δ^*) yields

$$\theta_M(v; D, M) = - \frac{(1 - \mu_1) \int_0^{\delta^*} X_M(\delta; M) D(\delta) d\delta}{\mu_1 + (1 - \mu_1) \int_0^{\delta^*} D(\delta) d\delta} < 0.$$

Now set $\theta_0(v) := \theta_\varepsilon(v)|_{\varepsilon=0}$ and $\delta^* := \delta^*(v; D^B, M_0)$. Differentiating (57) w.r.t. ε at $\varepsilon = 0$ gives

$$\theta'_0(v) = \theta_M(v; D^B, M_0) M'_0 + \frac{1 - \mu_1}{\mu_1 + (1 - \mu_1) \int_0^{\delta^*} D^B(\delta) d\delta} \int_0^{\bar{\delta}} (v - \theta_0(v) - X(\delta; M_0))_+ h(\delta) d\delta,$$

where $M'_0 = \left. \frac{dM_\varepsilon}{d\varepsilon} \right|_{\varepsilon=0} > 0$ by (56). The last integral is ≤ 0 because $\delta \mapsto (v - \theta_0(v) - X(\delta; M_0))_+$ is nonincreasing (Proposition 2) and h shifts mass from I_ℓ to the right-shifted I_h . Thus $\theta'_0(v) < 0$, so $\omega'_0(v) = -\theta'_0(v) > 0$; by continuity, $\omega_\varepsilon(v) > \omega_B(v)$ for all sufficiently small $\varepsilon > 0$. $\square$

D.2. Proof of Proposition 4

Lemma 7 gives $0 < \partial_M p^*(\delta; M) = \partial_M X(\delta; M) < 1$, hence $\left. \frac{dp^*(\delta; M_\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} = \partial_M p^*(\delta; M_0) M'_0 > 0$ by (56). $\square$

D.3. Proof of Proposition 5

Set $\tau = 1$.

(A) For any feasible D and mass-neutral $D_\varepsilon = D + \varepsilon h$, Appendix C.3 and (37) yield $\left. \frac{dR(D_\varepsilon)}{d\varepsilon} \right|_{\varepsilon=0} = \int_0^{\bar{\delta}} H(\delta; D) h(\delta) d\delta$, where $H(\delta; D) = C(M(D))r(\delta; M(D)) + \Gamma(D)X(\delta; M(D))$ (with $\Gamma(D)$ as in Appendix C.3). In the single-segment model, $r(\cdot; M)$ is strictly convex and $X(\cdot; M)$ is affine (Appendix C.5),

so $H(\cdot; D^B)$ is strictly convex and thus strictly increasing on the right tail. Let $B := \{\delta : D^B(\delta) < \bar{D}(\delta; D^B)\}$ and $\delta^{\max} := \sup B \leq \bar{\delta}$. Any feasible diversion from a binding region into B increases revenue to first order iff it moves mass to a point with larger $H(\cdot; D^B)$; the maximizer in B is therefore $\delta^{\max}$.

(B) When $m > m^\dagger$, $B \neq \emptyset$ and the diversion in (A) is feasible. In the thick-market limit $m \rightarrow \infty$, $D^B \Rightarrow \delta_0$, so profitability reduces to the following:

$$H(\bar{\delta}; D^B) > H(0; D^B). \quad (58)$$

Writing $H = K + \Gamma X$ with $K(\delta; M) := C(M)r(\delta; M)$ and using Lemma 5 (with $d^{\min} = 0$, $\bar{v}_1 = 1$) gives $X(0; M) = \underline{\omega}(M)$, $X(\bar{\delta}; M) = \bar{\omega}(M)$, $\Delta(M) = \bar{\omega}(M) - \underline{\omega}(M) = \underline{\omega}(M)$, $K(0; M) = C(M)\underline{\omega}(M)$, and $K(\bar{\delta}; M) = 0$, so (58) is equivalent to

$$\Gamma(D^B) > C(M). \quad (59)$$

Under $D^B = \delta_0$, $\Gamma(D^B) = \partial_M K(0; M)/(1 - \partial_M X(0; M))$; with $\underline{\omega}(M) = (\mu_1 + (1 - \mu_1)M)/2$ (Lemma 5), we have $\partial_M X(0; M) = (1 - \mu_1)/2$ and $\partial_M K(0; M) = C(M)(1 - \mu_1)$, hence $\Gamma(D^B) = 2C(M)(1 - \mu_1)/(1 + \mu_1)$ and $\Gamma(D^B) > C(M) \iff \mu_1 < 1/3$. $\square$

D.4. Proof of Proposition 6

Recall $CS(\varepsilon)$ and $Q(\varepsilon)$ from Section 5.

LEMMA 10 (CS equivalence). *For any ε , $CS(\varepsilon) = \int_{v \in (\varepsilon)} \Lambda_{in}(v) \pi_\varepsilon(v) dv$.*

Proof. Let $B_\varepsilon(v) := \mu_1 + (1 - \mu_1)q_{c,\varepsilon}(v)$. The Bellman equation gives $A_\varepsilon(v) = B_\varepsilon(v)\pi_\varepsilon(v)$ and steady-state flow balance gives $\Lambda_{in}(v) = c_{V,\varepsilon}(v)B_\varepsilon(v)$, so $CS(\varepsilon) = \int c_{V,\varepsilon}(v)A_\varepsilon(v) dv = \int \Lambda_{in}(v)\pi_\varepsilon(v) dv$.

Under $U[0, 1]$ arrivals, $\Lambda_{in}(v) = \lambda_1$, hence $CS(\varepsilon) = \lambda_1 \int_{v \in (\varepsilon)} \pi_\varepsilon(v) dv$ and

$$\left. \frac{dCS}{d\varepsilon} \right|_{\varepsilon=0} = \lambda_1 \int_{v \in (0)} \left. \frac{\partial \pi_\varepsilon(v)}{\partial \varepsilon} \right|_{\varepsilon=0} dv = \frac{\lambda_1}{1 - \mu_1} \int_{v \in (0)} \theta'_0(v) dv < 0,$$

since $\pi_\varepsilon(v) = \theta_\varepsilon(v)/(1 - \mu_1)$ (Lemma 1), $\pi_0(v \in (0)) = 0$ kills the boundary term, and $\theta'_0(v) < 0$ for $v \in \mathcal{V}^*$ by Proposition 3. Moreover, Theorem 2 and Lemma 5 imply $Q(\varepsilon) = \lambda_1(1 - M(\varepsilon))$, so $\left. \frac{dQ}{d\varepsilon} \right|_{\varepsilon=0} = -\lambda_1 M'_0 < 0$ by (56). $\square$

D.5. Derivations for Example 1

This subsection provides the closed-form equilibrium derivations for the two-type scarcity example in the main text. We work in a thick market ($m \rightarrow \infty$) with a single customer segment, valuations $v \sim U[0, 1]$, impatience $\mu = 0.2$, and two supplier types: Good ($\delta_g = 0$) and Bad ($\delta_b = 0.6$). The platform's recommendation policy $\alpha \in [0, 1]$ is the probability of showing the Bad type.

Preliminaries. By Theorem 2, the steady-state WTP distribution is uniform on an endogenous interval $[\underline{\omega}, \bar{\omega}]$. Customers accept an offer if the net utility exceeds their continuation value $\theta(v)$, so $\omega(v) = v - \theta(v)$ defines the WTP.

Supplier pricing. Consider the Good supplier ($\delta_g = 0$) choosing price p . Let $\phi_S(p) := \Pr(\omega \geq p)$ denote the sale probability conditional on being recommended. The supplier's expected profit per recommendation is $\pi_s(p) = (1 - \tau)p\phi_S(p)$. Under the uniform WTP distribution, $\phi_S(p) = (\bar{\omega} - p)/(\bar{\omega} - \underline{\omega})$ for $p \in [\underline{\omega}, \bar{\omega}]$, with derivative $\phi'_S(p) = -1/(\bar{\omega} - \underline{\omega})$. The first-order condition $\phi_S(p_g^*) + p_g^* \phi'_S(p_g^*) = 0$ yields $p_g^* = \bar{\omega}/2$.

Endogenous WTP support. Since the Bad type has $\delta_b = 0.6 > \bar{\omega}$ (verified below), it is never purchased: a customer would need $\omega \geq p_b + \delta_b \geq \delta_b$ to buy, which is impossible when $\delta_b > \bar{\omega}$. Consequently, active customers must have $\omega \geq p_g^*$; otherwise they would never purchase and would not enter. This pins down $\underline{\omega} = p_g^* = \bar{\omega}/2$, hence $\bar{\omega} = 2\underline{\omega}$.

REMARK 2 (ON THE “NEVER PURCHASED” EDGE CASE). This example treats both supplier types as exogenously available to isolate the search/continuation-value mechanism in closed form. In the full model, supplier entry is endogenous, and the high-misfit types used under “Scarcity by Design” are purchased with positive probability (earning strictly positive profit). The edge case where a displayed type is never transacted does not arise in equilibrium.

Customer continuation value (feedback identity). Consider the highest-valuation customer ($v = 1$). Her WTP is $\omega(1) = 1 - \theta(1)$, so $\bar{\omega} = 1 - \bar{\theta}$ where $\bar{\theta} := \theta(1)$ is her continuation value. With probability $1 - \alpha$, she sees the Good type and purchases, obtaining surplus $1 - p_g^* = 1 - \underline{\omega}$. With probability α , she sees the Bad type, does not purchase, and continues to the next period (if she does not exit). The Bellman equation is $\bar{\theta} = (1 - \mu)[(1 - \alpha)(1 - \underline{\omega} - \bar{\theta} + \bar{\theta}) + \alpha\bar{\theta}] = (1 - \mu)[\bar{\theta} + (1 - \alpha)(\bar{\omega} - \underline{\omega})]$, where we used $1 - \underline{\omega} = \bar{\omega} - \underline{\omega} + \bar{\theta}$. Rearranging gives $\mu\bar{\theta} = (1 - \mu)(1 - \alpha)(\bar{\omega} - \underline{\omega})$. Substituting $\bar{\omega} = 1 - \bar{\theta}$ yields the *feedback identity*:

$$\bar{\omega} = 1 - \frac{1 - \mu}{\mu}(1 - \alpha)(\bar{\omega} - \underline{\omega}). \quad (60)$$

Solving for equilibrium WTP. We have two equations: (i) $\bar{\omega} = 2\underline{\omega}$ and (ii) the feedback identity (60). Substituting (i) into (ii) gives $\bar{\omega} = 1 - \frac{1 - \mu}{\mu}(1 - \alpha) \cdot \frac{\bar{\omega}}{2}$. With $\mu = 0.2$, we have $(1 - \mu)/\mu = 4$, so $\bar{\omega} = 1 - 2(1 - \alpha)\bar{\omega}$, which yields:

$$\bar{\omega}(\alpha) = \frac{2}{6 - 4\alpha}, \quad \underline{\omega}(\alpha) = \frac{1}{6 - 4\alpha}, \quad p_g^*(\alpha) = \underline{\omega}(\alpha). \quad (61)$$

Baseline vs. Scarcity. Under the Baseline policy ($\alpha = 0$): $\bar{\omega}(0) = 1/3 \approx 0.333$, $\underline{\omega}(0) = 1/6 \approx 0.167$, and $p_g^*(0) = 1/6 \approx 0.167$. Under the Scarcity policy ($\alpha = 0.35$): $\bar{\omega}(0.35) = 10/23 \approx 0.435$ and $\underline{\omega}(0.35) = 5/23 \approx 0.217$. Note that $\bar{\omega}(0.35) < \delta_b = 0.6$, confirming that the Bad type is never purchased.

Aggregate outcomes. Since only the Good type transacts, the steady-state customer mass is $N = (\bar{\omega} - \underline{\omega})/\mu$ (by Theorem 2), and each customer purchases with probability $1 - \alpha$. Hence transaction volume is $Q = N(1 - \alpha)$, platform revenue is $R = \tau Q p_g^*$, and customer surplus is $CS = Q(1 - \underline{\omega})/2$, where $(1 - \underline{\omega})/2$ is the average surplus per transaction. With $\tau = 0.1$ and $\mu = 0.2$:

	Baseline ($\alpha = 0$)	Scarcity ($\alpha = 0.35$)
Q	0.833	0.707
R	0.0139	0.0154 (+10.6%)
CS	0.347	0.277 (-20.4%)

Platform revenue increases while customer surplus decreases, illustrating the welfare trade-off identified in Propositions 5 and 6.

Appendix E: Computational Methodology for Single-Segment Analysis

We detail the computational framework used to generate the numerical results presented in Section 6.3. We compute the unique stationary equilibrium and platform revenues under the Optimal (OPT), Best-Match (BM), and Uniform Random (U) policies.

E.1. Equilibrium Characterization and the Fixed-Point Problem

We establish the fundamental equations governing the single-segment equilibrium. We denote customer impatience by μ (dropping the subscript as $J = 1$).

E.1.1. WTP Structure and the Feedback Identity By Theorems 1 and 2, the steady-state WTP distribution is uniform over an endogenous support $[\underline{\omega}, \bar{\omega}]$. Let $\Delta = \bar{\omega} - \underline{\omega}$.

The equilibrium is governed by the Feedback Identity, derived from the customer’s Bellman equation (Section 3.1). It relates the maximum WTP ($\bar{\omega}$, corresponding to $v = 1$) to the Mean Effective Price (M):

$$\bar{\omega} = \mu + (1 - \mu)M. \quad (62)$$

Here, $M = \mathbb{E}_D[X^*(\delta)]$ is the expectation of the equilibrium effective prices $X^*(\delta)$ under the policy D .

E.1.2. Optimal Supplier Pricing A supplier of type δ chooses the effective price X to maximize static expected profit

$$\Pi_s(\delta, X) = (X - \delta)\sigma(X), \quad \sigma(X) = \frac{\bar{\omega} - X}{\Delta},$$

where $\sigma(X)$ is the sale probability induced by the uniform WTP support $[\underline{\omega}, \bar{\omega}]$. The first-order condition yields the closed-form effective price function

$$X^*(\delta) = \frac{\bar{\omega} + \delta}{2}. \quad (63)$$

E.1.3. The Fixed-Point Definition The equilibrium is the fixed point M^* of the operator $\mathcal{T}_D(M)$:

$$\mathcal{T}_D(M) = \int X^*(\delta; \bar{\omega}(M)) D(\delta) d\delta, \quad (64)$$

with $\bar{\omega}(M)$ given by (62) and X^* by (63). Computationally, we find M^* by solving $M - \mathcal{T}_D(M) = 0$. We employ Brent’s method over $M \in [0, 1]$.

E.2. Computation with Finite Market Thickness (Figure 1)

Figure 1 analyzes finite market thickness m . We utilize a normalized capacity parameter $Q \in [0, 1]$ representing the demand-to-supply ratio (the Supply/Demand Ratio in the figure is $1/Q$). The equilibrium simplifies: $\bar{\omega} = 2\Delta$ and $X^*(\delta) = \bar{\omega}/2 + \delta/2$.

E.2.1. Optimal Policy Computation (OPT) The optimal policy follows the Bind-Gap-Bind structure (Theorem 3(i)). Computing this requires solving the fixed-point problem where the policy D^* itself is optimized within the iteration. We employ a nested fixed-point optimization approach.

The Inner Loop: Optimization Given M (The Projection Method). For a fixed candidate M (and Q, μ), we determine the optimal structure (δ_0, δ_1) that maximizes revenue subject to the capacity constraint Q . This optimization is derived from the Karush–Kuhn–Tucker (KKT) conditions and corresponds to the `calculate_structure` function in the implementation.

1. Calculate derived quantities:

$$\begin{aligned} \bar{\omega}(M) &= \mu + (1 - \mu)M, \\ \Gamma(\mu, M) &= \frac{2(1 - \mu)\mu(1 - M)}{1 + \mu}. \end{aligned}$$

2. *Determine gap length and ideal center:*

$$L = \bar{\omega}(1 - Q),$$

$$C_{ideal} = \bar{\omega} - \Gamma.$$

3. *Project onto the feasible space:* The gap $[\delta_0, \delta_1]$ must lie within $[0, \bar{\omega}]$. The feasible range for the center is $[L/2, \bar{\omega} - L/2]$. The optimal center C_{struct} is the projection of C_{ideal} onto this interval:

$$C_{struct} = \max(L/2, \min(\bar{\omega} - L/2, C_{ideal})). \quad (65)$$

4. *Determine cutoffs:* $\delta_0 = C_{struct} - L/2$, $\delta_1 = C_{struct} + L/2$.

The Outer Loop: Solving the Fixed Point. Given the structure $(\delta_0(M), \delta_1(M))$, we calculate the induced Mean Effective Price $M' = \mathcal{T}_{D^*}(M)$. The policy D^* is uniform over the support $[0, \delta_0] \cup (\delta_1, \bar{\omega}]$ with density $1/(\bar{\omega}Q)$.

$$M' = \frac{1}{\bar{\omega}Q} \left(\int_0^{\delta_0} X^*(\delta) d\delta + \int_{\delta_1}^{\bar{\omega}} X^*(\delta) d\delta \right).$$

Using $X^*(\delta) = \bar{\omega}/2 + \delta/2$:

$$\int_a^b (\bar{\omega}/2 + \delta/2) d\delta = \left[\frac{\bar{\omega}\delta}{2} + \frac{\delta^2}{4} \right]_a^b = \frac{\bar{\omega}(b-a)}{2} + \frac{b^2 - a^2}{4}.$$

Applying this to the two intervals and simplifying yields:

$$M'(M) = \frac{3\bar{\omega}^2 + 2\bar{\omega}(\delta_0 - \delta_1) + (\delta_0^2 - \delta_1^2)}{4\bar{\omega}Q}. \quad (66)$$

The equilibrium M^* is the root of $Res(M) = M'(M) - M = 0$. We employ Brent's method (`scipy.optimize.brentq`) over $M \in [0, 1]$.

Revenue Calculation. Once M^* and the equilibrium structure are determined, revenue R^* is computed using:

$$R^* = \frac{1}{24\mu Q} ((\bar{\omega}^*)^3 - (\bar{\omega}^* - \delta_0^*)^3 + (\bar{\omega}^* - \delta_1^*)^3). \quad (67)$$

E.2.2. Benchmark Policies (BM and U) For the benchmark policies, the structure is fixed, allowing for analytical solutions to the fixed-point problem.

Best-Match (BM). The BM policy corresponds to a structure where the gap is at the upper boundary. The equilibrium $\bar{\omega}_{BM}$ is solved analytically:

$$\bar{\omega}_{BM} = \frac{4\mu}{2 - Q + \mu(2 + Q)}. \quad (68)$$

Uniform (U). The Uniform policy (assuming feasibility/concentrated attention) yields the analytical boundary:

$$\bar{\omega}_U = \frac{4\mu}{1 + 3\mu}. \quad (69)$$

The revenues R_{BM} and R_U are calculated based on these analytical equilibrium values.

Appendix F: Existence and Uniqueness of Stationary Equilibrium (Theorem 1)

We prove Theorem 1. The equilibrium feedback runs from the platform’s policy D to the induced effective-price distribution $X_j = p + d_j$, to customers’ continuation values (and hence WTP bounds), back to suppliers’ optimal prices. We collapse this feedback loop to a fixed point in the vector of *mean effective prices* $M = (M_1, \dots, M_J)$.

Segments with zero mass. We restrict to segments with $C_j(D) > 0$; if $C_j(D) = 0$, segment j never affects $\beta(\cdot)$ and can be dropped without changing the equilibrium of the remaining segments.

F.1. Fixed-point formulation

Fix a feasible recommendation profile $D = (D_1, \dots, D_J)$.

Step 1: Mean effective prices. For each segment j , define

$$M_j := \int_0^{\bar{\delta}} X_j(\delta; D) D_j(\delta) d\delta, \quad X_j(\delta; D) := p(\delta; D) + d_j(\delta). \quad (70)$$

Step 2: Endogenous WTP bounds and the Feedback Identity. Let $\bar{\omega}_j$ be the upper bound of segment- j WTP in a stationary equilibrium. By Lemma 5(i) (the *Feedback Identity*),

$$\bar{\omega}_j(M_j) = \mu_j \bar{v}_j + (1 - \mu_j) M_j. \quad (71)$$

(Importantly, Lemma 5(i) is a customer-side flow-balance result and does not invoke Segmental Exclusivity; SE is used only for Lemma 5(ii)–(iii).)

LEMMA 11 (WTP cannot exceed valuation). *In any stationary equilibrium, $\bar{\omega}_j \leq \bar{v}_j$ for all j . Hence (71) implies $M_j \leq \bar{v}_j$ for all j .*

Since a customer can always reject all future offers and obtain payoff 0, the continuation value $\theta_j(v) \geq 0$, so $\omega_j(v) = v - \theta_j(v) \leq v$. Taking maxima over $v \in [\underline{v}_j, \bar{v}_j]$ gives $\bar{\omega}_j \leq \bar{v}_j$. Plug into (71) and divide by $1 - \mu_j > 0$.

Step 3: Supplier best response and M -independent weights. Fix δ . Under the multi-segment equilibrium, the supplier faces segment mix $\beta_j(\delta)$ and segment- j linear demand (Appendix B). Using the effective-parameter representation (Definition 2 and Proposition 9), the best-response price can be written as a convex combination in segment-level WTP bounds with weights that depend only on $(D, \text{primitives})$.

LEMMA 12 (Weights for p^*). *Fix feasible D and define $\gamma_j := \lambda_j / (\mu_j L_j) > 0$. For δ with $\sum_k \gamma_k D_k(\delta) > 0$, let*

$$w_j(\delta) := \frac{\gamma_j D_j(\delta)}{\sum_{k=1}^J \gamma_k D_k(\delta)}, \quad j = 1, \dots, J, \quad (72)$$

and set $w(\delta) \equiv 0$ if $\sum_k \gamma_k D_k(\delta) = 0$. Then for any candidate $M \in \mathbb{R}^J$,

$$p^*(\delta; M) = \frac{1}{2} \sum_{j=1}^J w_j(\delta) (\bar{\omega}_j(M_j) - d_j(\delta)), \quad (73)$$

and the weights $w_j(\delta)$ are independent of M .

Proof. By Proposition 9, $p^*(\delta) = \frac{1}{2}(\bar{\omega}_{\text{eff}}(\delta) - d_{\text{eff}}(\delta))$ with $(\bar{\omega}_{\text{eff}}, d_{\text{eff}})$ from Definition 2. Hence $p^*(\delta) = \frac{1}{2} \sum_j \tilde{w}_j(\delta) (\bar{\omega}_j - d_j(\delta))$ where $\tilde{w}_j(\delta) \propto \beta_j(\delta) / \Delta_j$ and $\sum_j \tilde{w}_j(\delta) = 1$. Using $\beta_j(\delta) = \frac{C_j D_j(\delta)}{\sum_\ell C_\ell D_\ell(\delta)}$ and $C_j / \Delta_j = \lambda_j / (\mu_j L_j) = \gamma_j$ (Theorem 2), the normalizing denominator cancels and $\tilde{w}_j(\delta) = w_j(\delta)$ as in (72). Finally substitute $\bar{\omega}_j = \bar{\omega}_j(M_j)$ from (71).

Step 4: The equilibrium map. Given M , define $p(\delta; M) := p^*(\delta; M)$ by (73), and $X_j(\delta; M) := p(\delta; M) + d_j(\delta)$. Define $\mathbf{T}(\cdot; D)$ componentwise by

$$T_j(M; D) := \int_0^{\bar{\delta}} X_j(\delta; M) D_j(\delta) d\delta, \quad j = 1, \dots, J. \quad (74)$$

A stationary equilibrium induced by D must satisfy $M = \mathbf{T}(M; D)$.

F.2. Contraction

LEMMA 13 (Contraction of the equilibrium map). *Fix feasible D and let $\kappa := \frac{1}{2} \max_j (1 - \mu_j) \in (0, 1)$. Then for all $M, M' \in \mathbb{R}^J$,*

$$\|\mathbf{T}(M; D) - \mathbf{T}(M'; D)\|_\infty \leq \kappa \|M - M'\|_\infty,$$

so $\mathbf{T}(\cdot; D)$ is a contraction on $(\mathbb{R}^J, \|\cdot\|_\infty)$.

Proof. By (74) and $X_j(\delta; M) = p(\delta; M) + d_j(\delta)$, $T_j(M; D) - T_j(M'; D) = \int_0^{\bar{\delta}} (p(\delta; M) - p(\delta; M')) D_j(\delta) d\delta$. From (73) and (71),

$$p(\delta; M) - p(\delta; M') = \frac{1}{2} \sum_k w_k(\delta) (\bar{\omega}_k(M_k) - \bar{\omega}_k(M'_k)) = \frac{1}{2} \sum_k w_k(\delta) (1 - \mu_k) (M_k - M'_k).$$

Taking absolute values, using $\sum_k w_k(\delta) = 1$ on $\{\delta : D_j(\delta) > 0\}$, and $|M_k - M'_k| \leq \|M - M'\|_\infty$ yields $|p(\delta; M) - p(\delta; M')| \leq \kappa \|M - M'\|_\infty$. Integrate against D_j and use $\int_0^{\bar{\delta}} D_j = 1$ to obtain $|T_j(M; D) - T_j(M'; D)| \leq \kappa \|M - M'\|_\infty$. Taking $\max_j$ proves the claim.

REMARK 3 (BOUNDARY/TRUNCATION). Appendix B notes that outside the interior region the segment-level sale probability (and hence the best-response price) is obtained by truncation/projection. Since scalar projection is 1-Lipschitz, the bound in Lemma 13 continues to hold. In the present paper these boundary cases are ruled out by maintained interior-equilibrium conditions.

Proof of Theorem 1 Fix feasible D . Lemma 13 implies $\mathbf{T}(\cdot; D)$ is a contraction on the complete space $(\mathbb{R}^J, \|\cdot\|_\infty)$, hence admits a unique fixed point M^* .

Define $\bar{\omega}_j^* := \bar{\omega}_j(M_j^*)$ via (71), and define $p^*(\delta) := p^*(\delta; M^*)$ via (73). Let $X_j^*(\delta) := p^*(\delta) + d_j(\delta)$. Since M^* is a fixed point, $M_j^* = \int_0^{\bar{\delta}} X_j^*(\delta) D_j(\delta) d\delta$ for each j , i.e., mean effective prices are consistent with the prices they generate.

Given the induced effective-price environment $\{X_j^*\}$ and policy D , the remaining stationary equilibrium objects (customer thresholds/WTP distribution and steady-state masses) are determined as in the equilibrium definition and Appendix B; by construction, feedback is respected because $\bar{\omega}_j^*$ is pinned down by (71). Finally, Lemma 11 ensures $\bar{\omega}_j^* \leq \bar{v}_j$ (hence feasibility).

Uniqueness: if two stationary equilibria under D existed, their mean vectors would both be fixed points of $\mathbf{T}(\cdot; D)$, contradicting uniqueness of M^* . Given M^* , (71) and (73) pin down $(\bar{\omega}^*, p^*)$ uniquely, so the equilibria coincide.

Appendix G: Proof of Theorem 4

Roadmap. Theorem 4 establishes three structural properties of the optimal policy: (i) *Segmental Exclusivity (SE)*: On the set of suppliers where capacity binds, each supplier type is allocated exclusively to a single customer segment. (ii) *Personalized Structure*: Each segment’s allocation follows a Bind–Gap–Bind (BGB) structure, saturating residual capacity at the extremes of the type space. (iii) *Aggregate Structure*: The total exposure across all segments also follows a BGB structure.

The proof adapts the augmented-equilibrium approach developed for the single-segment case (Appendix C), treating equilibrium feedback via explicit constraints. However, the multi-segment setting introduces a new challenge: segment coupling. To resolve this, we proceed in three steps. First, we formulate an equivalent single-level optimization program where equilibrium objects are treated as explicit decision variables. This formulation reveals that the capacity constraint is linear in segment densities with a common pointwise multiplier. Second, we analyze the resulting KKT conditions, which define a “normalized marginal value” for each segment. Under a standard genericity condition (Assumption 1), we show that these values almost never tie. Consequently, for almost every supplier type, one segment dominates the others, proving Segmental Exclusivity (SE). Third, with SE established, the complex multi-segment problem locally collapses into a collection of single-segment pricing problems. We leverage the curvature results from Appendix B (Proposition 9) to show that the marginal value functions are strictly convex. This convexity, combined with the standard bind–or–zero logic (as in Appendix C.4), yields the polarized BGB structures for both individual segments and the aggregate flow, proving parts (ii) and (iii).

G.1. Standing primitives and equilibrium objects

We inherit primitives and notation from the main model and from Appendix C. We record here only the objects used repeatedly in this proof.

Misfit and recommendations. Segment- j misfit disutility is affine: $d_j(\delta) = a_j\delta + b_j$, $a_j \neq 0$. A recommendation policy is a profile $D = (D_1, \dots, D_J)$ where each D_j is a density on $[0, \bar{\delta}]$:

$$D_j(\delta) \geq 0 \text{ a.e.}, \quad \int_0^{\bar{\delta}} D_j(\delta) d\delta = 1, \quad j = 1, \dots, J. \quad (75)$$

Equilibrium induced by a policy. By Theorem 1, every feasible D induces a unique stationary equilibrium. We write $C_j(D)$ for the steady-state mass of active segment- j customers, $s(\delta; D)$ for the steady-state density of active suppliers of type δ , and $p(\delta; D)$ for the equilibrium posted price. The segment- j effective price is $X_j(\delta; D) := p(\delta; D) + d_j(\delta)$.

Exposure flow and capacity constraint. Define the total exposure flow allocated to type δ :

$$y(\delta; D) := \sum_{k=1}^J C_k(D) D_k(\delta). \quad (76)$$

Feasibility requires that exposures not exceed available supplier capacity: $y(\delta; D) \leq s(\delta; D)$ for a.e. $\delta \in [0, \bar{\delta}]$.

Per-supplier exposure probability. When useful, we interpret the per-supplier exposure probability as

$$\psi(\delta; D) := \min \left\{ 1, \frac{y(\delta; D)}{s(\delta; D)} \right\}, \quad (77)$$

with the convention $\psi(\delta; D) = 0$ if $s(\delta; D) = 0$.

Active types, hull, and binding set. Let $S(D) := \{\delta \in [0, \bar{\delta}] : s(\delta; D) > 0\}$ be the set of active supplier types. Let $[\delta(D), \bar{\delta}(D)]$ be the convex hull of $S(D)$. Let D^* be an optimal solution to the platform problem and write

$$S^* := S(D^*), \quad [\delta^*, \bar{\delta}^*] := [\delta(D^*), \bar{\delta}(D^*)], \quad C_j^* := C_j(D^*), \quad s^*(\delta) := s(\delta; D^*), \quad y^*(\delta) := y(\delta; D^*).$$

Define the *binding set among active types*:

$$B := \{\delta \in S^* \cap [\delta^*, \bar{\delta}^*] : y^*(\delta) = s^*(\delta)\}. \quad (78)$$

G.2. A single-level equilibrium-constrained formulation

In the original platform problem, the constraints are endogenous: both the available capacity $s(\delta; D)$ and the customer mass $C_j(D)$ shift as the recommendation policy D alters the market equilibrium. To capture the complex interactions resulting from this endogeneity rigorously, we adopt the same strategy used in the single-segment analysis (Appendix C): we enlarge the decision space. We treat the equilibrium objects Z (such as prices and steady-state masses) as explicit decision variables rather than fixed parameters. We then impose the equilibrium consistency conditions as formal constraints. This ensures that the KKT multipliers correctly price the marginal impact of recommendations on the market equilibrium.

Equilibrium conditions as constraints. Let Z denote the full collection of stationary equilibrium objects. Since every policy D induces a unique equilibrium (Theorem 1), the mapping from policy to equilibrium is well-defined. We represent this mapping as a system of equality constraints: $\mathcal{F}(D, Z) = 0$. This system ensures that for any chosen policy D , the variables Z satisfy all stationarity and optimality conditions.

Single-level program. Let $R(D, Z)$ denote the platform's steady-state per-period revenue computed from (D, Z) . Consider the program

$$\max_{D, Z} \quad R(D, Z) \quad (79)$$

$$\text{s.t.} \quad (75),$$

$$\sum_{k=1}^J C_k(Z) D_k(\delta) \leq s(\delta; Z) \quad \text{for a.e. } \delta, \quad (80)$$

$$\mathcal{F}(D, Z) = 0. \quad (81)$$

Here $C_k(Z)$ and $s(\delta; Z)$ are components of Z .

Equivalence to the original problem. Because (81) enforces $Z = \mathcal{Z}(D)$, a profile D is feasible for the original problem if and only if there exists a Z such that (D, Z) is feasible for (79)–(81); and for such (D, Z) the objective equals the original revenue $R(D)$. Hence any optimal policy D^* for the original problem corresponds to an optimal pair (D^*, Z^*) here.

Regularity. Assume the differentiability and constraint-qualification conditions needed for KKT multipliers to exist for (79)–(81) (e.g., Gateaux differentiability of the relevant mappings and a Slater-type condition for (80)). We now work with the KKT system at an optimal pair (D^*, Z^*) .

G.3. KKT system and the segment-level normalized marginal values

Let (D^*, Z^*) be optimal for (79)–(81). KKT yields multipliers $\nu_j^* \in \mathbb{R}$ (mass constraints), $\eta^*(\delta) \geq 0$ a.e. (capacity (80)), $\zeta_j^*(\delta) \geq 0$ a.e. (nonnegativity), and Λ^* (equilibrium constraint (81)).

Pointwise stationarity. Fix j . By the localized-perturbation argument in Appendix C.4, stationarity in D_j implies $\hat{H}_j^*(\delta) - \nu_j^* - C_j^* \eta^*(\delta) + \zeta_j^*(\delta) = 0$ for a.e. δ , where $C_j^* := C_j(Z^*)$ and

$$\hat{H}_j^*(\delta) := \frac{\delta}{\delta D_j(u)} \left(R(D, Z) + \langle \Lambda^*, \mathcal{F}(D, Z) \rangle \right) \Big|_{(D, Z) = (D^*, Z^*), u = \delta}.$$

(The Λ^* term ensures $\hat{H}_j^*$ internalizes equilibrium feedback.)

Complementary slackness. For each j and a.e. δ , $D_j^*(\delta) \geq 0$, $\zeta_j^*(\delta) \geq 0$, and $\zeta_j^*(\delta) D_j^*(\delta) = 0$. Writing $y^*(\delta) := y(\delta; D^*)$ as in (76) and $s^*(\delta) := s(\delta; Z^*)$, we have

$$\eta^*(\delta) \geq 0, \quad s^*(\delta) - y^*(\delta) \geq 0, \quad \eta^*(\delta)(s^*(\delta) - y^*(\delta)) = 0 \quad \text{a.e. } \delta. \quad (82)$$

Normalized marginal values. Define

$$F_j^*(\delta) := \frac{\hat{H}_j^*(\delta) - \nu_j^*}{C_j^*} \quad \text{a.e. } \delta. \quad (83)$$

Then $\eta^*(\delta) = F_j^*(\delta) + \zeta_j^*(\delta)/C_j^*$ a.e., hence $\eta^*(\delta) \geq F_j^*(\delta)$; moreover,

$$D_j^*(\delta) > 0 \implies \eta^*(\delta) = F_j^*(\delta) \quad \text{a.e. } \delta. \quad (84)$$

Minimal representative. Since multipliers are not unique on $\{D_j^*(\delta) = 0\}$, we take the a.e. minimal representative

$$\eta^*(\delta) = \max \left\{ \max_{j \in \{1, \dots, J\}} F_j^*(\delta), 0 \right\} \quad \text{a.e. } \delta, \quad (85)$$

implemented by setting $\zeta_j^*(\delta) := C_j^*(\eta^*(\delta) - F_j^*(\delta)) \geq 0$ whenever $D_j^*(\delta) = 0$.

G.4. Segmental Exclusivity (SE)

Genericity condition (No flat ties). To characterize the optimal policy, we must rule out “knife-edge” scenarios where the platform is perfectly indifferent between two segments over a continuous range of supplier types. We impose the following genericity assumption (formal justification provided in Appendix G.9):

ASSUMPTION 1 (No flat ties). *For every pair of distinct segments $i \neq j$, the set of supplier types where their normalized marginal values coincide,*

$$T_{ij} := \{\delta \in [\delta^*, \bar{\delta}^*] : F_i^*(\delta) = F_j^*(\delta)\},$$

has Lebesgue measure zero.

Intuitively, this assumption states that the marginal value functions for different segments—which are derived from distinct primitives—do not overlap perfectly on any non-degenerate interval. Such interval ties would require a precise, fragile alignment of parameters and would disappear under arbitrarily small perturbations to the model primitives. We provide a formal justification for this assumption using a perturbation argument in Appendix G.9.

LEMMA 14 (Segmental exclusivity on the binding set). *Under Assumption 1, for almost every $\delta \in B$ there exists a unique segment $j^*(\delta)$ such that*

$$D_{j^*(\delta)}^*(\delta) > 0, \quad \text{and} \quad D_k^*(\delta) = 0 \quad \forall k \neq j^*(\delta).$$

Proof. Fix any $\delta \in B$ where the KKT relations hold. By (78) and the definition of y^* in (76), we have $s^*(\delta) = y^*(\delta) = \sum_{k=1}^J C_k^* D_k^*(\delta)$ and $s^*(\delta) > 0$, so there exists at least one j with $D_j^*(\delta) > 0$. For any such j , (84) implies $\eta^*(\delta) = F_j^*(\delta)$. If two distinct segments $i \neq j$ were both positive at δ , then $F_i^*(\delta) = \eta^*(\delta) = F_j^*(\delta)$, so $\delta \in T_{ij}$. Assumption 1 implies each T_{ij} has Lebesgue measure zero; taking the finite union over pairs (i, j) yields that simultaneous activity of two segments occurs only on a null set within B . Hence, for almost every $\delta \in B$ there is a unique active segment. $\square$

G.5. Consequence of SE: single-segment reduction

Fix the optimal policy D^* . For almost every $\delta \in B$, the SE property gives a unique segment $j^*(\delta)$ with $D_{j^*(\delta)}^*(\delta) > 0$. At such a δ , a type- δ supplier is exposed only to customers from segment $j^*(\delta)$, so the supplier faces a single-segment demand environment.

Mean effective prices and endogenous willingness-to-pay. For each segment j , define its equilibrium mean effective price under D^* :

$$M_j^* := \int_0^{\bar{\delta}} X_j(\delta; D^*) D_j^*(\delta) d\delta. \quad (86)$$

The segment- j willingness-to-pay bounds are endogenous functions of M_j (via the customer fixed point); write $\underline{\omega}_j^* = \underline{\omega}_j(M_j^*)$, $\bar{\omega}_j^* = \bar{\omega}_j(M_j^*)$, and $\Delta_j^* = \bar{\omega}_j^* - \underline{\omega}_j^* = \Delta_j(M_j^*)$. These objects move with D through the induced change in M_j .

Supplier best response under single-segment exposure. Fix $\delta \in B$ where SE holds, and let $j = j^*(\delta)$. Then a type- δ supplier is exposed only to segment- j customers, so conditional on M_j^* it faces the single-segment linear demand environment of Appendix B. By Proposition 9 applied to this single-segment environment, the optimal posted price is $p_j^*(\delta) = \frac{\bar{\omega}_j(M_j^*) - d_j(\delta)}{2}$. Hence, the effective price and per-exposure revenue kernel are

$$X_j(\delta; M_j^*) = p_j^*(\delta) + d_j(\delta) = \frac{\bar{\omega}_j(M_j^*) + d_j(\delta)}{2}, \quad (87)$$

and

$$r_j(\delta; M_j^*) = p_j^*(\delta) \sigma_j(p_j^*(\delta), \delta; M_j^*) = \frac{(\bar{\omega}_j(M_j^*) - d_j(\delta))^2}{4 \Delta_j(M_j^*)}. \quad (88)$$

G.6. Reduced (D, M) formulation and explicit marginal values

After SE, the problem decomposes locally into J single-segment environments. We therefore apply the same equilibrium-constrained reduction as in Appendix C: we treat $M = (M_1, \dots, M_J)$ as explicit variables and impose the fixed-point identities as constraints. We record the resulting marginal-value expression used below.

Per-segment equilibrium identities. For each segment j , define $\Psi_j(D_j, M_j) := \int_0^{\bar{\delta}} X_j(\delta; M_j) D_j(\delta) d\delta - M_j$, where $X_j(\delta; M_j)$ has the form (87) with $\bar{\omega}_j = \bar{\omega}_j(M_j)$. Equilibrium requires $\Psi_j(D_j, M_j) = 0$ for all j .

KKT-relevant marginal value under explicit feedback. In this formulation, the objective can be written as $\sum_{j=1}^J C_j(M_j) \int_0^{\bar{\delta}} r_j(\delta; M_j) D_j(\delta) d\delta$, where $r_j(\delta; M_j)$ is given by (88) with $\bar{\omega}_j = \bar{\omega}_j(M_j)$ and $\Delta_j = \Delta_j(M_j)$. Let ξ_j^* be the multiplier on $\Psi_j(D_j, M_j) = 0$. Then stationarity with respect to $D_j(\delta)$ yields the marginal value density

$$\hat{H}_j^*(\delta) = C_j^* r_j(\delta; M_j^*) + \xi_j^* X_j(\delta; M_j^*), \quad (89)$$

where the $\xi_j^* X_j$ term is exactly the shadow value of equilibrium feedback (changing D_j shifts M_j and hence endogenous willingness-to-pay and prices).

Explicit normalized marginal values. Combining (83) with (89), define

$$F_j(\delta) = r_j(\delta; M_j^*) + \frac{\xi_j^*}{C_j^*} X_j(\delta; M_j^*) - \frac{\nu_j^*}{C_j^*}. \quad (90)$$

These F_j are the score functions appearing in (85).

G.7. Strict convexity and the aggregate Bind–Gap–Bind structure

Throughout this subsection, $X_j(\delta; M_j^*)$ and $r_j(\delta; M_j^*)$ are given by the single-segment formulas (87)–(88) derived from the pricing analysis in Appendix B. By Segmental Exclusivity (Lemma 14), these expressions coincide with the actual equilibrium effective prices and revenues at almost every $\delta \in B$ where segment j is active.

Step 1: strict convexity of each F_j^ .* Fix a segment j and work on the binding set B where SE holds a.e. by Lemma 14. Holding M_j^* fixed, $X_j(\delta; M_j^*)$ is affine in δ by (87). Moreover, since $d_j(\delta) = a_j\delta + b_j$ is affine with $a_j \neq 0$, the per-exposure revenue kernel $r_j(\delta; M_j^*)$ is strictly convex in δ by Lemma 6 in Appendix B. Therefore, (90) implies $F_j^*(\cdot)$ is strictly convex on $[\delta^*, \bar{\delta}^*]$.

Step 2: convexity of η^ and interval structure of the slack set.* By (85), we have that $\eta^*(\delta) = \max \left\{ \max_j F_j(\delta), 0 \right\}$. The pointwise maximum of convex functions is convex, so $\eta^*(\cdot)$ has a convex (hence continuous) representative on $[\delta^*, \bar{\delta}^*]$. Because $\eta^* \geq 0$, its zero set is convex and therefore an interval (up to null sets). Let $[\delta_0, \delta_1]$ denote the (closed) interval equal to $\{\eta^* = 0\}$ up to null sets:

$$\{\delta \in [\delta^*, \bar{\delta}^*] : \eta^*(\delta) = 0\} = [\delta_0, \delta_1] \cap [\delta^*, \bar{\delta}^*] \quad \text{a.e.} \quad (91)$$

Step 3: no exposure on the interior of the slack interval. Fix a segment j and $\delta \in (\delta_0, \delta_1)$ where KKT holds. Then $\eta^*(\delta) = 0$, so stationarity implies $F_j(\delta) = -\frac{\xi_j^*(\delta)}{C_j^*} \leq 0$. If $D_j^*(\delta) > 0$, complementary slackness forces $\xi_j^*(\delta) = 0$, hence $F_j(\delta) = 0$. Thus on the set $A_j := \{\delta \in (\delta_0, \delta_1) : D_j^*(\delta) > 0\}$ we have $F_j(\delta) = 0$.

If A_j had positive Lebesgue measure, it would contain three distinct points $\delta_1 < \delta_2 < \delta_3$ with $\delta_2 = t\delta_1 + (1-t)\delta_3$ for some $t \in (0, 1)$. Strict convexity of F_j would then imply $F_j(\delta_2) < tF_j(\delta_1) + (1-t)F_j(\delta_3) = 0$, which contradicts the equality $F_j(\delta_2) = 0$. Hence A_j has measure zero, i.e.

$$D_j^*(\delta) = 0 \quad \text{for a.e. } \delta \in (\delta_0, \delta_1). \quad (92)$$

Summing over j yields

$$y^*(\delta) = \sum_{j=1}^J C_j^* D_j^*(\delta) = 0 \quad \text{for a.e. } \delta \in (\delta_0, \delta_1). \quad (93)$$

Step 4: aggregate Bind–Gap–Bind. For $\delta \in [\delta^*, \delta_0) \cup (\delta_1, \bar{\delta}^*]$, we have $\eta^*(\delta) > 0$ a.e. by (91). By complementary slackness (82), $\eta^*(\delta) > 0$ implies the capacity constraint binds:

$$y^*(\delta) = s^*(\delta) \quad \text{for a.e. } \delta \in [\delta^*, \delta_0) \cup (\delta_1, \bar{\delta}^*]. \quad (94)$$

Combining (93) and (94) proves Theorem 4(iii).

COROLLARY 2 (Full exposure on the types that receive exposure under D^*). *Under the optimal policy D^* , $\psi(\delta; D^*) = 1$ for almost every δ such that $y^*(\delta) > 0$.*

Proof. By (93)–(94), for almost every δ with $y^*(\delta) > 0$ we have $y^*(\delta) = s^*(\delta)$. The claim follows from the definition (77). $\square$

G.8. Personalized Bind–Gap–Bind and residual-capacity saturation

Fix a segment j and let $[\delta_j^*, \bar{\delta}_j^*]$ be the convex hull of $\text{supp}(D_j^*)$. By (92), $\text{supp}(D_j^*)$ is contained (up to null sets) in $[\delta^*, \delta_0] \cup [\delta_1, \bar{\delta}^*]$. Define the segment- j cutoffs $\delta_{j,0}, \delta_{j,1}$ as the endpoints of the gap in the convex hull of $\text{supp}(D_j^*)$ (with the degenerate convention $\delta_{j,0} = \delta_{j,1}$ if $\text{supp}(D_j^*)$ lies in only one tail). Then $D_j^*(\delta) = 0$ for a.e. $\delta \in (\delta_{j,0}, \delta_{j,1})$, proving the segment-level BGB gap.

Now take δ with $D_j^*(\delta) > 0$ where KKT holds. Then $\delta \notin (\delta_0, \delta_1)$, so $\eta^*(\delta) > 0$ a.e., and complementary slackness gives binding:

$$s^*(\delta) = \sum_{k=1}^J C_k^* D_k^*(\delta).$$

By SE on B (Lemma G.1) and since $D_j^*(\delta) > 0$, we have $D_k^*(\delta) = 0$ for all $k \neq j$ at almost every such δ . Therefore,

$$s^*(\delta) = C_j^* D_j^*(\delta) \quad \text{for a.e. } \delta \text{ with } D_j^*(\delta) > 0,$$

which is equivalent to the residual-capacity saturation condition

$$C_j^* D_j^*(\delta) = s^*(\delta) - \sum_{k \neq j} C_k^* D_k^*(\delta) \quad \text{a.e. on } \text{supp}(D_j^*).$$

This proves Theorem 4(ii). Combined with (93)–(94) (part (iii)) and Lemma 14 (part (i)), this completes the proof of Theorem 4. ■

G.9. Discussion of Assumption 1

The role of Assumption 1. Lemma 14 establishes a necessary condition for the platform to mix segments: if two segments i and j are simultaneously active for a supplier type δ (i.e., $D_i^*(\delta) > 0$ and $D_j^*(\delta) > 0$), their associated scores must be identical, $F_i^*(\delta) = F_j^*(\delta)$. Assumption 1 ensures that this equality does not persist over a continuous range of supplier types. It allows us to conclude that while segments might share a specific supplier type at a crossover point, they cannot be mixed over a non-degenerate interval. Consequently, Segmental Exclusivity holds almost everywhere.

Mathematical definition of a “flat tie”. Consider any two distinct segments i and j . We define the difference in their scores as:

$$g_{ij}(\delta) := F_i^*(\delta) - F_j^*(\delta).$$

A *flat tie* occurs if this difference is zero throughout a non-degenerate interval $I \subseteq [\delta^*, \bar{\delta}^*]$. In such a case, the functions F_i^* and F_j^* would not merely cross at an intersection point; they would coincide perfectly over I .

Genericity. From a structural perspective, flat ties are “knife-edge” scenarios. For two distinct economic functions derived from different primitives (e.g., distinct valuations, arrival rates, or patience levels) to overlap perfectly over an interval requires a precise and fragile alignment of parameters. Such overlaps are non-generic: arbitrarily small perturbations to the market primitives would destroy the alignment, which resolves the indifference.

Formal justification via perturbation. To formalize this genericity, we can introduce a standard tie-breaking perturbation. Let $\gamma_1, \dots, \gamma_J$ be distinct constants for each segment. Consider a perturbed revenue objective that includes a vanishingly small linear preference for the location of recommendations:

$$R_\varepsilon(D, Z) := R(D, Z) + \varepsilon \sum_{j=1}^J \gamma_j \int_0^{\bar{\delta}} \delta D_j(\delta) d\delta, \quad \text{for some } \varepsilon > 0.$$

This perturbation does not alter the feasible set or equilibrium identities; it simply acts as a deterministic tie-breaker. In the resulting KKT system, the stationarity condition for the optimal policy shifts slightly, resulting in perturbed scores:

$$F_{j,\varepsilon}^*(\delta) = F_j^*(\delta) + \varepsilon \frac{\gamma_j}{C_j^*} \delta.$$

Now, consider the difference between the scores of any two segments i and j :

$$F_{i,\varepsilon}^*(\delta) - F_{j,\varepsilon}^*(\delta) = \underbrace{g_{ij}(\delta)}_{\text{Original Difference}} + \varepsilon \underbrace{\left(\frac{\gamma_i}{C_i^*} - \frac{\gamma_j}{C_j^*} \right)}_{\text{Perturbation Term}} \delta.$$

For a flat tie to persist in this perturbed setting, this expression would have to equal zero over an interval. This would require the original score difference $g_{ij}(\delta)$ to differ from zero by *exactly* the linear term $\varepsilon(\dots)\delta$. Since the original functions are fixed by the primitives and ε is an arbitrary parameter we introduced, this precise cancellation is impossible for a generic choice of γ . Thus, for any $\varepsilon > 0$, the set of suppliers where a tie occurs has Lebesgue measure zero. Hence, Assumption 1 essentially selects the robust solution that survives this perturbation argument.

Appendix H: Proof of Proposition 7: Value of Information Decomposition and Welfare Impact

Proof of Proposition 7. We use the decompositions in (4) and (5). Fix the benchmark equilibrium $E^{NP} = E(\mathbf{D}^{NP})$ (Theorem 1).

(i) *Allocative gain.* Since $\mathbf{D}^{NP}$ is feasible in (OPT-F), optimality implies $R^{\text{Re}} \geq R^{NP}$, hence $\text{VoI}^{\text{Allocative}} \geq 0$.

For the CS effect, evaluate all objects at E^{NP} and fix segment j . Let $X_j(\delta) := p^{NP}(\delta) + d_j(\delta)$ be the effective price and $\phi_j(\delta) := \Pr(\omega_j \geq X_j(\delta) \mid E^{NP})$ the sale probability when δ is shown. By Theorem 2, $\omega_j \mid E^{NP} \sim U[\underline{\omega}_j, \bar{\omega}_j]$ with span $\Delta_j := \bar{\omega}_j - \underline{\omega}_j$, hence $\phi_j(\delta) = (\bar{\omega}_j - X_j(\delta))_+ / \Delta_j$. Define the conditional average continuation value

$$\bar{\theta}_j(X) := \mathbb{E}[\theta_j(V) \mid \omega_j(V) \geq X, E^{NP}].$$

Using $V = \omega_j(V) + \theta_j(V)$, the per-impression expected realized surplus from type δ is

$$S_j(\delta) := \mathbb{E}[(V - X_j(\delta)) \mathbf{1}\{\omega_j(V) \geq X_j(\delta)\} \mid \delta, E^{NP}] = \frac{\Delta_j}{2} \phi_j(\delta)^2 + \phi_j(\delta) \bar{\theta}_j(X_j(\delta)). \quad (95)$$

In contrast, the per-impression revenue kernel in (OPT-F) is $K_j(\delta) := p^{NP}(\delta) \phi_j(\delta)$. Since S_j and K_j are not aligned in general, the revenue-maximizing reallocation in (OPT-F) can reduce CS. For instance, take two segments H, L and a type δ with $p^{NP}(\delta) = 1$ and

$$(\phi_H, \Delta_H, \bar{\theta}_H) = (0.9, 0.2, 0), \quad (\phi_L, \Delta_L, \bar{\theta}_L) = (0.6, 1, 0.5).$$

Then $K_H(\delta) > K_L(\delta)$ but, by (95), $S_H(\delta) < S_L(\delta)$. Shifting an ε -mass of exposure for δ from L to H therefore increases frozen revenue while decreasing frozen CS, so $\Delta CS^{\text{Allocative}}$ is sign-ambiguous.

(ii) *Feedback gain.* The sign of $\text{VoI}^{\text{Feedback}} = R^P - R^{\text{Re}}$ is not pinned down because R^{Re} evaluates policies at the fixed environment E^{NP} , whereas R^P is evaluated at the endogenous equilibrium induced by $\mathbf{D}^P$ (Theorem 1).

For customer surplus, the equilibrium adjustment from the Frozen benchmark to E^P operates through the equilibrium-feedback channel. By Segmental Exclusivity (Theorem 4), each active SE block is a single-segment market, so the comparative statics in Propositions 3, 4, and 6 apply blockwise. In particular, any nontrivial “scarcity injection” (a feasible shift of recommendation mass from lower- to higher-misfit supplier types) raises the corresponding mean effective price ($dM_j/d\varepsilon > 0$, cf. (56)) and strictly lowers per-period customer surplus ($dCS_j/d\varepsilon < 0$, Proposition 6 via Lemma 10). Hence $\Delta CS^{\text{Feedback}} \leq 0$, with strict inequality whenever the adjustment is nontrivial. Moreover, the magnitude of this loss is increasing in patience: the threshold sensitivity $\partial\theta_j/\partial M_j < 0$ is larger for smaller μ_j (Appendix D), so a given increase in M_j induces a larger decline in lifetime surplus and hence in CS.

(iii) *Total gain.* Feasibility of the non-personalized policy in the personalized regime implies $R^P \geq R^{NP}$, hence $\text{VoI}^{\text{Total}} \geq 0$. Finally, (5) gives $\Delta CS^{\text{Total}} = \Delta CS^{\text{Allocative}} + \Delta CS^{\text{Feedback}}$; since $\Delta CS^{\text{Allocative}}$ is ambiguous and $\Delta CS^{\text{Feedback}} \leq 0$, the sign of ΔCS^{Total} is ambiguous.

Appendix I: Equilibrium Analysis for Experiment 1 (Thick Market, Patience Heterogeneity)

We study Experiment 1 in the thick-market limit ($m \rightarrow \infty$) with segment impatience μ_j and common valuation bound $\bar{v}_j = 1$. Under the personalized two-point policy, segment j is routed to $\delta = 0$ w.p. $1 - \alpha_j$ and to the market boundary $\delta = \bar{\delta}$ w.p. α_j (i.e., scarcity level α_j).

I.1. Environment and equilibrium characterization

By Theorems 1 and 2, $\omega_j \sim U[\underline{\omega}_j, \bar{\omega}_j]$. With $d_j^{\min} = 0$, Lemma 5(ii) gives $\underline{\omega}_j = \bar{\omega}_j/2$ and hence $\bar{\omega}_j = 2\Delta_j$ with $\Delta_j := \bar{\omega}_j - \underline{\omega}_j$. A type- δ supplier exposed to segment j therefore sets effective price $X_j^*(\delta) = (\bar{\omega}_j + \delta)/2 = \Delta_j + \delta/2$ by (63). Let

$$M_j := \mathbb{E}_{\delta \sim D_j} [X_j^*(\delta)].$$

Specializing the feedback identity (62) to $\bar{v}_j = 1$ and $\bar{\omega}_j = 2\Delta_j$ yields

$$2\Delta_j = \mu_j + (1 - \mu_j)M_j. \quad (96)$$

Let $j^* \in \arg \max_k \Delta_k$ and define the market boundary $\bar{\delta} := 2\Delta_{j^*}$.

Top segment j^ .* Under the two-point recommendation distribution, $M_{j^*} = (1 - \alpha_{j^*})X_{j^*}^*(0) + \alpha_{j^*}X_{j^*}^*(\bar{\delta}) = (1 - \alpha_{j^*})\Delta_{j^*} + \alpha_{j^*}\bar{\delta} = \Delta_{j^*}(1 + \alpha_{j^*})$, so (96) implies

$$\Delta_{j^*}(\alpha_{j^*}) = \frac{\mu_{j^*}}{1 + \mu_{j^*} - (1 - \mu_{j^*})\alpha_{j^*}}. \quad (97)$$

Other segments $j \neq j^$.* At $\delta = \bar{\delta}$, the boundary type is marginal for segment j^* and satisfies $X_{j^*}^*(\bar{\delta}) = \bar{\delta}$, hence it posts $p^*(\bar{\delta}) = X_{j^*}^*(\bar{\delta}) - \bar{\delta} = 0$; consequently, every segment faces effective price $\bar{\delta}$ on the bad arm. Moreover, since $\bar{\delta} > \bar{\omega}_j$ for $j \neq j^*$, these segments never purchase from $\delta = \bar{\delta}$ at any nonnegative posted price, so they do not affect the boundary type’s myopic best response. Thus $M_j = (1 - \alpha_j)\Delta_j + \alpha_j\bar{\delta}$, and (96) gives

$$\Delta_j(\alpha_j | \bar{\delta}) = \frac{\mu_j + (1 - \mu_j)\alpha_j\bar{\delta}}{1 + \mu_j + (1 - \mu_j)\alpha_j}. \quad (98)$$

Equations (97)–(98) with $\bar{\delta} = 2\Delta_{j^*}$ characterize the coupled equilibrium used in Experiment 1.

I.2. Revenue optimization

Per-segment revenue. By Theorem 2 with $v_j \sim U[0, 1]$, the stationary mass of active customers is $C_j = \lambda_j \Delta_j / \mu_j$. Since $p_j^*(0) = X_j^*(0) = \Delta_j$ and all active customers purchase on the good arm (whereas the bad arm yields no trade), the expected commission per interaction is $\mathbb{E}[K_j] = \tau(1 - \alpha_j)\Delta_j$, so

$$R_j(\alpha) = \frac{\tau \lambda_j}{\mu_j} (1 - \alpha_j) \Delta_j(\alpha)^2. \quad (99)$$

Personalized (P). For the top segment, substituting (97) into (99) reduces to maximizing $(1 - \alpha)(1 + \mu_{j^*} - (1 - \mu_{j^*})\alpha)^{-2}$ on $[0, 1]$, with interior critical point $\alpha_{j^*}^* = (1 - 3\mu_{j^*})/(1 - \mu_{j^*})$; hence $\alpha_{j^*}^P = \max\{0, \alpha_{j^*}^*\}$ and $\bar{\delta}^P = 2\Delta_{j^*}(\alpha_{j^*}^P)$. Given $\bar{\delta}^P$, each $j \neq j^*$ solves the scalar maximization of (99) using $\Delta_j(\cdot | \bar{\delta}^P)$ from (98) (FOC quadratic; solved numerically in the code).

Non-personalized (NP). Impose $\alpha_j \equiv \alpha$ and evaluate $R^{NP}(\alpha) = \sum_j R_j(\alpha)$ using (97)–(98) with $\bar{\delta}(\alpha) = 2\Delta_{j^*}(\alpha)$; solve $\max_{\alpha \in [0, 1]} R^{NP}(\alpha)$ numerically.

Appendix J: Capacity-Constrained Analysis (Experiment 2) and Proof of Proposition 8

We consider the stylized two-point market used in Experiment 2 with homogeneous patience $\mu \in (0, 1)$ and segment-specific valuation caps $\bar{v}_j$. Let $D(\alpha) := 1 + \mu - (1 - \mu)\alpha$. Under a segment- j policy α_j , the steady-state customer mass and best-match exposure are $C_j(\alpha_j) = \lambda_j / D(\alpha_j)$ and $Exp_j(\alpha_j) := (1 - \alpha_j)C_j(\alpha_j) = \lambda_j(1 - \alpha_j) / D(\alpha_j)$. Platform revenue from segment j is $R_j(\alpha_j) = \tau \lambda_j \mu (1 - \mu) \bar{v}_j (1 - \alpha_j) / D(\alpha_j)^2$. The “maximum potential demand” for best-match capacity is $\sum_j Exp_j(0) = \sum_j \lambda_j / (1 + \mu)$, so the relative capacity parameter can be written as $M_{\text{rel}} := \bar{M} / \sum_j Exp_j(0) = \bar{M}(1 + \mu) / \sum_j \lambda_j$. The platform solves

$$\max_{\alpha_j \in [0, 1]} \sum_j R_j(\alpha_j) \quad \text{s.t.} \quad \sum_j Exp_j(\alpha_j) \leq \bar{M}.$$

Let $\eta \geq 0$ be the multiplier on the capacity constraint. Since $D'(\alpha) = -(1 - \mu)$, we have $Exp_j'(\alpha) = -2\mu \lambda_j / D(\alpha)^2$ and $R_j'(\alpha) = \tau \lambda_j \mu (1 - \mu) \bar{v}_j ((1 - 3\mu) - (1 - \mu)\alpha) / D(\alpha)^3$. Hence any interior optimizer satisfies $R_j'(\alpha_j) - \eta Exp_j'(\alpha_j) = 0$, which yields

$$\alpha_j(\eta) = \frac{\tau(1 - \mu)\bar{v}_j(1 - 3\mu) + 2\eta(1 + \mu)}{(1 - \mu)(\tau(1 - \mu)\bar{v}_j + 2\eta)}. \quad (100)$$

Let η^* satisfy complementary slackness and define the implemented policy as $\alpha_j^P := \min\{1, \max\{0, \alpha_j(\eta^*)\}\}$.

Proof of Proposition 8. If the capacity constraint binds, then $\eta^* > 0$. From (100), $\alpha_j(\eta^*)$ is linear-fractional in $\bar{v}_j$: $\alpha_j(\eta^*) = (a\bar{v}_j + b)/(c\bar{v}_j + d)$ with $a = \tau(1 - \mu)(1 - 3\mu)$, $b = 2\eta^*(1 + \mu)$, $c = \tau(1 - \mu)^2$, and $d = 2\eta^*(1 - \mu)$. Thus

$$\frac{\partial \alpha_j(\eta^*)}{\partial \bar{v}_j} = \frac{ad - bc}{(c\bar{v}_j + d)^2} = -\frac{8\eta^* \tau \mu}{(2\eta^* + \tau(1 - \mu)\bar{v}_j)^2} < 0.$$

Therefore $\bar{v}_G > \bar{v}_B$ implies $\alpha_G(\eta^*) < \alpha_B(\eta^*)$. Since the clipping map $x \mapsto \min\{1, \max\{0, x\}\}$ is nondecreasing, the ordering is preserved for the implemented policies: $\alpha_G^P \leq \alpha_B^P$ (strict whenever both are interior). If instead the constraint is slack, then $\eta^* = 0$ and the interior solution is valuation-independent, so the two segments receive the same scarcity level. $\square$

Appendix K: Robustness Check: Endogenous Customer Departure

In our baseline model, we assume the per-period departure probability μ is exogenous. This assumption is crucial for analytical tractability and underpins the uniform willingness-to-pay result in Theorem 2. However, one might expect that if we allow customers to react to a worsening search environment, then degrading recommendations causes customers to exit the market more frequently. Consequently, the platform might lose the incentive to induce scarcity. We address this by endogenizing departure, assuming that customers are more likely to leave when the equilibrium search environment deteriorates. This extension reveals a tension between two opposing economic forces. On one hand, a higher departure rate reduces the total mass of active customers, which we term the *attrition effect*. On the other hand, a higher risk of search termination reduces the option value of waiting. This makes the remaining customers more impatient and willing to pay higher prices, which we term the *impatience effect*. Our analysis of the thick-market limit shows that the revenue gains from the impatience effect strictly dominate the losses from attrition. Consequently, endogenizing departure actually strengthens the profitability of the Scarcity by Design strategy. This suggests that the results from our baseline model are conservative.

Model Setup. We focus on a single segment ($J = 1$) with valuations $v \sim U[0, 1]$. Let $M \equiv \mathbb{E}_D[p^*(\delta) + \delta]$ denote the equilibrium mean effective price induced by policy D (i.e., the expected posted price plus misfit). We set $\mu = \mu(M)$ and assume $\mu'(M) = \eta > 0$.

K.1. Impact on the Profitability of Scarcity ($m \rightarrow \infty$)

In the thick-market limit ($d^{\min} = 0$), the baseline is $D^B = \delta_0$. Following the derivation leading to (59) in Appendix D.3, the deviation toward $\bar{\delta}$ is locally profitable iff (59) holds, where Γ is defined as in Appendix D.3 but with *total* derivatives in M (capturing $M \mapsto \mu(M)$): $\Gamma = \frac{dK(0; M)/dM}{1 - dX(0; M)/dM}$. Under D^B , $X(0; M) = \underline{\omega}(M)$ and $K(0; M) = C(M)\underline{\omega}(M)$.

Differentiating Lemma 5(iii) (with $\bar{v}_1 = 1$ and $d^{\min} = 0$) yields

$$\underline{\omega}'(M) = \frac{dX(0; M)}{dM} = \frac{(1 - \mu(M)) + \mu'(M)(1 - M)}{2}. \quad (101)$$

Moreover, by Theorem 2 and Lemma 5(ii), $C(M) = \lambda \underline{\omega}(M)/\mu(M)$, so $K(0; M) = \lambda \underline{\omega}(M)^2/\mu(M)$ and thus $dK(0; M)/dM = K(0; M)(2\underline{\omega}'(M)/\underline{\omega}(M) - \mu'(M)/\mu(M))$. Assume $\underline{\omega}'(M) < 1$ (equivalently, $\mu'(M)(1 - M) < 1 + \mu(M)$) so $1 - dX(0; M)/dM > 0$. Substituting into (59) and cancelling $C(M) > 0$ gives

$$\underline{\omega}(M) \left(2 \frac{\underline{\omega}'(M)}{\underline{\omega}(M)} - \frac{\mu'(M)}{\mu(M)} \right) > 1 - \underline{\omega}'(M),$$

i.e.,

$$3\underline{\omega}'(M) > 1 + \underline{\omega}(M) \frac{\mu'(M)}{\mu(M)}. \quad (102)$$

When $\mu'(M) = 0$, (102) reduces to the benchmark $\mu < 1/3$ in Proposition 5. Under D^B , $M = X(0; M) = \underline{\omega}(M)$, hence Lemma 5(iii) implies $2M = \mu(M) + (1 - \mu(M))M$, so $M = \mu(M)/(1 + \mu(M))$ and $1 - M = \underline{\omega}(M)/\mu(M)$. Evaluated at this baseline, the μ' -increment in the left-hand side of (102) equals $\frac{3}{2} \mu'(M)(1 - M)$, while the right-hand side increases by $\mu'(M)\underline{\omega}(M)/\mu(M) = \mu'(M)(1 - M)$. Thus endogenizing departure strictly relaxes the profitability condition, so the critical impatience threshold exceeds $1/3$.

K.2. Implications for the Main Model

This extension preserves the structural properties of our main equilibrium analysis. Because the departure rate $\mu(M)$ depends on aggregate market conditions rather than individual valuations, the uniform willingness-to-pay property (Theorem 2) remains valid, as do the results regarding existence, uniqueness, and the optimal policy structure. In the thick-market limit, making departure endogenous renders the scarcity strategy even more attractive. The mechanism operates through a feedback loop where raising scarcity increases the mean effective price M . This deterioration in the search environment triggers an increase in the departure rate. While this reduces the mass of active customers in the market, it simultaneously increases the impatience of the remaining pool, leading to higher equilibrium willingness-to-pay and prices. Our analysis demonstrates that the marginal revenue gain from this price increase strictly outweighs the marginal volume loss from attrition around the baseline policy. As a result, the threshold of customer patience required for scarcity to be profitable is strictly higher than in the benchmark case. Hence, our results in this sections suggests that, by treating departure as exogenous, our main model effectively *understates* the platform’s incentive and the potential revenue gains from strategically degrading recommendation quality.